Nile Bank

Enterprise Banking Infrastructure Lab

Complete A-to-Z Infrastructure Project Documentation

Hyper-V | Windows Server | Active Directory | DNS | DHCP | Group Policy
File & Print | Linux | Monitoring | Backup & DR | Security Hardening

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Project Type: Personal Project / Home Lab
Environment: Hyper-V + Windows Server + Linux
Domain: corp.nilebank.lab (NetBIOS: NILEBANK)

Simulated learning environment. Not an actual banking production system.

Table of Contents

1. Introduction, Scope and Conventions . . . 7

1.1 Project objective . . . 7

1.2 Document conventions . . . 7

1.3 Two environments: do not mix them . . . 7

1.4 Reference platform versions . . . 8

2. Host Machine and Resource Strategy . . . 9

2.1 Host specification . . . 9

2.2 Enable Hyper-V . . . 9

2.3 Memory budget . . . 9

2.4 Dynamic Memory recommendations . . . 9

2.5 Run the lab in profiles . . . 10

2.6 Startup priority and order . . . 10

2.7 Ways to reduce resource consumption . . . 10

3. Lab Storage Structure . . . 11

3.1 Antivirus exclusions on the host (optional) . . . 11

4. Hyper-V Network . . . 12

4.1 Switch types . . . 12

4.2 Chosen design . . . 12

4.3 Create the switches (exact steps) . . . 12

4.4 How Internet access flows . . . 13

4.5 Test . . . 13

5. Enterprise Network Design . . . 14

5.1 Logical network architecture . . . 14

5.2 Why segmentation matters (and how it improves security in a real bank) . . . 14

5.3 How VLANs are implemented on one Hyper-V switch . . . 14

5.4 Build RTR01 (Linux router) . . . 14

6. Server Inventory . . . 16

7. Domain Design . . . 17

7.1 Namespace . . . 17

7.2 Concepts . . . 17

7.3 Design decisions . . . 17

8. DC01 — Primary Domain Controller: VM Creation . . . 18

8.1 Design parameters . . . 18

8.2 GUI steps . . . 18

8.3 PowerShell equivalent . . . 18

9. Windows Server Installation — DC01 . . . 19

9.1 DNS configuration: before and after promotion . . . 19

9.2 Time . . . 19

10. Verify DC01 Networking . . . 20

11. Install Active Directory Domain Services and Promote DC01 . . . 21

11.1 Install the role (GUI) . . . 21
11.2 Promote DC01 — Active Directory Domain Services Configuration Wizard . . . 21
12. Verify Active Directory . . . 23
13. DNS Implementation . . . 24
13.1 Architecture . . . 24
13.2 Why domain controllers use internal AD DNS as primary DNS . . . 24
13.3 Configure DNS . . . 24
13.4 GUI check . . . 24
13.5 DNS validation and troubleshooting . . . 25
14. Active Directory Organizational Units . . . 26
14.1 Design . . . 26
14.2 Design rationale: administration and Group Policy, not the org chart . . . 26
14.3 Create the OUs . . . 26
14.3a Redirect default container for new computers . . . 26
14.4 Verification (GUI) . . . 27
15. Users and Security Groups . . . 28
15.1 Group concepts . . . 28
15.2 Test users . . . 28
15.3 Security groups (AGDLP) . . . 28
15.4 Practical permission design . . . 29
16. Service Accounts . . . 30
16.1 Principles . . . 30
16.2 Create accounts . . . 30
17. CLIENT01 — Windows Client and Domain Join . . . 31
17.1 Create the VM . . . 31
17.2 Install Windows 11 Pro . . . 31
17.3 Network (static until DHCP01 exists) . . . 31
17.4 Join the domain (GUI) . . . 31
17.5 Verification . . . 31
18. Group Policy . . . 33
18.1 Concepts . . . 33
18.2 GPO plan . . . 33
18.3 Create the GPOs and links . . . 33
18.4 Inheritance, block and loopback — lab exercise . . . 35
18.5 Apply and verify . . . 35
19. DHCP01 — DHCP Server . . . 37
19.1 Which systems are static and which use DHCP . . . 37
19.2 VM and network . . . 37
19.3 Install and authorize . . . 37
19.4 Scopes . . . 37
19.5 DHCP security . . . 38
19.6 Test . . . 38

20. FILE01 — File Server . . . 39
20.1 Build . . . 39
20.2 Install File and Storage Services . . . 39
20.3 Concepts . . . 39
20.4 Create the folders, shares and permissions . . . 39
20.5 Test procedures . . . 40
21. PRINT01 — Print Server . . . 41
21.1 Build . . . 41
21.2 Install the role . . . 41
21.3 Configure the printer . . . 41
21.4 Client deployment . . . 41
21.5 Test from CLIENT01 . . . 41
21.6 Print troubleshooting . . . 42
22. LINUX01 — Linux Monitoring Server . . . 43
22.1 VM creation . . . 43
22.2 Static IP, SSH, users, updates, firewall . . . 43
22.3 Monitoring options for a home lab . . . 43
22.4 Install Zabbix (recommended) — outline with commands . . . 44
22.5 What to monitor . . . 44
23. DC02 — Secondary Domain Controller . . . 45
23.1 DC02 VM . . . 45
23.2 Install Windows Server and configure the network . . . 45
23.3 Pre-promotion tests . . . 45
23.4 Join DC02 to the domain . . . 46
23.5 Install AD DS and promote to additional domain controller . . . 46
23.6 After promotion: DNS client settings and DHCP . . . 46
24. DC02 Post-Promotion Validation . . . 47
25. Active Directory Replication . . . 48
25.1 Concepts . . . 48
25.2 Commands . . . 48
25.3 What healthy looks like . . . 48
26. Domain Controller Failover Test . . . 50
26.1 Prerequisites . . . 50
26.2 Procedure . . . 50
26.3 Reverse test . . . 50
26.4 Documentation template . . . 50
27. Backup and Disaster Recovery . . . 52
27.1 Architecture . . . 52
27.2 What is backed up . . . 52
27.3 Configure the jobs . . . 52
27.4 RPO and RTO . . . 53
27.5 Recovery scenarios and restore tests . . . 53

27.6 Restore test log . . . 54
28. Security Hardening . . . 55
28.1 Why a bank needs defense in depth . . . 55
28.2 Controls implemented in this lab . . . 55
28.3 Hardening commands . . . 55
28.4 Event Viewer security queries . . . 56
29. Monitoring . . . 57
29.1 Monitoring matrix . . . 57
29.2 Alert rules . . . 57
30. Troubleshooting Runbook . . . 58
30.1 AD login failure . . . 58
30.2 DNS failure . . . 58
30.3 Domain join failure . . . 59
30.4 File access failure . . . 59
30.5 Printer failure . . . 59
31. Event Viewer . . . 60
How an IT Support / System Administrator uses Event Viewer . . . 60
32. Failure and Recovery Exercises . . . 61
32.1 DNS unavailable . . . 61
32.2 DC01 unavailable . . . 61
32.3 DC02 unavailable . . . 61
32.4 DHCP unavailable . . . 61
32.5 File server unavailable . . . 62
32.6 Print Spooler stopped . . . 62
32.7 Wrong NTFS permission . . . 62
32.8 Broken GPO . . . 62
32.9 Domain join failure . . . 63
32.10 Incorrect DNS configuration . . . 63
33. Documentation and Evidence . . . 64
33.1 Naming convention . . . 64
33.2 Evidence checklist . . . 64
33.3 Evidence log template . . . 65
34. Project Acceptance Checklist . . . 66
Acceptance criteria . . . 66
35. Complete Command Reference . . . 67
Additional useful commands . . . 67
36. Architecture Diagram . . . 69
37. IP Address Table . . . 70
38. VM Resource Table . . . 71
39. Real-World Banking Context . . . 72
40. Career Value: IT Support → System Administrator → Cloud Engineer . . . 73

41. CV Section . . . 74

42. LinkedIn Project Description . . . 75

43. Interview Answers . . . 76

43.1 30-second answer . . . 76

43.2 Two-minute answer . . . 76

43.3 Detailed technical answers . . . 76

44. GitHub Repository Structure . . . 78

45. PowerShell Automation (Optional) . . . 79

45.1 New-NileBankOUs.ps1 . . . 79

45.2 New-NileBankGroups.ps1 . . . 79

45.3 New-NileBankUsers.ps1 . . . 79

45.4 Check-NileBankServices.ps1 . . . 80

45.5 Check-NileBankDNS.ps1 . . . 80

45.6 Check-NileBankADHealth.ps1 and Check-NileBankReplication.ps1 . . . 80

45.7 Export-NileBankSystemInfo.ps1 . . . 81

46. Project Phases . . . 82

Phase 1 — Hyper-V . . . 82

Phase 2 — DC01 . . . 82

Phase 3 — AD DS + DNS . . . 82

Phase 4 — OUs + Users + Groups . . . 83

Phase 5 — CLIENT01 . . . 83

Phase 6 — GPO . . . 83

Phase 7 — DHCP01 . . . 84

Phase 8 — FILE01 . . . 84

Phase 9 — PRINT01 . . . 85

Phase 10 — LINUX01 . . . 85

Phase 11 — DC02 . . . 85

Phase 12 — Replication + Failover . . . 86

Phase 13 — Monitoring . . . 86

Phase 14 — Backup . . . 86

Phase 15 — Disaster Recovery . . . 87

Phase 16 — Security Hardening . . . 87

Phase 17 — Failure Testing . . . 87

Phase 18 — Documentation . . . 88

Phase 19 — Portfolio / CV / LinkedIn . . . 88

47. Final Project Report . . . 90

47.1 Executive summary . . . 90

47.2 Business scenario . . . 90

47.3 Requirements . . . 90

47.4 Architecture, network, server, AD and security design . . . 90

47.5 Implementation, testing and failure testing . . . 90

47.6 Backup, DR, monitoring and troubleshooting . . . 90

47.7 Lessons learned . . . 90
47.8 Future improvements, career value, conclusion . . . 91
48. Future Improvements . . . 92
Project Completion Statement . . . 93

1. Introduction, Scope and Conventions

1.1 Project objective

Nile Bank is a **simulated** enterprise banking IT infrastructure built as a hands-on home lab on a single Windows 11 Pro workstation running Microsoft Hyper-V. The lab reproduces the building blocks an enterprise infrastructure team operates: virtualization, network segmentation, centralized identity (Active Directory), core network services (DNS, DHCP), endpoint management (Group Policy), file and print services, Linux administration, monitoring, backup, disaster recovery, security hardening, redundancy (a second domain controller), failure testing and documentation.

This document is written so another IT professional can reproduce the lab from an empty host to a fully validated environment. Every phase lists the exact steps, commands, validation checks, common errors and evidence to capture.

> **WARNING:** This is a learning environment. It is NOT a real banking system, contains no real customer data, and must never be described as production banking infrastructure. Where lab settings differ from what a regulated bank would do, the text is marked **LAB vs PRODUCTION**.

1.2 Document conventions

Marker	Meaning
CMD block (dark title bar)	Commands to run. Prompt is shown only where the shell matters: PS> = elevated PowerShell, CMD> = Command Prompt, $ = Linux shell.
NOTE	Clarification or helpful context.
WARNING	Action that can break the lab, lose data or cause a security problem.
LAB vs PRODUCTION	Marks a deliberate lab simplification and states what a real bank would do instead.
OPTIONAL / ADVANCED	Not required for the core lab. Implement only after the core phases are complete and validated.
TBD — To Be Defined During Implementation	A value deliberately left open (see Section 6). Never guess an address; pick it from the proposed plan in Section 37 and record it.

1.3 Two environments: do not mix them

An earlier throw-away test lab may exist using the domain `lab.local`. That is the **temporary test domain**. The final project uses `corp.nilebank.lab` (NetBIOS `NILEBANK`). The two must never share a network, DNS server, VM or snapshot chain.

Item	Test lab (temporary)	Final project
DNS domain	lab.local	corp.nilebank.lab
NetBIOS	(test value)	NILEBANK
Switch	any test switch	NILEBANK-LAB (+ NILEBANK-WAN)
VM storage	anywhere	D:\NileBank-Lab\
Action	Shut down, export if wanted, then delete before the final build	Build fresh from Phase 1

> **WARNING:** Never reuse a VM, VHDX or checkpoint from the lab.local environment. Re-using old Active Directory VHDX files carries over the old domain, SIDs and DNS records and will corrupt the new design. Build every VM from the ISO.

1.4 Reference platform versions

- **Windows Server 2022 Standard (Desktop Experience)** for DC01, DC02, DHCP01, FILE01, PRINT01, BACKUP01. Windows Server 2019/2025 behave the same for every procedure in this document except where noted.
- **Windows 11 Pro** (Generation 2 VM with virtual TPM) for CLIENT01. Windows 10 Pro is an acceptable lighter alternative.
- **Ubuntu Server 22.04/24.04 LTS** for LINUX01 and RTR01.
- Evaluation ISOs from Microsoft Evaluation Center are valid for a lab (180-day evaluation; rearm limits apply).

2. Host Machine and Resource Strategy

2.1 Host specification

Component	Value	Lab implication
OS	Windows 11 Pro	Includes Hyper-V role (Home edition does not).
CPU	AMD Ryzen 5 2600X (6 cores / 12 threads)	Supports SLAT and virtualization (SVM). Allows ~12 vCPUs total with light over-commit.
RAM	16 GB	The binding constraint. Windows 11 host needs ~4 GB. About 10–11 GB is usable for VMs.
GPU	NVIDIA RTX 3060 Ti	Not used by the lab. Do not attempt GPU partitioning.
Storage	~1.1 TB usable (drive D: used)	Ample. Use dynamically expanding VHDX. Prefer SSD for DC and client VHDX.
Hypervisor	Microsoft Hyper-V	Type-1; Windows host runs as parent partition.

2.2 Enable Hyper-V

Enable Hyper-V (elevated PowerShell)

```
PS> Get-ComputerInfo -Property "HyperV*"
PS> Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All
PS> Restart-Computer
```

If `HyperVRequirementVirtualizationFirmwareEnabled` is False, enable **SVM Mode** in the BIOS (Advanced → CPU Configuration) and retry.

2.3 Memory budget

Budget: 16 GB total − ~4 GB host/Hyper-V overhead = **~12 GB maximum**, but plan to keep running VMs under **10 GB of assigned memory** so the host never pages heavily. This is why the lab is run in **profiles** rather than with all VMs powered on at once.

2.4 Dynamic Memory recommendations

VM	Startup	Minimum	Maximum	Notes
RTR01 (Linux router)	512 MB	256 MB	768 MB	Very light.
DC01	2048 MB	1024 MB	3072 MB	AD DS caches in RAM; do not go below 1 GB.
DC02	2048 MB	1024 MB	3072 MB	Same as DC01.
DHCP01	1024 MB	768 MB	1536 MB	Can be Server Core to save RAM (Optional).
FILE01	1536 MB	1024 MB	2048 MB	SMB caching benefits from headroom.
PRINT01	1536 MB	1024 MB	2048 MB	
LINUX01	1024 MB	512 MB	2048 MB	Monitoring stack (Zabbix/Prometheus) needs ≥1.5 GB under load.
BACKUP01	1536 MB	1024 MB	2048 MB	Power on only for backup windows.
CLIENT01	3072 MB	2048 MB	4096 MB	Windows 11 officially needs 4 GB; dynamic range works in a lab.
MAIL01 (OPTIONAL)	6144 MB	—	8192 MB	Exchange needs ≥ 6–8 GB. Run only alone with DC01. Static memory (no Dynamic Memory).

NOTE: Enable Dynamic Memory only on Windows guests. Exchange and SQL Server should use static memory. Linux guests support Dynamic Memory with Hyper-V integration services (included in the Ubuntu kernel).

Example: apply Dynamic Memory to a VM

```
PS> Set-VM -Name DC01 -DynamicMemory -MemoryStartupBytes 2GB -MemoryMinimumBytes 1GB -MemoryMaximumBytes 3GB
PS> Set-VMProcessor -VMName DC01 -Count 2
PS> Set-VM -Name DC01 -AutomaticStartAction Nothing -AutomaticStopAction ShutDown
```

2.5 Run the lab in profiles

Profile	VMs powered on	Approx. RAM (typical)	Used in phases
A — Core Identity	RTR01, DC01, CLIENT01	~4.5 GB	1–6
B — Core + Redundancy	RTR01, DC01, DC02, CLIENT01	~6.5 GB	11–12, 17
C — Services	RTR01, DC01, DHCP01, FILE01, PRINT01, CLIENT01	~8.5 GB	7–9
D — Operations	RTR01, DC01, DC02, LINUX01, CLIENT01	~8 GB	10, 13
E — Backup / DR	RTR01, DC01, DC02, BACKUP01, FILE01	~8.5 GB	14–15
F — Mail (OPTIONAL)	RTR01, DC01, MAIL01	~9 GB	Optional

2.6 Startup priority and order

Order	VM	Reason	Delay before next
1	RTR01	Routing and Internet NAT; all VLAN gateways.	30 s
2	DC01	AD, DNS, time source. Everything depends on it.	120 s (wait for 'Active Directory Domain Services' to start)
3	DC02	Redundant AD/DNS.	90 s
4	DHCP01	Clients need leases.	30 s
5	FILE01, PRINT01	Need AD and DNS before they register and authenticate.	30 s
6	LINUX01 / BACKUP01	Monitoring and backup start last.	—
7	CLIENT01	Last: needs DNS, DHCP, DCs.	—

Shutdown order is the reverse. Shut down CLIENT01 → services → DC02 → DC01 → RTR01. Never power off a DC with the host power button; use Shut Down from the guest or `Stop-VM`.

Ordered start / stop script (lab)

```
PS> $order = 'RTR01','DC01','DC02','DHCP01','FILE01','PRINT01','LINUX01','CLIENT01'
PS> foreach ($v in $order) { Start-VM -Name $v; Start-Sleep -Seconds 60 }

# Reverse order graceful shutdown
PS> [array]::Reverse($order); foreach ($v in $order) { Stop-VM -Name $v }
```

2.7 Ways to reduce resource consumption

- Power off VMs not needed in the current phase (profiles above). A powered-off VM costs only disk.
- Use **Checkpoints** sparingly. Standard checkpoints of domain controllers are dangerous (USN rollback risk). Use **Production checkpoints** only on non-DC VMs; for DCs use System State backups (Section 27).
- Use Windows Server **Core** for DHCP01 (OPTIONAL) and BACKUP01 where GUI is not required.
- Disable visual effects on server VMs: System → Advanced → Performance → Adjust for best performance.
- Close Hyper-V Manager 'Connect' windows; use Enhanced `Session` only when needed.
- Disable Windows Defender real-time scanning on the *host's* VM folders only if you understand the risk; prefer adding an exclusion for D:\NileBank-Lab\VHDX\ instead.
- Run Windows Update inside VMs one at a time; simultaneous updates cause heavy disk and CPU use.
- Use **Resource Metering** to observe real use: `Enable-VMResourceMetering -VMName DC01`, then `Measure-VM -VMName DC01`.

LAB vs PRODUCTION: A bank runs domain controllers on dedicated, redundant hosts with static, reserved memory and multiple sites. Here, Dynamic Memory and small allocations are acceptable only because the lab holds no real workload.

3. Lab Storage Structure

Directory tree

```
D:\NileBank-Lab\
├── Backups\   (backup targets, exports, System State copies)
├── ISOs\      (Windows Server, Windows 11, Ubuntu installation media)
├── VHDX\      (virtual hard disks, one subfolder per VM)
└── VMs\       (Hyper-V VM configuration: one folder per VM)
```

Directory	Purpose	Notes
D:\NileBank-Lab\Backups	Backup repository: Windows Server Backup targets, VM exports (Export-VM), configuration exports (DNS/DHCP/GPO), script outputs.	Grant access to administrators only. Keep out of any VM's own disk to avoid 'backup on the same disk' failure.
D:\NileBank-Lab\ISOs	Read-only installation media.	Verify SHA-256 hashes after download (Get-FileHash).
D:\NileBank-Lab\VHDX	Virtual disks. Convention: VHDX\DC01\DC01-OS.vhdx, VHDX\FILE01\FILE01-Data.vhdx.	Dynamically expanding. Large I/O stays here.
D:\NileBank-Lab\VMs	Per-VM configuration, snapshots metadata, runtime state, smart paging.	One folder per VM: VMs\DC01, VMs\DC02 ...

Create the structure and set Hyper-V defaults

```
PS> $root = 'D:\NileBank-Lab'
PS> 'Backups','ISOs','VHDX','VMs' | ForEach-Object { New-Item -ItemType Directory -Path "$root\$_" -Force }
PS> 'RTR01','DC01','DC02','DHCP01','FILE01','PRINT01','LINUX01','MAIL01','BACKUP01','CLIENT01' |
      ForEach-Object { New-Item -ItemType Directory -Path "$root\VMs\$_","$root\VHDX\$_" -Force | Out-Null }
PS> Set-VMHost -VirtualMachinePath 'D:\NileBank-Lab\VMs' -VirtualHardDiskPath 'D:\NileBank-Lab\VHDX'
PS> Get-FileHash 'D:\NileBank-Lab\ISOs\*.iso' -Algorithm SHA256
```

NOTE: `Set-VMHost` makes the New Virtual Machine wizard default to these paths. In the wizard, still confirm the path for each VM explicitly: VMs\DC01 for the configuration and VHDX\DC01 for the disk.

3.1 Antivirus exclusions on the host (optional)

Reduce host scan overhead

```
PS> Add-MpPreference -ExclusionPath 'D:\NileBank-Lab\VHDX'
PS> Add-MpPreference -ExclusionProcess 'vmms.exe','vmwp.exe'
```

LAB vs PRODUCTION: Exclusions reduce security on the host. Acceptable in an isolated lab; a bank would use hypervisor-aware endpoint protection configured by the vendor's guidance.

4. Hyper-V Network

4.1 Switch types

Type	Connects	Host access	Internet	Use in this lab
External	VMs ↔ physical NIC ↔ home LAN	Host shares NIC	Yes (same LAN as host)	Not used for the core lab. Exposes VMs (incl. DHCP, DCs) to the home LAN and risks rogue DHCP/DNS.
Internal	VMs ↔ each other ↔ host (virtual NIC)	Yes	No by default; yes via host NAT	**NILEBANK-LAB** (VM LAN) and **NILEBANK-WAN** (NAT side).
Private	VMs ↔ VMs only	No	No	Maximum isolation. Alternative for NILEBANK-LAB if host management access is not needed.

4.2 Chosen design

NILEBANK-LAB (Internal) carries all three VLANs. The Windows host has a virtual NIC on this switch tagged to VLAN 10 so the administrator's PC acts as the management station. **NILEBANK-WAN (Internal + Windows NAT)** connects only the router VM RTR01, giving the lab controlled outbound Internet through two layers (RTR01 NAT, then host NAT). No inbound path from the Internet exists.

- Why isolated: a banking lab contains authentication services (AD), name services and DHCP. A DHCP server or DNS server leaking onto a home LAN breaks other devices; inbound exposure of a Windows domain to the Internet is a serious risk (RDP/SMB/Kerberos attacks).
- Internet access is needed only for Windows Update, package installs and time. Provide it outbound only, through RTR01, and drop all unsolicited inbound traffic.
- If you want a fully air-gapped lab, simply do not connect RTR01's WAN adapter. Download updates through ISO/offline media instead.

4.3 Create the switches (exact steps)

GUI

1. Open **Hyper-V Manager** → right-pane **Virtual Switch Manager**.
2. Select **New virtual network switch** → **Internal** → **Create Virtual Switch**.
3. Name: **NILEBANK-LAB**. Connection type: Internal network. Leave 'Enable virtual LAN identification for management operating system' unchecked here (it is set in PowerShell below because it needs a specific VLAN). Apply.
4. Repeat: Internal switch named **NILEBANK-WAN**. Apply → OK.

PowerShell (equivalent and repeatable)

Create switches, management NIC and NAT

```
PS> New-VMSwitch -Name 'NILEBANK-LAB' -SwitchType Internal
PS> New-VMSwitch -Name 'NILEBANK-WAN' -SwitchType Internal

# Host management NIC on VLAN 10 (access mode)
PS> Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName 'NILEBANK-LAB' -Access -VlanId 10
PS> New-NetIPAddress -InterfaceAlias 'vEthernet (NILEBANK-LAB)' -IPAddress 10.10.10.5 -PrefixLength 24

# NAT side for outbound Internet (host = 192.168.250.1)
PS> New-NetIPAddress -InterfaceAlias 'vEthernet (NILEBANK-WAN)' -IPAddress 192.168.250.1 -PrefixLength 24
PS> New-NetNat -Name 'NILEBANK-NAT' -InternalIPInterfaceAddressPrefix 192.168.250.0/24

PS> Get-VMSwitch | Format-Table Name,SwitchType
PS> Get-NetNat
```

WARNING: Do not set a default gateway on the host's vEthernet (NILEBANK-LAB) adapter. The host already has its own Internet route; a second default gateway breaks host networking.

> **NOTE:** Windows NAT supports one NetNat per host. If an earlier test lab created a NetNat, run `Get-NetNat` and remove it first (`Remove-NetNat`).

4.4 How Internet access flows

Outbound path

```
VM (10.10.30.x) → RTR01 gateway 10.10.30.1 → RTR01 NAT (192.168.250.2)
   → host vEthernet (NILEBANK-WAN) 192.168.250.1 → Windows NAT → physical NIC → home router → Internet
```

4.5 Test

Verify on the host

```
PS> Get-VMSwitch
PS> Get-NetIPAddress -InterfaceAlias 'vEthernet (NILEBANK-LAB)'
PS> Test-NetConnection 192.168.250.1
```

5. Enterprise Network Design

5.1 Logical network architecture

VLAN	Name	Network	Gateway	Purpose
10	Management	10.10.10.0/24	10.10.10.1 (RTR01)	IT administrators, management systems, infrastructure administration. Host vNIC 10.10.10.5.
20	Users / Clients	10.10.20.0/24	10.10.20.1 (RTR01)	Employee workstations, laptops, standard users. Addresses from DHCP.
30	Servers	10.10.30.0/24	10.10.30.1 (RTR01)	Domain controllers, DNS, DHCP, file, print, Linux monitoring, application servers, other infrastructure.

5.2 Why segmentation matters (and how it improves security in a real bank)

- **Blast-radius reduction.** A compromised workstation in the Users segment cannot talk freely to servers; it can reach only the specific ports published to it (DNS, Kerberos, SMB, print).
- **Least-privilege networking.** Only the Management VLAN can reach administrative ports (RDP, WinRM, SSH) on servers. Users cannot attempt RDP to a domain controller.
- **Monitoring and detection.** Inter-VLAN traffic crosses a firewall where it can be logged, rate-limited and inspected; unusual lateral movement stands out.
- **Compliance alignment.** Banking standards (for example PCI DSS network segmentation, regulator IT-risk guidance) expect separation of user, server, management and payment/core networks.
- **Operational stability.** Broadcast traffic and DHCP stay inside their segment, so faults are contained.

5.3 How VLANs are implemented on one Hyper-V switch

Hyper-V virtual switches support 802.1Q VLAN tagging per vNIC. A VM vNIC set in **Access mode** with a VLAN ID can talk only to other members of that VLAN. A **router VM** vNIC set to **Trunk mode** receives all VLANs and routes between them. This design adds one infrastructure VM that is not in the original server list: **RTR01** (Ubuntu Server, 1 vCPU, 512 MB). Without a router, hosts in different VLANs cannot communicate, so CLIENT01 (VLAN 20) could not reach DC01 (VLAN 30).

VM	vNIC	Switch	VLAN mode
RTR01	WAN	NILEBANK-WAN	Untagged
RTR01	LAN	NILEBANK-LAB	**Trunk**, allowed 10,20,30
DC01, DC02, DHCP01, FILE01, PRINT01, LINUX01, MAIL01, BACKUP01	NIC	NILEBANK-LAB	**Access** VLAN 30
CLIENT01	NIC	NILEBANK-LAB	**Access** VLAN 20
Host (management)	vEthernet	NILEBANK-LAB	**Access** VLAN 10

Apply VLAN settings (run after each VM is created)

```
PS> Set-VMNetworkAdapterVlan -VMName DC01     -Access -VlanId 30
PS> Set-VMNetworkAdapterVlan -VMName CLIENT01 -Access -VlanId 20
PS> Set-VMNetworkAdapterVlan -VMName RTR01 -VMNetworkAdapterName LAN -Trunk -AllowedVlanIdList "10,20,30" -NativeVlanId 0
PS> Get-VMNetworkAdapterVlan
```

5.4 Build RTR01 (Linux router)

1. Create Gen 2 VM RTR01: 1 vCPU, 512 MB Dynamic, 8 GB VHDX at D:\NileBank-Lab\VHDX\RTR01\RTR01.vhdx, two vNICs: **WAN** on NILEBANK-WAN and **LAN** on NILEBANK-LAB.
2. Set Secure Boot template: `Set-VMFirmware -VMName RTR01 -SecureBootTemplate MicrosoftUEFICertificateAuthority`.
3. Install Ubuntu Server (minimal). Create admin user `labadmin`. Enable OpenSSH.

4. Identify NIC names with `ip -br link` (typically eth0 = first NIC, eth1 = second; match by MAC in Hyper-V Manager → NIC → Advanced Features if unsure).
5. Write netplan (below), apply, enable forwarding and the firewall.

RTR01 netplan

```
# /etc/netplan/01-rtr01.yaml   (eth0 = WAN, eth1 = LAN trunk)
network:
  version: 2
  ethernets:
    eth0:
      addresses: [192.168.250.2/24]
      routes: [{to: default, via: 192.168.250.1}]
      nameservers: {addresses: [1.1.1.1, 9.9.9.9]}
    eth1: {}
  vlans:
    vlan10: {id: 10, link: eth1, addresses: [10.10.10.1/24]}
    vlan20: {id: 20, link: eth1, addresses: [10.10.20.1/24]}
    vlan30: {id: 30, link: eth1, addresses: [10.10.30.1/24]}
```

Apply and test

```
$ sudo chmod 600 /etc/netplan/01-rtr01.yaml && sudo netplan apply
$ echo 'net.ipv4.ip_forward=1' | sudo tee /etc/sysctl.d/99-forward.conf && sudo sysctl --system
$ ip -br addr
$ ping -c2 192.168.250.1 ; ping -c2 1.1.1.1
```

Baseline firewall and NAT (nftables)

/etc/nftables.conf

```
$ sudo apt install -y nftables && sudo nano /etc/nftables.conf

table inet filter {
  chain input   { type filter hook input priority 0; policy drop;
                  ct state established,related accept; iif lo accept
                  iifname "vlan10" tcp dport 22 accept        # SSH from Management only
                  icmp type echo-request accept }
  chain forward { type filter hook forward priority 0; policy drop;
                  ct state established,related accept
                  iifname "vlan10" accept                      # Management -> everywhere
                  iifname "vlan30" oifname "eth0" accept       # Servers -> Internet (updates)
                  iifname "vlan20" oifname "vlan30" accept     # LAB BASELINE: tightened in Section 28
                  iifname "vlan30" oifname "vlan20" accept
                  iifname "vlan20" oifname "eth0" accept }
}
table ip nat { chain post { type nat hook postrouting priority 100;
                            oifname "eth0" masquerade } }

$ sudo systemctl enable --now nftables && sudo nft list ruleset
```

LAB vs PRODUCTION: The baseline permits all VLAN 20 ↔ 30 traffic to get the lab working first. Section 28 replaces it with a port-specific ruleset. A bank would use a stateful, logged next-generation firewall pair with change-controlled rule sets.

DHCP relay (needed because DHCP01 lives in VLAN 30)

Relay DHCP broadcasts from VLAN 20 to DHCP01

```
$ sudo apt install -y isc-dhcp-relay
  # servers: 10.10.30.12   interfaces: vlan20 vlan30 vlan10   options: (blank)
$ sudo systemctl enable --now isc-dhcp-relay
$ sudo systemctl status isc-dhcp-relay
```

NOTE: If the relay package is unavailable on your release, ISC Kea or dnsmasq relay may be substituted. The relay forwards broadcast DHCPDISCOVER from VLAN 20 to 10.10.30.12 as unicast; DHCP01 chooses the scope by the relay's 10.10.20.1 address.

6. Server Inventory

The inventory below matches the project specification. Addresses not yet defined are shown as **TBD — To Be Defined During Implementation**. A proposed non-conflicting plan is given in Section 37 and should be confirmed when each server is built.

Hostname	Role	IP	Network
DC01	Primary Domain Controller + DNS + GC	10.10.30.10	VLAN 30
DC02	Secondary Domain Controller + DNS + GC	10.10.30.11	VLAN 30
DHCP01	DHCP Server	10.10.30.x — TBD	VLAN 30
FILE01	File Server	10.10.30.x — TBD	VLAN 30
PRINT01	Print Server	10.10.30.x — TBD	VLAN 30
LINUX01	Linux Monitoring Server	10.10.30.x — TBD	VLAN 30
MAIL01	Mail / Exchange Lab (OPTIONAL / ADVANCED)	10.10.30.x — TBD	VLAN 30
BACKUP01	Backup / Recovery	10.10.30.x — TBD	VLAN 30
CLIENT01	Windows Client	10.10.20.x — DHCP	VLAN 20
RTR01 (added)	Inter-VLAN router / NAT / DHCP relay (infrastructure support VM)	10.10.x.1 per VLAN	VLAN 10/20/30 trunk

NOTE: DC01 and DC02 are the only fixed addresses. The reserved address ranges inside VLAN 30 are 10.10.30.1 (gateway), 10.10.30.10–.19 (identity: DCs/DHCP), 10.10.30.20–.49 (file/print/app), 10.10.30.50–.99 (operations/backup/monitoring).

7. Domain Design

7.1 Namespace

Item	Value
DNS name of root domain	corp.nilebank.lab
Forest root domain	corp.nilebank.lab
NetBIOS (pre-Windows 2000) name	NILEBANK
Functional level	Windows Server 2016 (or highest supported by chosen OS)
DNS	Active Directory-integrated zone on DC01 and DC02

WARNING: Never use a name you do not control (for example corp.nilebank.com if you do not own it) and avoid the single-label names and '.local' for production. '.lab' is used here because it is unrouted and clearly not a real domain. Production banks use a registered parent domain with an internal subdomain, e.g. ad.examplebank.com.

7.2 Concepts

Term	Explanation
Domain	A security and administrative boundary holding users, computers and groups, sharing one directory database (NTDS.dit) and policy.
Forest	The top-level security boundary: one or more domains sharing a schema, configuration and Global Catalog. corp.nilebank.lab is the single-domain forest.
Tree	A hierarchy of domains sharing a contiguous DNS namespace (e.g. corp.nilebank.lab → eu.corp.nilebank.lab). This lab has one domain, so one tree.
NetBIOS name	Legacy 15-character flat name (NILEBANK) used in NILEBANK\username and older protocols.
Organizational Unit (OU)	A container inside a domain used to delegate administration and link Group Policy. Not a security boundary.
Global Catalog (GC)	A DC holding a full copy of its domain plus a partial read-only replica of every other domain. Required for UPN logon and universal-group membership lookup. Make every DC a GC in a small forest.
Domain Controller	A server running AD DS that authenticates users (Kerberos/NTLM), serves LDAP and replicates changes.
AD-integrated DNS	DNS zone data stored in Active Directory and replicated with it, with secure dynamic updates, instead of a standalone zone file.

7.3 Design decisions

- Single forest, single domain — appropriate for one organization with one security boundary. More domains add complexity without benefit.
- Two domain controllers, each DNS + GC, so a single failure never prevents logon or name resolution.
- AD-integrated DNS with secure dynamic updates only.
- DC01 holds the five FSMO roles initially (default). They are documented in Section 25 and may be redistributed.

8. DC01 — Primary Domain Controller: VM Creation

8.1 Design parameters

Setting	Value	Reason
Name	DC01	Naming standard: ROLE + two-digit number.
Generation	2	UEFI, Secure Boot, faster boot. Supported by Windows Server 2012 R2+.
vCPU	2	AD DS is I/O and memory bound; 2 vCPUs avoid single-thread stalls.
Memory	2 GB startup, Dynamic 1–3 GB	See Section 2.4.
Disk	60 GB dynamically expanding VHDX	OS ~25 GB + NTDS/SYSVOL/logs growth. Dynamic saves space.
Secure Boot	On, template 'Microsoft Windows'	Prevents unsigned boot loaders. Keep enabled for Windows guests.
Network	NILEBANK-LAB, VLAN 30 access	Servers segment.
Checkpoints	Disabled for DCs	A restored DC checkpoint causes USN rollback and replication divergence.

8.2 GUI steps

1. Hyper-V Manager → **Action → New → Virtual Machine**. Next.
2. Name: **DC01**. Tick 'Store the virtual machine in a different location' → `D:\NileBank-Lab\VMs\DC01`. Next.
3. Generation: **Generation 2**. Next.
4. Startup memory: 2048 MB; tick **Use Dynamic Memory**. Next.
5. Networking: Connection **NILEBANK-LAB**. Next.
6. Virtual hard disk: Create → Name `DC01-OS.vhdx`, Location `D:\NileBank-Lab\VHDX\DC01\`, Size 60 GB. Next.
7. Installation Options: 'Install an operating system from a bootable image file' → browse to `D:\NileBank-Lab\ISOs\` Windows Server ISO. Finish.
8. Right-click DC01 → Settings → Processor: **2** virtual processors. Memory: set Minimum 1024 / Maximum 3072. Security: Secure Boot enabled.
9. Settings → Checkpoints: uncheck **Enable checkpoints**. Management → Integration Services: leave Time synchronization enabled for the lab.
10. Settings → Firmware: confirm the DVD drive is first in the boot order.

8.3 PowerShell equivalent

Create DC01

```
PS> New-VM -Name DC01 -Generation 2 -MemoryStartupBytes 2GB -Path 'D:\NileBank-Lab\VMs' `
      -NewVHDPath 'D:\NileBank-Lab\VHDX\DC01\DC01-OS.vhdx' -NewVHDSizeBytes 60GB -SwitchName 'NILEBANK-LAB'
PS> Set-VMProcessor -VMName DC01 -Count 2
PS> Set-VMMemory -VMName DC01 -DynamicMemoryEnabled $true -MinimumBytes 1GB -StartupBytes 2GB -MaximumBytes 3GB
PS> Set-VM -VMName DC01 -CheckpointType Disabled
PS> Set-VMNetworkAdapterVlan -VMName DC01 -Access -VlanId 30
PS> Add-VMDvdDrive -VMName DC01 -Path 'D:\NileBank-Lab\ISOs\WindowsServer2022.iso'
PS> $dvd = Get-VMDvdDrive -VMName DC01; Set-VMFirmware -VMName DC01 -FirstBootDevice $dvd
PS> Start-VM DC01; vmconnect.exe localhost DC01
```

NOTE: The ISO file name will differ; adjust the path. In the VM console, press any key at 'Press any key to boot from CD or DVD' within five seconds or the VM will attempt network boot.

9. Windows Server Installation — DC01

1. **Boot the ISO.** Press a key at the prompt. Windows Setup loads.
2. Choose language, time and keyboard → **Next → Install now**.
3. **Select edition:** 'Windows Server 2022 Standard (Desktop Experience)'. *Desktop Experience* gives the GUI used throughout this guide. Server Core (no GUI) is smaller but harder to learn on.
4. Accept the license terms. Choose **Custom: Install Windows only**.
5. Select the 60 GB unallocated drive → Next. Setup copies files and restarts several times.
6. **Set the local Administrator password** when prompted (strong, 14+ characters; record it in a lab password vault, never in this document).
7. Sign in: Hyper-V **Action → Ctrl+Alt+Delete**.
8. **Rename the server** (command below). Restart.
9. Install Windows Updates (Settings → Windows Update, or sconfig option 6). Reboot as required.
10. Configure the static IP (below).

Rename and configure the network

```
PS> Rename-Computer -NewName DC01 -Restart
# after reboot
PS> Get-NetAdapter
PS> Rename-NetAdapter -Name 'Ethernet' -NewName 'LAN'
PS> New-NetIPAddress -InterfaceAlias 'LAN' -IPAddress 10.10.30.10 -PrefixLength 24 -DefaultGateway 10.10.30.1
PS> Set-DnsClientServerAddress -InterfaceAlias 'LAN' -ServerAddresses 1.1.1.1   # TEMPORARY, for Windows Update only
PS> Set-TimeZone -Id 'Egypt Standard Time'     # use your own time zone ID: Get-TimeZone -ListAvailable
```

9.1 DNS configuration: before and after promotion

Stage	Preferred DNS	Alternate DNS	Reason
Before promotion (updates)	1.1.1.1 (temporary)	—	Server must reach Windows Update. No AD DNS exists yet.
Immediately before promotion	**10.10.30.10 (itself)**	—	The promotion wizard installs DNS locally and the server must resolve its own SRV records. Public DNS here will break registration.
After promotion, DC01 only	10.10.30.10	—	Single DC: points to itself.
After DC02 exists (Section 23)	10.10.30.11 (partner DC)	10.10.30.10 (itself)	Partner first avoids an isolated 'DNS island' if DC01's DNS is in a bad state; self second keeps the DC resolvable if the partner is down.

Set preferred DNS before promoting

```
PS> Set-DnsClientServerAddress -InterfaceAlias 'LAN' -ServerAddresses 10.10.30.10   # run just before promotion
```

WARNING: A domain controller must never use only a public DNS server (1.1.1.1, 8.8.8.8, the home router). Public DNS cannot answer _ldap._tcp.dc._msdcs.corp.nilebank.lab, so replication, logon and DC location fail. Public resolvers belong only in the DNS **forwarders** list of the DC's DNS server (Section 13).

9.2 Time

Kerberos tolerates only 5 minutes of clock skew. DC01 holds the PDC Emulator role and is the authoritative time source. In the lab the Hyper-V host time and the VM Time Synchronization service are sufficient; optionally point DC01 at an external NTP source after promotion:

OPTIONAL: external NTP on DC01 (PDC Emulator) after promotion

```
CMD> w32tm /config /manualpeerlist:"pool.ntp.org,0x8" /syncfromflags:manual /reliable:yes /update
CMD> w32tm /resync
CMD> w32tm /query /status
```

10. Verify DC01 Networking

Run on DC01

```
CMD> ipconfig /all
CMD> ping 10.10.30.10
CMD> ping 10.10.30.1
CMD> hostname
CMD> whoami
CMD> systeminfo
```

Command	What it validates	Expected result
ipconfig /all	IP, mask, gateway, DNS servers, DHCP state, MAC address, host name.	IPv4 10.10.30.10 (Preferred), mask 255.255.255.0, gateway 10.10.30.1, DHCP Enabled: No, DNS as per stage in 9.1.
ping 10.10.30.10	The local TCP/IP stack and NIC answer on the assigned address.	Replies, TTL=128, 0% loss.
ping 10.10.30.1	Layer-2 reachability to the gateway on VLAN 30 (confirms VLAN tag and RTR01).	Replies. Failure = wrong VLAN ID or RTR01 off.
hostname	The computer name.	DC01
whoami	Current security context.	dc01\administrator before promotion; nilebank\administrator after.
systeminfo	OS version, hotfixes, domain/workgroup, memory, virtualization.	Host Name DC01; Domain WORKGROUP before promotion, corp.nilebank.lab after; System Manufacturer Microsoft Corporation (Hyper-V).

11. Install Active Directory Domain Services and Promote DC01

11.1 Install the role (GUI)

1. **Server Manager → Manage → Add Roles and Features**.
2. Before You Begin → Next. Installation Type: **Role-based or feature-based installation** → Next.
3. Server Selection: select **DC01** from the server pool → Next.
4. Server Roles: tick **Active Directory Domain Services**. In the pop-up click **Add Features** (adds Group Policy Management and AD administration tools) → Next.
5. Features → Next → AD DS page → Next → Confirmation → **Install**. Do not tick 'Restart automatically' (a restart is not required here).
6. When complete, click the yellow flag in Server Manager → **Promote this server to a domain controller**.

PowerShell equivalent

```
PS> Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
PS> Get-WindowsFeature AD-Domain-Services
```

11.2 Promote DC01 — Active Directory Domain Services Configuration Wizard

Wizard page	Choice	Explanation
Deployment Configuration	**Add a new forest**. Root domain name: **corp.nilebank.lab**	Creates the first domain of a new forest. The other options (add DC to existing domain, add child domain) are used for DC02 and future expansion.
Domain Controller Options	Forest and domain functional level: Windows Server 2016. Tick **Domain Name System (DNS) server** and **Global Catalog (GC)**. RODC unavailable (first DC must be writable). Enter the **DSRM password**.	Functional level decides which features are available and the lowest DC OS allowed. GC cannot be unticked on the first DC. DSRM (Directory Services Restore Mode) password is the offline local admin used to restore or repair AD; it is separate from the domain Administrator password — store it securely.
DNS Options	A warning 'A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found' is **expected**. Click Next.	There is no parent zone (nilebank.lab) to delegate from. In a real enterprise, a delegation would be created in the parent zone; in an isolated lab it is not needed.
Additional Options	NetBIOS domain name: **NILEBANK**	Accept the proposed value.
Paths	Database: C:\Windows\NTDS · Log files: C:\Windows\NTDS · SYSVOL: C:\Windows\SYSVOL	NTDS.dit is the AD database; logs are transaction logs; SYSVOL stores GPO files and logon scripts and replicates by DFSR. Lab: defaults are fine. Production: place NTDS/logs on a separate disk from the OS.
Review Options	Optionally click **View script**	Shows the generated PowerShell (Install-ADDSForest) — useful for documentation and repeatability.
Prerequisites Check	Wait for 'All prerequisite checks passed successfully' (warnings about cryptography/NT 4.0 and delegation are normal)	Click **Install**.

The server restarts automatically. Log in as `NILEBANK\Administrator` (the former local Administrator becomes the first domain administrator).

> **OPTIONAL / ADVANCED:** Scripted promotion (equivalent). Run the command in an elevated PowerShell on DC01. You will be prompted for the DSRM password.

Install-ADDSForest

```
PS> Install-ADDSForest -DomainName corp.nilebank.lab -DomainNetbiosName NILEBANK `
      -ForestMode WinThreshold -DomainMode WinThreshold -InstallDns:$true `
      -DatabasePath 'C:\Windows\NTDS' -LogPath 'C:\Windows\NTDS' -SysvolPath 'C:\Windows\SYSVOL' `
      -SafeModeAdministratorPassword (Read-Host -AsSecureString 'DSRM password') -Force
```

LAB vs PRODUCTION: A bank protects the DSRM password in a privileged-access vault, enforces dual control, and restricts who can physically or virtually console a domain controller.

12. Verify Active Directory

Run on DC01

```
PS> whoami
PS> hostname
PS> Get-WindowsFeature AD-Domain-Services
PS> Get-ADDomain
PS> Get-ADForest
PS> Get-ADDomainController
PS> Get-Service NTDS,DNS,Netlogon,KDC,DFSR | Format-Table Name,Status,StartType
PS> net share
```

Command	Expected output
whoami	nilebank\administrator
hostname	DC01
Get-WindowsFeature AD-Domain-Services	[X] Active Directory Domain Services — Install State: Installed
Get-ADDomain	DNSRoot corp.nilebank.lab · NetBIOSName NILEBANK · DomainMode Windows2016Domain · PDCEmulator DC01.corp.nilebank.lab · InfrastructureMaster DC01… · RIDMaster DC01…
Get-ADForest	Name corp.nilebank.lab · ForestMode Windows2016Forest · SchemaMaster DC01… · DomainNamingMaster DC01… · GlobalCatalogs {DC01.corp.nilebank.lab}
Get-ADDomainController	HostName DC01.corp.nilebank.lab · IPv4Address 10.10.30.10 · IsGlobalCatalog True · Site Default-First-Site-Name · OperationMasterRoles {all five}
Get-Service …	NTDS, DNS, Netlogon, KDC, DFSR all Running/Automatic.
net share	NETLOGON and SYSVOL shares exist (confirms the SYSVOL is published and the DC is functional).

NOTE: If Get-ADDomain returns 'Unable to find a default server with Active Directory Web Services running', wait two to three minutes after boot and retry; ADWS starts late. If it persists, check `Get-Service ADWS`.

13. DNS Implementation

13.1 Architecture

The DNS Server role was installed during promotion. The zone `corp.nilebank.lab` is **Active Directory-integrated**: its records live in the directory partition `DomainDnsZones`, and `_msdcs.corp.nilebank.lab` lives in `ForestDnsZones`. Replication of DNS therefore rides on AD replication, supports multi-master updates and allows **secure dynamic updates** (only authenticated, domain-joined machines can register or change their records).

Concept	Description
Forward Lookup Zone	Maps names to IP addresses: DC01.corp.nilebank.lab → 10.10.30.10.
Reverse Lookup Zone	Maps IP to names via PTR records in in-addr.arpa. Needed for logging, some applications and troubleshooting. Zones: 30.10.10.in-addr.arpa, 20.10.10.in-addr.arpa, 10.10.10.in-addr.arpa.
A record	Name → IPv4 address.
PTR record	IPv4 address → name.
SRV record	Service locator, e.g. _ldap._tcp.dc._msdcs.corp.nilebank.lab lists DCs; _kerberos._tcp for Kerberos KDCs. Clients use these to find DCs. Created by the Netlogon service.
Forwarders	Where DNS sends queries it cannot answer (Internet names). Lab: 1.1.1.1, 9.9.9.9.
Root hints	Fallback to the Internet root servers if no forwarders respond.

13.2 Why domain controllers use internal AD DNS as primary DNS

- AD depends on SRV records to locate DCs and services. Only the AD DNS zones hold them. A public resolver returns NXDOMAIN and the DC cannot find its replication partners or its own services.
- Netlogon registers and refreshes the DC's records through dynamic update to the DNS server configured on the NIC. Public DNS would reject those updates.
- Internal DNS keeps internal names private and allows split-brain designs; forwarders handle external names without exposing the domain.
- Using the internal DNS server first and public DNS only as a forwarder gives resolution of both, with logging and security control at the DNS server.

13.3 Configure DNS

DNS configuration

```
PS> Get-WindowsFeature DNS
PS> Get-Service DNS
PS> Get-DnsServerZone

# Forwarders
PS> Set-DnsServerForwarder -IPAddress 1.1.1.1,9.9.9.9
PS> Get-DnsServerForwarder

# Secure dynamic updates on the AD-integrated zone
PS> Set-DnsServerPrimaryZone -Name corp.nilebank.lab -DynamicUpdate Secure

# Reverse lookup zones (AD-integrated, replicated to all DNS servers in the domain)
PS> Add-DnsServerPrimaryZone -NetworkId 10.10.30.0/24 -ReplicationScope Domain -DynamicUpdate Secure
PS> Add-DnsServerPrimaryZone -NetworkId 10.10.20.0/24 -ReplicationScope Domain -DynamicUpdate Secure
PS> Add-DnsServerPrimaryZone -NetworkId 10.10.10.0/24 -ReplicationScope Domain -DynamicUpdate Secure

# Create static A + PTR records for servers not yet joined (example)
PS> Add-DnsServerResourceRecordA -ZoneName corp.nilebank.lab -Name 'MONITOR' -IPv4Address 10.10.30.30 -CreatePtr
PS> Get-DnsServerResourceRecord -ZoneName corp.nilebank.lab | Sort-Object RecordType | Format-Table HostName,RecordType,@

# Scavenging: remove stale dynamic records (lab values)
PS> Set-DnsServerScavenging -ScavengingState $true -ScavengingInterval 7.00:00:00 -RefreshInterval 7.00:00:00 -NoRefreshI
```

13.4 GUI check

1. Server Manager → Tools → **DNS**. Expand DC01 → Forward Lookup Zones.

2. Confirm **corp.nilebank.lab** (Type: Active Directory-Integrated) with folders _msdcs, _sites, _tcp, _udp and host record DC01.
3. Expand Reverse Lookup Zones; confirm the three zones. Right-click DC01 → Properties → Forwarders tab to see the forwarders.

13.5 DNS validation and troubleshooting

DNS tests

```
CMD> nslookup corp.nilebank.lab
CMD> nslookup DC01.corp.nilebank.lab
CMD> nslookup 10.10.30.10
CMD> nslookup -type=SRV _ldap._tcp.dc._msdcs.corp.nilebank.lab
CMD> nslookup www.microsoft.com
CMD> ipconfig /all
CMD> ipconfig /flushdns
CMD> ipconfig /registerdns
PS> Clear-DnsServerCache -Force
PS> dcdiag /test:dns /v
```

Command	Purpose
nslookup corp.nilebank.lab	Resolves the domain name to the IP of every DC (10.10.30.10 now; .11 after DC02). Confirms the server answers authoritatively.
nslookup -type=SRV _ldap._tcp.dc._msdcs…	Proves the DC locator records exist.
nslookup www.microsoft.com	Proves forwarders/Internet resolution work.
ipconfig /flushdns	Clears the client resolver cache (use after changing records).
ipconfig /registerdns	Forces the client to re-register its A/PTR records with its DNS server.
Clear-DnsServerCache	Clears the DNS server cache (not the zone data).
dcdiag /test:dns /v	Verbose DNS health test for the DC (delegation, dynamic update, forwarders, record registration).

Diagnostic sequence for any name resolution problem: (1) Is the client's DNS server correct (`ipconfig /all`)? (2) Can the client reach the DNS server (`ping`, `Test-NetConnection 10.10.30.10 -Port 53`)? (3) Is the DNS service running (`Get-Service DNS`)? (4) Does the record exist (DNS Manager or `Get-DnsServerResourceRecord`)? (5) Flush caches and retry. (6) Check the DNS Server event log.

14. Active Directory Organizational Units

14.1 Design

OU tree

```
corp.nilebank.lab
└── NILEBANK
    ├── Users
    │   ├── IT
    │   ├── Finance
    │   ├── HR
    │   ├── Operations
    │   └── Management
    ├── Computers
    │   ├── Workstations
    │   └── Laptops
    ├── Servers
    │   ├── Domain Controllers   (placeholder — see note)
    │   ├── Application
    │   ├── File
    │   └── Print
    ├── Groups
    └── Service Accounts
```

14.2 Design rationale: administration and Group Policy, not the org chart

- OUs exist for two reasons: **delegating administration** (who may reset passwords or join computers here) and **linking Group Policy** (which settings apply here). Design around those needs.
- Org charts change (reorganizations, renamed departments). Rebuilding OUs and re-linking GPOs after each change is costly and error-prone.
- Separate object types (users, workstations, servers) because they need different policies: user settings (drive maps, desktop restrictions) vs computer settings (firewall, update, security baseline). Servers must not receive workstation lockdown policies and vice versa.
- Department sub-OUs under Users are kept only because this lab uses them for user-targeted policies and delegation (e.g. HR helpdesk may reset HR users). Access to files is granted by **security groups**, not by OU membership.

> **NOTE:** The built-in **Domain Controllers** OU at the domain root must remain the home of real domain controllers: the Default Domain Controllers Policy is linked there and many tools assume it. The OU *Servers\Domain Controllers* requested in the design is a placeholder for documentation/future use (for example staging an RODC or a delegated sub-scope). Do **not** move DC01 or DC02 into it.

14.3 Create the OUs

Create OUs

```
PS> $root = 'DC=corp,DC=nilebank,DC=lab'
PS> New-ADOrganizationalUnit -Name 'NILEBANK' -Path $root -ProtectedFromAccidentalDeletion $true
PS> $top = "OU=NILEBANK,$root"
PS> 'Users','Computers','Servers','Groups','Service Accounts' | ForEach-Object { New-ADOrganizationalUnit -Name $_ -Path
PS> 'IT','Finance','HR','Operations','Management' | ForEach-Object { New-ADOrganizationalUnit -Name $_ -Path 'OU=Users,$t
PS> 'Workstations','Laptops'                        | ForEach-Object { New-ADOrganizationalUnit -Name $_ -Path "OU=Compute
PS> 'Domain Controllers','Application','File','Print' | ForEach-Object { New-ADOrganizationalUnit -Name $_ -Path "OU=Serv
PS> Get-ADOrganizationalUnit -Filter * -SearchBase $top | Sort-Object DistinguishedName | Format-Table Name,Distinguished
```

14.3a Redirect default container for new computers

By default new computers land in the *Computers* container, where most GPOs cannot be linked. Redirect new computer and user objects to the proper OUs.

Default locations

```
PS> redircmp 'OU=Workstations,OU=Computers,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
PS> redirusr 'OU=Users,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
```

14.4 Verification (GUI)

Server Manager → Tools → **Active Directory Users and Computers** → View → Advanced Features. Expand corp.nilebank.lab → NILEBANK. Confirm every OU. Right-click one OU → Properties → Object tab → 'Protect object from accidental deletion' is ticked.

15. Users and Security Groups

15.1 Group concepts

Concept	Explanation
Security group	Has a SID; can be used in permissions (ACLs) and can also receive email if mail-enabled.
Distribution group	Email lists only; no SID in tokens; cannot be used for permissions.
Global group	Members from the same domain; can be used in any domain of the forest. Use for *who* (people by role).
Domain Local group	Members from any domain; used only in ACLs in its own domain. Use for *what access* (permission on a resource).
Universal group	Members from any domain, used anywhere; stored in the Global Catalog. Useful in multi-domain forests; unnecessary in this single-domain lab.
AGDLP	**A**ccounts → **G**lobal groups → **D**omain **L**ocal groups → **P**ermissions. Put users into role (global) groups; put role groups into resource (domain local) groups; assign the permission only to the domain local group.

15.2 Test users

All accounts are fictitious. Sign-in format: `NILEBANK\sAMAccountName` or `sAMAccountName@corp.nilebank.lab`. Passwords are entered interactively at run time; do not store them in the script or in this document.

Department (OU)	Display name	sAMAccountName	Title
IT	Ahmed Hassan	a.hassan	IT Support Engineer (standard account)
IT	Ahmed Hassan (Admin)	adm.a.hassan	Administrative account (Domain Admins) — used only for administration
IT	Salma Adel	s.adel	Help Desk Analyst
Finance	Mona Farouk	m.farouk	Accountant
Finance	Samir Nabil	s.nabil	Finance Manager
HR	Layla Mostafa	l.mostafa	HR Specialist
HR	Dina Salem	d.salem	HR Manager
Operations	Omar Kamal	o.kamal	Branch Operations Officer
Operations	Rania Ibrahim	r.ibrahim	Teller
Management	Karim El-Sayed	k.elsayed	Branch Manager

Create users

```
PS> $base = 'OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
PS> $pw = Read-Host -AsSecureString 'Initial password for lab users'
PS> $users = @(
  @('Ahmed','Hassan','a.hassan','IT','IT Support Engineer'),
  @('Ahmed','Hassan (Admin)','adm.a.hassan','IT','Administrative account'),
  @('Salma','Adel','s.adel','IT','Help Desk Analyst'),
  @('Mona','Farouk','m.farouk','Finance','Accountant'),
  @('Samir','Nabil','s.nabil','Finance','Finance Manager'),
  @('Layla','Mostafa','l.mostafa','HR','HR Specialist'),
  @('Dina','Salem','d.salem','HR','HR Manager'),
  @('Omar','Kamal','o.kamal','Operations','Branch Operations Officer'),
  @('Rania','Ibrahim','r.ibrahim','Operations','Teller'),
  @('Karim','El-Sayed','k.elsayed','Management','Branch Manager'))
PS> foreach ($u in $users) {
      New-ADUser -GivenName $u[0] -Surname $u[1] -Name "$($u[0]) $($u[1])" -SamAccountName $u[2] `
        -UserPrincipalName "$($u[2])@corp.nilebank.lab" -Department $u[3] -Title $u[4] `
        -Path "OU=$($u[3]),OU=Users,$base" -AccountPassword $pw -Enabled $true -ChangePasswordAtLogon $true
    }
PS> Get-ADUser -Filter * -SearchBase "OU=Users,$base" | Select-Object Name,SamAccountName,Enabled
```

15.3 Security groups (AGDLP)

Type	Group name	Purpose / members
Global (role)	GG-IT-Users	All IT staff (a.hassan, s.adel).
Global (role)	GG-IT-Admins	Administrative accounts (adm.a.hassan).
Global (role)	GG-Finance-Users	m.farouk, s.nabil
Global (role)	GG-HR-Users	l.mostafa, d.salem
Global (role)	GG-Operations-Users	o.kamal, r.ibrahim
Global (role)	GG-Management-Users	k.elsayed
Global (role)	GG-AllStaff	Nested: all department groups (for Public share)
Domain Local (resource)	DL-Share-Finance-RW / -R	Modify / Read on \\FILE01\Finance
Domain Local (resource)	DL-Share-HR-RW / -R	Modify / Read on \\FILE01\HR
Domain Local (resource)	DL-Share-IT-RW	Modify on \\FILE01\IT
Domain Local (resource)	DL-Share-Operations-RW / -R	Modify / Read on \\FILE01\Operations
Domain Local (resource)	DL-Share-Public-R / -RW	Read for all staff; Modify for IT
Domain Local (resource)	DL-Print-Finance, DL-Print-General	Print permission on specific printers

Create groups and memberships

```
PS> $g = 'OU=Groups,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
PS> 'IT-Users','IT-Admins','Finance-Users','HR-Users','Operations-Users','Management-Users','AllStaff' |
      ForEach-Object { New-ADGroup -Name "GG-$_" -GroupScope Global -GroupCategory Security -Path $g }
PS> 'Share-Finance-RW','Share-Finance-R','Share-HR-RW','Share-HR-R','Share-IT-RW','Share-Operations-RW','Share-Operations
      ForEach-Object { New-ADGroup -Name "DL-$_" -GroupScope DomainLocal -GroupCategory Security -Path $g }

# Accounts -> Global groups
PS> Add-ADGroupMember GG-IT-Users         a.hassan,s.adel
PS> Add-ADGroupMember GG-IT-Admins        adm.a.hassan
PS> Add-ADGroupMember GG-Finance-Users    m.farouk,s.nabil
PS> Add-ADGroupMember GG-HR-Users         l.mostafa,d.salem
PS> Add-ADGroupMember GG-Operations-Users o.kamal,r.ibrahim
PS> Add-ADGroupMember GG-Management-Users k.elsayed
PS> Add-ADGroupMember GG-AllStaff GG-IT-Users,GG-Finance-Users,GG-HR-Users,GG-Operations-Users,GG-Management-Users

# Global -> Domain Local (resource) groups
PS> Add-ADGroupMember DL-Share-Finance-RW   GG-Finance-Users
PS> Add-ADGroupMember DL-Share-Finance-R    GG-Management-Users
PS> Add-ADGroupMember DL-Share-HR-RW        GG-HR-Users
PS> Add-ADGroupMember DL-Share-IT-RW        GG-IT-Users
PS> Add-ADGroupMember DL-Share-Operations-RW GG-Operations-Users
PS> Add-ADGroupMember DL-Share-Operations-R GG-Management-Users
PS> Add-ADGroupMember DL-Share-Public-R     GG-AllStaff
PS> Add-ADGroupMember DL-Share-Public-RW    GG-IT-Users
PS> Add-ADGroupMember DL-Print-General      GG-AllStaff
PS> Add-ADGroupMember DL-Print-Finance      GG-Finance-Users,GG-Management-Users

# Administrative account separation
PS> Add-ADGroupMember 'Domain Admins' adm.a.hassan
PS> Add-ADGroupMember 'Protected Users' adm.a.hassan
```

15.4 Practical permission design

Example: **Finance share.** The NTFS ACL on D:\Shares\Finance contains only DL-Share-Finance-RW (Modify) and DL-Share-Finance-R (Read), plus Administrators and SYSTEM. Karim (Management) receives Read through GG-Management-Users → DL-Share-Finance-R. When Mona leaves Finance, remove her from GG-Finance-Users once; every resource that group touches updates automatically. No ACL on any server ever lists an individual user.

> **WARNING:** Do not add the administrative account to the same groups as the standard account. Do not use adm.a.hassan for email, web browsing or normal desktop work. Do not add normal users to Domain Admins, Enterprise Admins, Schema Admins or Administrators.

16. Service Accounts

16.1 Principles

- **Separate OU.** NILEBANK\Service Accounts holds only service identities. This allows a distinct GPO/password policy, targeted auditing and clear ownership. They are not users and must not mix with the Users OU.
- **Least privilege.** Grant only the rights the application needs. Never put a service account in Domain Admins for convenience; a service compromise would hand over the domain.
- **Password management.** Long random passwords (25+ characters), no interactive logon rights, 'Deny log on locally' and 'Deny log on through Remote Desktop Services' via GPO, ownership documented.
- **Managed Service Accounts (MSA)** let one server use an account whose password AD rotates automatically. **Group Managed Service Accounts (gMSA)** extend this to several servers (clusters, farms). gMSAs use 240-character passwords rotated automatically (default 30 days) and nobody ever knows or types them.
- Prefer gMSA where the application supports it (Windows services, scheduled tasks, IIS app pools, SQL Server). Use ordinary service accounts only when the software cannot use gMSA.

16.2 Create accounts

Service accounts and gMSA

```
PS> $sa = 'OU=Service Accounts,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
PS> $pw = Read-Host -AsSecureString 'Long random password for svc-monitor'
PS> New-ADUser -Name 'svc-monitor' -SamAccountName svc-monitor -UserPrincipalName svc-monitor@corp.nilebank.lab -Path $sa
      -AccountPassword $pw -Enabled $true -PasswordNeverExpires $true -CannotChangePassword $true `
      -Description 'Read-only LDAP bind for Linux monitoring (LINUX01). Owner: IT Infrastructure'

# gMSA for the backup service (OPTIONAL / ADVANCED)
PS> Add-KdsRootKey -EffectiveTime ((Get-Date).AddHours(-10))     # LAB ONLY: skips the 10-hour replication wait
PS> New-ADGroup -Name 'GG-Servers-Backup' -GroupScope Global -Path 'OU=Groups,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
PS> New-ADServiceAccount -Name 'gmsa-backup' -DNSHostName 'gmsa-backup.corp.nilebank.lab' `
      -PrincipalsAllowedToRetrieveManagedPassword 'GG-Servers-Backup' -Path $sa
PS> Add-ADGroupMember GG-Servers-Backup 'BACKUP01$'      # after BACKUP01 is joined (Section 27)

# On BACKUP01:
PS> Install-WindowsFeature RSAT-AD-PowerShell
PS> Install-ADServiceAccount gmsa-backup
PS> Test-ADServiceAccount gmsa-backup            # must return True
```

LAB vs PRODUCTION: In production, the KDS root key's effective time must be left to replicate (10 hours). The back-dated command above is a lab shortcut and should not be used in a real forest.

17. CLIENT01 — Windows Client and Domain Join

17.1 Create the VM

Create CLIENT01 (VLAN 20)

```
PS> New-VM -Name CLIENT01 -Generation 2 -MemoryStartupBytes 3GB -Path 'D:\NileBank-Lab\VMs' `
      -NewVHDPath 'D:\NileBank-Lab\VHDX\CLIENT01\CLIENT01-OS.vhdx' -NewVHDSizeBytes 60GB -SwitchName 'NILEBANK-LAB'
PS> Set-VMProcessor CLIENT01 -Count 2
PS> Set-VMMemory CLIENT01 -DynamicMemoryEnabled $true -MinimumBytes 2GB -StartupBytes 3GB -MaximumBytes 4GB
PS> Set-VMNetworkAdapterVlan -VMName CLIENT01 -Access -VlanId 20
PS> Set-VMKeyProtector -VMName CLIENT01 -NewLocalKeyProtector
PS> Enable-VMTPM -VMName CLIENT01                    # Windows 11 requires a TPM
PS> Add-VMDvdDrive -VMName CLIENT01 -Path 'D:\NileBank-Lab\ISOs\Win11.iso'
PS> Set-VMFirmware CLIENT01 -FirstBootDevice (Get-VMDvdDrive -VMName CLIENT01)
PS> Set-VM CLIENT01 -CheckpointType Production
```

17.2 Install Windows 11 Pro

1. Boot ISO, install **Windows 11 Pro**. At the network page there is no Internet unless RTR01 is up; choose 'I don't have internet' or press **Shift+F10** and run OOBE\BYPASSNRO (if offered by the build).
2. Choose **Set up for work or school → Sign-in options → Domain join instead**, create a local account `localadmin`.
3. After first sign-in rename the computer: `Rename-Computer -NewName CLIENT01 -Restart`.

17.3 Network (static until DHCP01 exists)

CLIENT01 networking

```
PS> New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress 10.10.20.50 -PrefixLength 24 -DefaultGateway 10.10.20.1
PS> Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses 10.10.30.10
PS> ping 10.10.20.1
PS> ping 10.10.30.10
PS> nslookup corp.nilebank.lab
```

> **NOTE:** Pinging DC01 from CLIENT01 proves VLAN 20 → VLAN 30 routing through RTR01. If it fails, check the VLAN IDs (`Get-VMNetworkAdapterVlan`), the RTR01 trunk, and 'ip_forward' on RTR01 before touching Windows. The client DNS **must** be the domain controller, not the home router and not 1.1.1.1.

17.4 Join the domain (GUI)

1. Settings → System → About → **Domain or workgroup** (or run `sysdm.cpl`) → **Change…**.
2. Select **Domain**, enter `corp.nilebank.lab` → OK.
3. Credentials prompt: `adm.a.hassan` (or NILEBANK\Administrator) and password. 'Welcome to the corp.nilebank.lab domain' appears.
4. Restart. At the sign-in screen choose **Other user** and sign in as `NILEBANK\m.farouk`; change the password when forced.

PowerShell equivalent (places computer in the Workstations OU)

```
PS> Add-Computer -DomainName corp.nilebank.lab `
      -OUPath 'OU=Workstations,OU=Computers,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab' `
      -Credential (Get-Credential NILEBANK\adm.a.hassan) -Restart
```

17.5 Verification

Run on CLIENT01 as a domain user

```
CMD> whoami
CMD> systeminfo | findstr /B /C:"Domain"
CMD> nslookup corp.nilebank.lab
CMD> gpresult /r
CMD> echo %LOGONSERVER%
CMD> nltest /dsgetdc:corp.nilebank.lab
```

Command	Expected
whoami	nilebank\m.farouk
systeminfo | findstr /B /C:"Domain"	Domain: corp.nilebank.lab
nslookup corp.nilebank.lab	Server: dc01.corp.nilebank.lab; Address 10.10.30.10
gpresult /r	Lists the 'Computer Settings' and 'User Settings', the site, the DC used and applied GPOs (Default Domain Policy at minimum).
%LOGONSERVER%	\\DC01 (the DC that authenticated the user)
nltest /dsgetdc:...	Returns DC name DC01, address, flags: GC DS LDAP KDC TIMESERV WRITABLE

On DC01 confirm the computer account: `Get-ADComputer CLIENT01 | Format-List Name,DistinguishedName,Enabled` shows it in OU=Workstations.

18. Group Policy

18.1 Concepts

Concept	Explanation
GPMC	Group Policy Management Console (Server Manager → Tools → Group Policy Management; gpmc.msc). Create, link, back up and report on GPOs.
GPO	A set of settings stored in AD (Group Policy Container) and SYSVOL (Group Policy Template).
Linking / OU linking	A GPO takes effect only where it is linked: site, domain or OU. Processing order: Local → Site → Domain → OU (LSDOU); the last applied wins on conflict.
Security filtering	Restricts a GPO to specific users/computers/groups through the 'Authenticated Users' entry in the Scope tab. Apply = 'Read' + 'Apply group policy' permission.
Inheritance	Child OUs inherit GPOs linked to parents.
Enforced	Marks a link so lower-level GPOs cannot override it and it overrides Block Inheritance. Use rarely (security baselines).
Block Inheritance	OU-level setting that stops inherited GPOs (except Enforced). Use sparingly: it hides policy and complicates troubleshooting.
Loopback processing	Makes the *computer's* OU decide user settings (Merge or Replace). Used on kiosks, shared PCs, RDS servers.
WMI filters	Dynamic targeting by OS version, etc. Slower; avoid unless needed.

18.2 GPO plan

#	GPO name	Linked to	Type	Key settings
1	GPO-Domain-PasswordPolicy	Domain root (account policies only apply here)	Computer	Min length 14, history 24, max age 90 d, min age 1 d, complexity on. *Alternative: Fine-Grained Password Policy (PSO).*
2	GPO-Domain-AccountLockout	Domain root	Computer	Threshold 5 invalid attempts, duration 15 min, reset counter 15 min
3	GPO-Users-ScreenLock	OU=Users	User	Screen saver on, timeout 600 s, password-protected on resume. Also 'Interactive logon: Machine inactivity limit' 900 s (computer).
4	GPO-Computers-Defender	OU=Computers, OU=Servers	Computer	Real-time protection on, cloud protection, scheduled scan, PUA block.
5	GPO-Computers-WindowsUpdate	OU=Computers	Computer	Auto download + install, active hours 8–18, no auto restart with logged-on users. (Servers: separate GPO with scheduled install).
6	GPO-Users-Restrictions	OU=Users (not IT)	User	Deny Control Panel/Settings, block cmd/registry editor, block removable storage write, hide drives.
7	GPO-Users-DriveMaps	OU=Users	User (Preferences)	Item-level targeting by security group: F: Finance, H: HR, I: IT, O: Operations, P: Public.
8	GPO-Computers-SecurityBaseline	OU=Computers, OU=Servers (Enforced)	Computer	Firewall on all profiles, SMBv1 disabled, NTLMv2 only, audit policy, Remote Desktop with NLA, LLMNR disabled, local admin controls.

18.3 Create the GPOs and links

Use Group Policy Management (GPMC): right-click the target → **Create a GPO in this domain, and Link it here…** → name → right-click the new link → **Edit**. The PowerShell below creates and links the GPOs; settings are then configured in the editor or by cmdlet.

Create and link

```
PS> Import-Module GroupPolicy
PS> $dom = 'DC=corp,DC=nilebank,DC=lab'; $top = "OU=NILEBANK,$dom"
PS> New-GPO -Name 'GPO-Domain-PasswordPolicy'     | New-GPLink -Target $dom
PS> New-GPO -Name 'GPO-Domain-AccountLockout'     | New-GPLink -Target $dom
PS> New-GPO -Name 'GPO-Users-ScreenLock'          | New-GPLink -Target "OU=Users,$top"
PS> New-GPO -Name 'GPO-Computers-Defender'        | New-GPLink -Target "OU=Computers,$top"
PS> New-GPO -Name 'GPO-Computers-WindowsUpdate'   | New-GPLink -Target "OU=Computers,$top"
PS> New-GPO -Name 'GPO-Users-Restrictions'        | New-GPLink -Target "OU=Users,$top"
PS> New-GPO -Name 'GPO-Users-DriveMaps'           | New-GPLink -Target "OU=Users,$top"
PS> New-GPO -Name 'GPO-Computers-SecurityBaseline'| New-GPLink -Target "OU=Computers,$top" -Enforced Yes
PS> Get-GPO -All | Format-Table DisplayName,GpoStatus
```

GPO 1 and 2 — Password and lockout (Computer Configuration)

1. GPMC → right-click **GPO-Domain-PasswordPolicy** → Edit.
2. Computer Configuration → Policies → Windows Settings → Security Settings → **Account Policies → Password Policy**.
3. Set: Enforce password history **24**; Maximum password age **90** days; Minimum password age **1** day; Minimum password length **14**; Password must meet complexity requirements **Enabled**.
4. Edit **GPO-Domain-AccountLockout** → Account Policies → **Account Lockout Policy**: Threshold **5**, Duration **15** min, Reset counter after **15** min.

> **NOTE:** Domain account policies only take effect from a GPO linked at the **domain root** (the Default Domain Policy normally holds them). Having separate GPOs at the root is valid for clarity, but make sure only one GPO defines each setting — the highest precedence wins. Best practice for differentiated rules (e.g. admins stricter) is a Fine-Grained Password Policy:

OPTIONAL: PSO for administrators

```
PS> New-ADFineGrainedPasswordPolicy -Name 'PSO-Admins' -Precedence 10 -MinPasswordLength 16 -PasswordHistoryCount 24 `
      -ComplexityEnabled $true -LockoutThreshold 3 -LockoutDuration 00:30:00 -LockoutObservationWindow 00:30:00 `
      -MaxPasswordAge 60.00:00:00 -ReversibleEncryptionEnabled $false
PS> Add-ADFineGrainedPasswordPolicySubject 'PSO-Admins' -Subjects 'GG-IT-Admins'
```

GPO 3 — Screen lock (User Configuration)

User Configuration → Policies → Administrative Templates → Control Panel → Personalization: **Enable screen saver** = Enabled; **Screen saver timeout** = 600; **Password protect the screen saver** = Enabled; **Force specific screen saver** = scrnsave.scr. Computer Configuration → Windows Settings → Security Settings → Local Policies → Security Options: **Interactive logon: Machine inactivity limit** = 900.

GPO 4 — Microsoft Defender

Computer Configuration → Administrative Templates → Windows Components → Microsoft Defender Antivirus: **Turn off Microsoft Defender Antivirus** = Not configured/Disabled; Real-time Protection → **Turn on real-time protection** = Enabled; MAPS → **Join Microsoft MAPS** = Advanced (optional); Scan → **Specify the scan type to use for a scheduled scan** = Quick scan; **Configure detection for potentially unwanted applications** = Block.

GPO 5 — Windows Update

Computer Configuration → Administrative Templates → Windows Components → Windows Update → **Manage end user experience → Configure Automatic Updates** = Enabled, 4 – Auto download and schedule the install; **No auto-restart with logged on users for scheduled automatic updates installations** = Enabled; **Specify active hours** 08:00–18:00. Servers: use a separate GPO on OU=Servers with installs on a maintenance day and no automatic restart on DCs.

> **LAB vs PRODUCTION:** A bank patches through WSUS/SCCM/Intune or a patch-management platform with test rings, change approval and maintenance windows; DCs and critical servers are patched in controlled sequence. Auto-update directly from Microsoft is a lab simplification.

GPO 6 — Standard user restrictions

User Configuration → Administrative Templates: **Control Panel → Prohibit access to Control Panel and PC settings** = Enabled; **System → Prevent access to the command prompt** = Enabled (script disable = No); **System → Prevent access to registry editing tools** = Enabled; **System → Removable Storage**

Access → All Removable Storage classes: Deny all access = Enabled. **Security filtering:** in the Scope tab remove 'Authenticated Users', add **GG-AllStaff** with Read+Apply, then add **GG-IT-Users** with **Deny → Apply group policy** (Delegation → Advanced) so IT staff are exempt.

GPO 7 — Drive mapping (Group Policy Preferences)

User Configuration → Preferences → Windows Settings → Drive Maps → New → Mapped Drive. Action **Update**; Location `\\FILE01.corp.nilebank.lab\Finance`; Reconnect on; Drive letter F; Label 'Finance'. On the **Common** tab tick **Item-level targeting** → Targeting → New Item → **Security Group** = GG-Finance-Users, 'User in group'. Repeat for H (HR), I (IT), O (Operations), P (Public → GG-AllStaff). Do this after FILE01 exists (Section 20).

GPO 8 — Security baseline

Computer Configuration → Policies → Windows Settings → Security Settings: **Windows Defender Firewall with Advanced Security** — Domain, Private and Public profiles: Firewall state On, inbound default Block. **Local Policies → Security Options**: 'Network security: LAN Manager authentication level' = Send NTLMv2 response only, refuse LM & NTLM; 'Interactive logon: Do not display last user name' = Enabled. **Advanced Audit Policy**: enable Success+Failure for Account Logon, Logon/Logoff, Account Management, Directory Service Changes, Policy Change, Privilege Use, Object Access (File System). Administrative Templates → Network → DNS Client → **Turn off multicast name resolution** (LLMNR) = Enabled. Disable SMBv1: `Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol` (or Preferences → Registry). Consider Microsoft's Security Compliance Toolkit baselines (OPTIONAL / ADVANCED) imported with `Import-GPO`.

18.4 Inheritance, block and loopback — lab exercise

1. Link **GPO-Users-Restrictions** to OU=Users. Sign in as m.farouk on CLIENT01: Control Panel is blocked.
2. Sign in as s.adel (IT): the Deny entry exempts IT. Control Panel opens.
3. **Block Inheritance** on OU=Users\Management (right-click → Block Inheritance). Sign in as k.elsayed: the restrictions disappear, demonstrating how a blocked OU escapes inherited policy. Run `gpresult /r` to see that GPO-Users-Restrictions is no longer applied.
4. Mark the security baseline link as **Enforced** and observe that Block Inheritance on a child OU no longer removes it.
5. **Loopback:** create GPO-Computers-KioskLoopback on OU=Computers\Laptops with User Configuration → Administrative Templates → System → Group Policy → **Configure user Group Policy loopback processing mode** = Enabled, Mode: Merge. Verify with `gpresult /h` that user settings from the computer's OU apply to every user who logs on there.

18.5 Apply and verify

Client / DC

```
CMD> gpupdate /force
CMD> gpresult /r
CMD> mkdir C:\Temp
CMD> gpresult /h C:\Temp\gpresult.html
CMD> start C:\Temp\gpresult.html
CMD> rsop.msc
PS> Get-GPOReport -Name 'GPO-Users-Restrictions' -ReportType Html -Path C:\Temp\GPO-Users-Restrictions.html
PS> Backup-GPO -All -Path 'D:\NileBank-Lab\Backups\GPO'
```

Command	Purpose
gpupdate /force	Reapplies all policies immediately (computer and user). Some settings need logoff/restart.
gpresult /r	Text summary of applied/denied GPOs, security groups and the DC used.
gpresult /h file.html	Detailed HTML report including the winning GPO per setting; best evidence for documentation.
rsop.msc	Resultant Set of Policy GUI (shows effective settings; some editions limit it to local data).
Backup-GPO -All	Exports every GPO to a backup folder for recovery (Section 27).

Common GPO error	Cause	Fix
GPO not applying	Linked to wrong OU; computer/user in different OU; security filtering excludes object; GPO link disabled; replication delay.	gpresult /r → 'Denied GPOs'. Check object OU, filtering and Delegation tab.
User setting in Computer GPO	Wrong configuration half or loopback absent.	Move setting or enable loopback.
'Authenticated Users' removed → GPO fails	After MS16-072 patch, computers need Read right on user GPOs.	Always keep **Authenticated Users: Read** (without Apply) when using group-based filtering, or add Domain Computers Read.
Password policy not changing	Defined in a GPO not linked at domain root, or Default Domain Policy has higher precedence.	Set in one GPO at the root; check Link Order.

19. DHCP01 — DHCP Server

19.1 Which systems are static and which use DHCP

Static (fixed)	DHCP (dynamic)
Domain Controllers (DC01, DC02), DNS servers, DHCP servers themselves	User workstations (CLIENT01 and future desktops)
All servers (FILE01, PRINT01, LINUX01, BACKUP01, MAIL01)	Laptops
Network infrastructure (RTR01 interfaces, switches, firewalls)	Non-critical clients and guest devices
Printers that users map by IP (or DHCP reservations)	Anything that moves or needs no inbound access

19.2 VM and network

Create DHCP01 (then install Windows Server as in Section 9)

```
PS> New-VM -Name DHCP01 -Generation 2 -MemoryStartupBytes 1GB -Path 'D:\NileBank-Lab\VMs' `
      -NewVHDPath 'D:\NileBank-Lab\VHDX\DHCP01\DHCP01-OS.vhdx' -NewVHDSizeBytes 50GB -SwitchName 'NILEBANK-LAB'
PS> Set-VMNetworkAdapterVlan -VMName DHCP01 -Access -VlanId 30
PS> Set-VMMemory DHCP01 -DynamicMemoryEnabled $true -MinimumBytes 768MB -MaximumBytes 1536MB
```

Static IP and domain join (IP proposed in Section 37)

```
# On DHCP01 after install:
PS> Rename-Computer DHCP01 -Restart
PS> New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress 10.10.30.12 -PrefixLength 24 -DefaultGateway 10.10.30.1
PS> Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses 10.10.30.10,10.10.30.11
PS> Add-Computer -DomainName corp.nilebank.lab -OUPath 'OU=Application,OU=Servers,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
```

> **NOTE:** DHCP01 is built before DC02 in the phase order. Until DC02 exists, set DNS to 10.10.30.10 only and add 10.10.30.11 in Phase 11.

19.3 Install and authorize

1. Server Manager → Add Roles and Features → **DHCP Server** → Add Features → Install.
2. Click the post-deployment notification **Complete DHCP configuration**. The wizard creates the local groups *DHCP Administrators* and *DHCP Users* and **authorizes the server in Active Directory** (use credentials of a Domain/Enterprise Admin: NILEBANK\adm.a.hassan). **Authorization** stops rogue DHCP servers from answering clients in the domain.
3. Verify in DHCP console (Tools → DHCP): server icon shows a green arrow (authorized).

Install, create groups, authorize in AD

```
PS> Install-WindowsFeature DHCP -IncludeManagementTools
PS> netsh dhcp add securitygroups
PS> Restart-Service DHCPServer
PS> Add-DhcpServerInDC -DnsName DHCP01.corp.nilebank.lab -IPAddress 10.10.30.12
PS> Set-ItemProperty -Path HKLM:\SOFTWARE\Microsoft\ServerManager\Roles\12 -Name ConfigurationState -Value 2
PS> Get-DhcpServerInDC
```

19.4 Scopes

Setting	VLAN 20 — Users	VLAN 10 — Management (OPTIONAL)
Scope name	Users-VLAN20	Mgmt-VLAN10
Network	10.10.20.0/24	10.10.10.0/24
Range	10.10.20.100 – 10.10.20.199	10.10.10.100 – 10.10.10.119
Exclusions	10.10.20.1 – 10.10.20.99 (gateway, static, reservations); 10.10.20.200 – .254 (future)	10.10.10.1 – 10.10.10.99
Option 003 Router	10.10.20.1	10.10.10.1
Option 006 DNS	10.10.30.10, 10.10.30.11	10.10.30.10, 10.10.30.11
Option 015 DNS domain	corp.nilebank.lab	corp.nilebank.lab

Setting	VLAN 20 — Users	VLAN 10 — Management (OPTIONAL)
Lease duration	8 days (default) — workstations on a stable network. Use 1–8 hours for guest/high-turnover networks.	1 day
Reservations	Printers and special clients (by MAC)	Admin devices

Create scope, options, exclusions and reservation

```
PS> Add-DhcpServerv4Scope -Name 'Users-VLAN20' -StartRange 10.10.20.100 -EndRange 10.10.20.199 -SubnetMask 255.255.255.0
PS> Set-DhcpServerv4OptionValue -ScopeId 10.10.20.0 -Router 10.10.20.1 -DnsServer 10.10.30.10,10.10.30.11 -DnsDomain corp
PS> Add-DhcpServerv4ExclusionRange -ScopeId 10.10.20.0 -StartRange 10.10.20.1 -EndRange 10.10.20.99

# Reservation example (use CLIENT01's MAC from: Get-VMNetworkAdapter -VMName CLIENT01)
PS> Add-DhcpServerv4Reservation -ScopeId 10.10.20.0 -IPAddress 10.10.20.50 -ClientId '00-15-5D-xx-xx-xx' -Description 'CL

# DNS dynamic update credentials (DHCP registers client records on behalf of clients that cannot)
PS> Set-DhcpServerv4DnsSetting -DynamicUpdates OnClientRequest -DeleteDnsRROnLeaseExpiry $true
PS> Set-DhcpServerDnsCredential -Credential (Get-Credential NILEBANK\svc-dhcp-dns)    # dedicated low-privilege account (
PS> Get-DhcpServerv4Scope | Format-Table ScopeId,Name,State
PS> Get-DhcpServerv4Lease -ScopeId 10.10.20.0
```

19.5 DHCP security

- **Authorize** the server in AD; never run unauthorized DHCP on production networks. Enable *DHCP Snooping* on physical switches (a bank would) to block rogue DHCP; Hyper-V equivalent: `Set-VMNetworkAdapter -VMName CLIENT01 -DhcpGuard On` on every non-DHCP VM.
- Use a **dedicated low-privilege account** for DHCP DNS credentials (not Domain Admin); place it in Service Accounts. Avoid putting the DHCP server in 'DnsUpdateProxy' unless understood (it permits record hijack on stale registrations).
- Restrict DHCP administration to *DHCP Administrators*. Enable audit logging (default %windir%\System32\dhcp).
- Back up the database: `Backup-DhcpServer -Path D:\NileBank-Lab\Backups\DHCP`.
- Use **DHCP failover** (OPTIONAL / ADVANCED) with a second server for resilience: `Add-DhcpServerv4Failover`. Not implemented in the baseline lab with only one DHCP server; DHCP01 downtime is demonstrated in Section 32.

19.6 Test

Switch CLIENT01 to DHCP and verify

```
CMD> ipconfig /release && ipconfig /renew
CMD> ipconfig /all
PS> Get-DhcpServerv4Lease -ScopeId 10.10.20.0
PS> Get-WinEvent -LogName Microsoft-Windows-DHCP-Server/Operational -MaxEvents 20
```

Change CLIENT01 to automatic addressing (`Set-NetIPInterface -InterfaceAlias Ethernet -Dhcp Enabled; Set-DnsClientServerAddress -InterfaceAlias Ethernet -ResetServerAddresses; Remove-NetRoute -InterfaceAlias Ethernet -NextHop 10.10.20.1 -Confirm:$false`) then renew. Expect an address in 10.10.20.100–199, gateway 10.10.20.1, DNS 10.10.30.10/.11 and DHCP Server 10.10.30.12. If no lease: check the relay on RTR01, scope state Active, VLAN IDs and DHCP authorization.

20. FILE01 — File Server

20.1 Build

1. Create VM FILE01 (Gen 2, 2 vCPU, 1.5 GB Dynamic, VLAN 30). Add **second VHDX** 100 GB dynamic for data: `D:\NileBank-Lab\VHDX\FILE01\FILE01-Data.vhdx`.
2. Install Windows Server, rename FILE01, static IP **10.10.30.20** (proposed), DNS 10.10.30.10/.11, join domain into OU=File.
3. Initialize the data disk: Disk Management → Online → Initialize (GPT) → New Simple Volume, NTFS, letter **D:**, label Shares.

Add and format data disk

```
PS> New-VHD -Path 'D:\NileBank-Lab\VHDX\FILE01\FILE01-Data.vhdx' -SizeBytes 100GB -Dynamic
PS> Add-VMHardDiskDrive -VMName FILE01 -Path 'D:\NileBank-Lab\VHDX\FILE01\FILE01-Data.vhdx'
# On FILE01
PS> Get-Disk | Where PartitionStyle -eq 'RAW' | Initialize-Disk -PartitionStyle GPT -PassThru |
      New-Partition -DriveLetter D -UseMaximumSize | Format-Volume -FileSystem NTFS -NewFileSystemLabel Shares
```

20.2 Install File and Storage Services

Roles

```
PS> Install-WindowsFeature FS-FileServer,FS-Resource-Manager,FS-DFS-Namespace -IncludeManagementTools
PS> Install-WindowsFeature FS-SMB1 -Remove    # confirm SMB1 is absent
PS> Get-SmbServerConfiguration | Select EnableSMB1Protocol,EnableSMB2Protocol,EncryptData
```

NOTE: *File and Storage Services → File Server* is the core role. FSRM (quotas, file screening) and DFS Namespaces are optional additions for demonstrating enterprise features.

20.3 Concepts

Term	Explanation
SMB	Server Message Block: the Windows file-sharing protocol (TCP 445). Use SMB 2/3 only; SMB 1 is insecure and must stay disabled.
Share permissions	Apply only over the network. The most restrictive of share + NTFS wins. Best practice: set share permission to **Full Control for Authenticated Users/Everyone-equivalent** or the same groups, and control with NTFS. Here shares grant Full Control to the same domain local groups; NTFS enforces Modify/Read.
NTFS permissions	Apply locally and over the network; granular (Read, Modify, Full Control, Special). The real access control.
Inheritance	Child folders inherit parent ACEs. Break inheritance on the share root so only the intended groups remain.
Explicit Deny	Deny overrides Allow. Use sparingly: it complicates troubleshooting. Prefer simply not granting access. A Deny ACE is used here only as a demonstration/safety net.
Least privilege	Grant the minimum access necessary: Read for most, Modify for owners, Full Control only for administrators.

20.4 Create the folders, shares and permissions

Folders, NTFS, shares

```
PS> 'Finance','HR','IT','Operations','Public' | ForEach-Object { New-Item -ItemType Directory -Path "D:\Shares\$_" -Force

# --- Helper: reset NTFS ACL per share (inheritance removed, only listed groups + admins) ---
PS> function Set-ShareAcl($path, $rw, $r) {
      icacls $path /inheritance:r | Out-Null
      icacls $path /grant 'NT AUTHORITY\SYSTEM:(OI)(CI)F' 'BUILTIN\Administrators:(OI)(CI)F' 'NILEBANK\GG-IT-Admins:(OI)(
      if ($rw) { icacls $path /grant "NILEBANK\${rw}:(OI)(CI)M" | Out-Null }
      if ($r)  { icacls $path /grant "NILEBANK\${r}:(OI)(CI)RX" | Out-Null }
    }
PS> Set-ShareAcl 'D:\Shares\Finance'    'DL-Share-Finance-RW'    'DL-Share-Finance-R'
PS> Set-ShareAcl 'D:\Shares\HR'         'DL-Share-HR-RW'         'DL-Share-HR-R'
PS> Set-ShareAcl 'D:\Shares\IT'         'DL-Share-IT-RW'         $null
PS> Set-ShareAcl 'D:\Shares\Operations' 'DL-Share-Operations-RW' 'DL-Share-Operations-R'
PS> Set-ShareAcl 'D:\Shares\Public'     'DL-Share-Public-RW'     'DL-Share-Public-R'

# --- SMB shares: share-level Full Control to the same groups; NTFS does the real limiting ---
PS> New-SmbShare -Name Finance    -Path D:\Shares\Finance    -FullAccess 'NILEBANK\DL-Share-Finance-RW','NILEBANK\DL-Shar
PS> New-SmbShare -Name HR         -Path D:\Shares\HR         -FullAccess 'NILEBANK\DL-Share-HR-RW','NILEBANK\DL-Share-HR-
PS> New-SmbShare -Name IT         -Path D:\Shares\IT         -FullAccess 'NILEBANK\DL-Share-IT-RW','NILEBANK\GG-IT-Admins
PS> New-SmbShare -Name Operations -Path D:\Shares\Operations -FullAccess 'NILEBANK\DL-Share-Operations-RW','NILEBANK\DL-S
PS> New-SmbShare -Name Public     -Path D:\Shares\Public     -FullAccess 'NILEBANK\DL-Share-Public-RW','NILEBANK\DL-Share
PS> Get-SmbShare | Format-Table Name,Path,EncryptData
PS> icacls D:\Shares\Finance
```

NOTE: **Access-Based Enumeration** hides folders the user cannot open. **SMB encryption** (-EncryptData) protects data in transit for sensitive shares (Finance, HR); it requires SMB 3 clients (Windows 8+/Server 2012+).

Optional explicit Deny demonstration

To demonstrate the order of precedence, add an explicit Deny for HR on the Finance folder, then remove it after testing:

Add / remove explicit Deny

```
PS> icacls D:\Shares\Finance /deny 'NILEBANK\GG-HR-Users:(OI)(CI)F'
PS> icacls D:\Shares\Finance /remove:d 'NILEBANK\GG-HR-Users'
```

20.5 Test procedures

#	Test (on CLIENT01)	Expected
1	Log on as m.farouk (Finance) → open \\FILE01\Finance → create *test.txt*	**Allow**: create, edit, delete
2	m.farouk → \\FILE01\HR	**Deny**: 'You do not have permission' (or the share is hidden by ABE)
3	l.mostafa (HR) → \\FILE01\Finance	**Deny**
4	l.mostafa → \\FILE01\HR create a file	Allow
5	k.elsayed (Management) → \\FILE01\Finance	Read-only: open a file OK; creating/deleting a file is **Denied**
6	adm.a.hassan or GG-IT-Admins member → every share	Full Control (administrative access)
7	s.adel (IT) → \\FILE01\IT write; \\FILE01\Public write	Allow on IT and Public (Modify)
8	r.ibrahim (Operations) → \\FILE01\Public write	Read-only; write Denied
9	Add m.farouk to GG-HR-Users, log off/on, retest #2	Token updates at next logon; access follows group membership, not the user

Diagnostics on FILE01

```
PS> Get-ADPrincipalGroupMembership m.farouk | Select Name
PS> Get-SmbShareAccess -Name Finance
PS> (Get-Acl D:\Shares\Finance).Access | Format-Table IdentityReference,FileSystemRights,AccessControlType,IsInherited
PS> icacls D:\Shares\Finance
# Effective access: Properties → Security → Advanced → Effective Access → Select a user
```

Group membership changes are only visible in a **new logon token**. Always log off and on (or run `klist purge` then reopen the session) after changing group membership. Capture screenshots for Section 33.

21. PRINT01 — Print Server

21.1 Build

Create PRINT01 like FILE01 (Gen 2, 2 vCPU, 1.5 GB Dynamic, 50 GB VHDX, VLAN 30), static IP **10.10.30.21** (proposed), DNS 10.10.30.10/.11, join the domain into *OU=Print*.

21.2 Install the role

Print and Document Services

```
PS> Install-WindowsFeature Print-Server -IncludeManagementTools
PS> Get-Service Spooler
```

21.3 Configure the printer

A lab has no physical printer, so use a **Standard TCP/IP port to a non-existent address** or the built-in **Microsoft PS Class Driver** with the *Microsoft Print to PDF*/XPS style virtual printer. The management and sharing procedure is identical.

1. Server Manager → Tools → **Print Management**. Expand *Print Servers → PRINT01 → Drivers* and check that a Class Driver (e.g. Microsoft enhanced Point and Print / Microsoft PS Class Driver) exists.
2. Right-click **Printers → Add Printer…** → 'Add a new printer using an existing port' / 'Create a new port: Standard TCP/IP Port' → IP 10.10.30.250 (example virtual printer) → driver 'Microsoft PS Class Driver'.
3. Name **PRN-FIN-01**. Tick **Share this printer**, share name PRN-FIN-01. Tick **List in the directory** (publishes it to Active Directory so users can find it).
4. Printer Properties → **Security**: remove Everyone → add **DL-Print-Finance** (Print), **GG-IT-Admins** (Manage this printer, Manage documents, Print). Repeat for **PRN-GEN-01** with **DL-Print-General**.
5. Printer Properties → **Sharing → Additional Drivers**: add x64 (and ARM64 if needed) drivers so clients download automatically.

PowerShell equivalent

```
PS> Add-PrinterPort -Name 'IP_10.10.30.250' -PrinterHostAddress 10.10.30.250
PS> Add-PrinterDriver -Name 'Microsoft PS Class Driver'
PS> Add-Printer -Name 'PRN-FIN-01' -DriverName 'Microsoft PS Class Driver' -PortName 'IP_10.10.30.250' -Shared -ShareName
PS> Add-Printer -Name 'PRN-GEN-01' -DriverName 'Microsoft PS Class Driver' -PortName 'IP_10.10.30.250' -Shared -ShareName
PS> Get-Printer | Format-Table Name,ShareName,Published,DriverName
```

21.4 Client deployment

- **Manual:** CLIENT01 → Settings → Bluetooth & devices → Printers & scanners → Add device → 'Add manually' → `\\PRINT01\PRN-FIN-01`.
- **Group Policy Preferences:** User Configuration → Preferences → Control Panel Settings → **Printers → New → Shared Printer**, Path \\PRINT01.corp.nilebank.lab\PRN-GEN-01, Action Update, set as default, with Item-level targeting to the appropriate group (or via GPO-Users-DriveMaps).
- **Point and Print restrictions** (security): GPO 'Point and Print Restrictions' = Enabled, restrict to servers PRINT01.corp.nilebank.lab, 'Do not show warning or elevation prompt' only for that server (PrintNightmare hardening).

21.5 Test from CLIENT01

On CLIENT01 (as m.farouk)

```
PS> Add-Printer -ConnectionName '\\PRINT01\PRN-GEN-01'
PS> Get-Printer | Format-Table Name,PrinterStatus
PS> Out-Printer -InputObject 'Nile Bank print test' -Name '\\PRINT01\PRN-GEN-01'
```

Expected: m.farouk can add both printers; r.ibrahim (Operations) can add PRN-GEN-01 but receives 'Access denied' for PRN-FIN-01. The job is visible in Print Management → PRINT01 → Printers → Right-click → **Open Printer Queue**.

21.6 Print troubleshooting

Symptom	Check	Fix
All jobs stuck	`Get-Service Spooler`; queue folder C:\Windows\System32\spool\PRINTERS	`Stop-Service Spooler; Remove-Item C:\Windows\System32\spool\PRINTERS\* -Force; Start-Service Spooler`
Single document stuck	Print queue → job status 'Error' / 'Deleting'	Cancel job; if it will not delete, restart the Spooler.
Driver problem	Print Management → Drivers; Event ID 372/808 in PrintService	Replace with Class Driver; remove and re-add the driver package; check architecture (x64).
Cannot connect to \\PRINT01	`ping PRINT01`; `nslookup PRINT01`; `Test-NetConnection PRINT01 -Port 445`	Fix DNS/firewall (File and Printer Sharing rule enabled on PRINT01).
Access denied	Printer → Security tab; user's group membership	Add the correct DL- group, log off/on client.
Print server unavailable	Server status, Spooler, network	Start/restore PRINT01; clients cache connections but cannot print until it returns.
Printer not found in AD search	Printer 'List in the directory' flag	Enable it or reshare the printer.

Diagnostics

```
PS> Get-WinEvent -LogName 'Microsoft-Windows-PrintService/Admin' -MaxEvents 20 | Format-Table TimeCreated,Id,Message -Wra
PS> Get-PrintJob -PrinterName 'PRN-GEN-01'
```

22. LINUX01 — Linux Monitoring Server

22.1 VM creation

Create LINUX01

```
PS> New-VM -Name LINUX01 -Generation 2 -MemoryStartupBytes 1GB -Path 'D:\NileBank-Lab\VMs' `
      -NewVHDPath 'D:\NileBank-Lab\VHDX\LINUX01\LINUX01-OS.vhdx' -NewVHDSizeBytes 40GB -SwitchName 'NILEBANK-LAB'
PS> Set-VMNetworkAdapterVlan -VMName LINUX01 -Access -VlanId 30
PS> Set-VMProcessor LINUX01 -Count 2
PS> Set-VMMemory LINUX01 -DynamicMemoryEnabled $true -MinimumBytes 512MB -StartupBytes 1GB -MaximumBytes 2GB
PS> Set-VMFirmware LINUX01 -EnableSecureBoot On -SecureBootTemplate MicrosoftUEFICertificateAuthority
PS> Add-VMDvdDrive LINUX01 -Path 'D:\NileBank-Lab\ISOs\ubuntu-24.04-live-server-amd64.iso'
PS> Set-VMFirmware LINUX01 -FirstBootDevice (Get-VMDvdDrive LINUX01)
```

Install **Ubuntu Server LTS** (minimal, OpenSSH server enabled). Hostname `linux01`, admin user `labadmin`.

22.2 Static IP, SSH, users, updates, firewall

Static IP with AD DNS

```
# /etc/netplan/01-linux01.yaml
network:
  version: 2
  ethernets:
    eth0:
      addresses: [10.10.30.30/24]
      routes: [{to: default, via: 10.10.30.1}]
      nameservers: {addresses: [10.10.30.10, 10.10.30.11], search: [corp.nilebank.lab]}

$ sudo netplan apply && ip -br addr && resolvectl status
$ ping -c2 10.10.30.10 && nslookup dc01.corp.nilebank.lab
```

Baseline hardening

```
# SSH hardening
$ sudo apt update && sudo apt -y full-upgrade && sudo apt -y install unattended-upgrades ufw chrony
$ ssh-keygen -t ed25519      # on the admin PC; copy with: ssh-copy-id labadmin@10.10.30.30
$ sudo nano /etc/ssh/sshd_config.d/99-hardening.conf
      PermitRootLogin no
      PasswordAuthentication no     # only after key login works
      AllowUsers labadmin
$ sudo systemctl restart ssh

# Users
$ sudo adduser monitoruser
$ sudo usermod -aG sudo labadmin            # administrators only
$ sudo passwd -l root

# Firewall (SSH only from Management VLAN; monitoring web UI from Management/Users)
$ sudo ufw default deny incoming && sudo ufw default allow outgoing
$ sudo ufw allow from 10.10.10.0/24 to any port 22 proto tcp
$ sudo ufw allow from 10.10.10.0/24 to any port 3000 proto tcp   # Grafana
$ sudo ufw allow from 10.10.10.0/24 to any port 80,443 proto tcp # Zabbix web (if used)
$ sudo ufw enable && sudo ufw status verbose

# Time sync from DC01
$ sudo sed -i 's/^pool.*/server 10.10.30.10 iburst/' /etc/chrony/chrony.conf && sudo systemctl restart chrony && chronyc
```

22.3 Monitoring options for a home lab

Tool	Type	Strengths	Cost on 16 GB host	Fit
Zabbix	All-in-one monitoring (server + DB + web)	Agent-based and agentless; templates for Windows, Linux, AD, DNS, DHCP, SNMP; alerting, dashboards, triggers.	~1.5–2 GB RAM	**Recommended primary tool**: broadest enterprise-style coverage

Tool	Type	Strengths	Cost on 16 GB host	Fit
Prometheus + Grafana + node/windows exporter	Metrics + dashboards	Modern, cloud-native skills; Alertmanager; PromQL.	~1–1.5 GB	Excellent for Cloud Engineer portfolio
Netdata	Real-time per-host	Single command install; instant dashboards.	~200 MB per host	Quick start/optional
Checkmk Raw	Agent + auto-discovery	Strong Windows/Linux checks, easy.	~1.5 GB	Alternative to Zabbix
Windows Admin Center	Windows server management	Perfmon, events, services, updates for Windows hosts.	Runs on a Windows VM	Complement for Windows

22.4 Install Zabbix (recommended) — outline with commands

Use the official Zabbix repository for Ubuntu LTS (version numbers change, so confirm the current LTS from zabbix.com/download). The sequence below is correct for any Zabbix 7.0 LTS install on Ubuntu with MySQL + Apache.

Zabbix on LINUX01

```
$ wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_latest_7.0+ubuntu24.04_all.deb
$ sudo dpkg -i zabbix-release_latest_7.0+ubuntu24.04_all.deb && sudo apt update
$ sudo apt install -y zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent2 mysql-s
$ sudo mysql -e "create database zabbix character set utf8mb4 collate utf8mb4_bin;"
$ sudo mysql -e "create user zabbix@localhost identified by '<STRONG-DB-PASSWORD>'; grant all privileges on zabbix.* to z
$ zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql --default-character-set=utf8mb4 -uzabbix -p zabbix
$ sudo mysql -e "set global log_bin_trust_function_creators = 0;"
$ sudo sed -i 's/^# DBPassword=.*/DBPassword=<STRONG-DB-PASSWORD>/' /etc/zabbix/zabbix_server.conf
$ sudo systemctl restart zabbix-server zabbix-agent2 apache2 && sudo systemctl enable zabbix-server zabbix-agent2 apache2
# Browse to http://10.10.30.30/zabbix and finish the setup wizard (default login Admin; change the password immediately).
```

NOTE: Package URLs and versions change between releases. Always follow the current instructions at zabbix.com/download, run the steps on LINUX01 and screenshot the result as evidence.

22.5 What to monitor

Category	Items	Method
CPU	Utilization, load average, steal	Agent (system.cpu.util, load)
RAM	Used %, swap	Agent (vm.memory.utilization)
Disk	Free space per volume, I/O latency	Agent / WMI (vfs.fs.size)
Network	Interface throughput, errors, latency to gateway	Agent, ICMP ping, SNMP (RTR01)
Services	NTDS, DNS, Netlogon, DHCPServer, Spooler, LanmanServer; SSH on Linux	Windows service state checks
Uptime	System uptime / unexpected reboot	system.uptime trigger when < 600 s
Event logs	Directory Service errors, DNS server errors, failed logons (4625)	eventlog[] item or log forwarding

Windows agents: install the Zabbix Agent 2 MSI on DC01, DC02, DHCP01, FILE01, PRINT01 and BACKUP01 (Server=10.10.30.30), allow TCP 10050 from 10.10.30.30 in Windows Firewall, add the hosts in Zabbix with the templates *Windows by Zabbix agent*, and add *Active Directory/DNS* checks as needed.

23. DC02 — Secondary Domain Controller

Purpose: provide Active Directory and DNS redundancy so that authentication, Group Policy, name resolution and the Global Catalog survive the loss of DC01. **Dependencies:** DC01 healthy with AD DS and DNS (Sections 11–13); RTR01 routing; DC01 reachable on 10.10.30.10.

Setting	Value
Hostname	DC02
Role	Additional Domain Controller + DNS + Global Catalog (existing domain corp.nilebank.lab)
IP / mask / gateway	10.10.30.11 / 255.255.255.0 / 10.10.30.1
Preferred DNS before promotion	10.10.30.10 (DC01)
VLAN	30
VM resources	2 vCPU, 2 GB startup (Dynamic 1–3 GB), 60 GB VHDX, Gen 2, checkpoints disabled

23.1 DC02 VM

1. Hyper-V Manager → **New → Virtual Machine**. Name **DC02**; store at `D:\NileBank-Lab\VMs\DC02`. Generation 2.
2. Startup memory 2048 MB with Dynamic Memory; network **NILEBANK-LAB**; new VHDX `DC02-OS.vhdx` in `D:\NileBank-Lab\VHDX\DC02\`, 60 GB.
3. Install from the ISO in `D:\NileBank-Lab\ISOs\`. Finish.
4. Settings → Processor 2; Memory min 1024/max 3072; Checkpoints off; set VLAN 30 (command below).

Create DC02

```
PS> New-VM -Name DC02 -Generation 2 -MemoryStartupBytes 2GB -Path 'D:\NileBank-Lab\VMs' `
      -NewVHDPath 'D:\NileBank-Lab\VHDX\DC02\DC02-OS.vhdx' -NewVHDSizeBytes 60GB -SwitchName 'NILEBANK-LAB'
PS> Set-VMProcessor DC02 -Count 2
PS> Set-VMMemory DC02 -DynamicMemoryEnabled $true -MinimumBytes 1GB -StartupBytes 2GB -MaximumBytes 3GB
PS> Set-VM DC02 -CheckpointType Disabled
PS> Set-VMNetworkAdapterVlan -VMName DC02 -Access -VlanId 30
PS> Add-VMDvdDrive DC02 -Path 'D:\NileBank-Lab\ISOs\WindowsServer2022.iso'
PS> Set-VMFirmware DC02 -FirstBootDevice (Get-VMDvdDrive DC02)
PS> Start-VM DC02
```

23.2 Install Windows Server and configure the network

1. Install Windows Server 2022 Standard (Desktop Experience) exactly as in Section 9.
2. Rename and restart; install updates; set the static IP and **preferred DNS = DC01**; restart.

DC02 network

```
PS> Rename-Computer -NewName DC02 -Restart
PS> Rename-NetAdapter -Name 'Ethernet' -NewName 'LAN'
PS> New-NetIPAddress -InterfaceAlias 'LAN' -IPAddress 10.10.30.11 -PrefixLength 24 -DefaultGateway 10.10.30.1
PS> Set-DnsClientServerAddress -InterfaceAlias 'LAN' -ServerAddresses 10.10.30.10
PS> Restart-Computer
```

23.3 Pre-promotion tests

Run on DC02

```
CMD> ipconfig /all
CMD> ping 10.10.30.10
CMD> nslookup corp.nilebank.lab
CMD> whoami
CMD> systeminfo | findstr /B /C:"Domain"
PS>  Test-NetConnection 10.10.30.10 -Port 389
PS>  Test-NetConnection 10.10.30.10 -Port 445
```

Command	Expected result	If not
ipconfig /all	IP 10.10.30.11, DNS 10.10.30.10	Fix NIC settings.

Command	Expected result	If not
ping 10.10.30.10	Replies from DC01	Check VLAN 30, DC01 powered on, Windows Firewall ICMP.
nslookup corp.nilebank.lab	Server dc01.corp.nilebank.lab, returns 10.10.30.10	DNS wrong or DC01 DNS service down — stop and fix before proceeding.
whoami	dc02\administrator (local, not yet in domain)	—
systeminfo ... Domain	Domain: WORKGROUP	—
Test-NetConnection :389 / :445	TcpTestSucceeded True	Firewall/DC01 services.

23.4 Join DC02 to the domain

Join and verify

```
PS> Add-Computer -DomainName corp.nilebank.lab -Credential (Get-Credential NILEBANK\adm.a.hassan) -Restart
# after reboot, log on as NILEBANK\adm.a.hassan
PS> (Get-WmiObject Win32_ComputerSystem).Domain        # corp.nilebank.lab
PS> systeminfo | findstr /B /C:"Domain"
```

NOTE: The computer account is created in the default location (Workstations OU if redircmp was used). It moves automatically to the built-in **Domain Controllers** OU during promotion. Check in Section 24.

23.5 Install AD DS and promote to additional domain controller

1. **Server Manager → Manage → Add Roles and Features → Role-based installation → DC02 → Active Directory Domain Services → Add Features → Install**.
2. Click the notification flag → **Promote this server to a domain controller**.
3. Deployment Configuration: choose **Add a domain controller to an existing domain**. **Do NOT select 'Add a new forest' or 'Add a new domain to an existing forest'.** Domain: corp.nilebank.lab. Click Change... and supply NILEBANK\adm.a.hassan (member of Domain Admins).
4. Domain Controller Options: keep **Domain Name System (DNS) server** and **Global Catalog (GC)** ticked; leave **Read only domain controller** unticked; Site name Default-First-Site-Name; enter the **DSRM password** (different from DC01 is acceptable and preferred).
5. DNS Options: delegation warning is expected for the same reason as on DC01; continue.
6. Additional Options: Replicate from **DC01.corp.nilebank.lab** (or 'Any domain controller').
7. Paths: default NTDS/SYSVOL (C:\Windows\...). Review (optionally 'View script'). Prerequisites Check must pass. **Install**.
8. The server restarts automatically. Sign in as NILEBANK\adm.a.hassan.

Scripted equivalent (Install-ADDSDomainController)

```
PS> Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
PS> Install-ADDSDomainController -DomainName corp.nilebank.lab -InstallDns:$true `
      -Credential (Get-Credential NILEBANK\adm.a.hassan) -SiteName 'Default-First-Site-Name' `
      -ReplicationSourceDC 'DC01.corp.nilebank.lab' -NoRebootOnCompletion:$false `
      -DatabasePath 'C:\Windows\NTDS' -LogPath 'C:\Windows\NTDS' -SysvolPath 'C:\Windows\SYSVOL' `
      -SafeModeAdministratorPassword (Read-Host -AsSecureString 'DSRM password') -Force
```

Global Catalog is enabled by default in `Install-ADDSDomainController` (parameter `-NoGlobalCatalog:$false`).

23.6 After promotion: DNS client settings and DHCP

Set redundant DNS

```
# On DC01
PS> Set-DnsClientServerAddress -InterfaceAlias 'LAN' -ServerAddresses 10.10.30.11,10.10.30.10
# On DC02
PS> Set-DnsClientServerAddress -InterfaceAlias 'LAN' -ServerAddresses 10.10.30.10,10.10.30.11
# DHCP options already hand out both DNS servers (Section 19). Update statically configured servers (FILE01, PRINT01, DHC
PS> Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses 10.10.30.10,10.10.30.11
PS> ipconfig /registerdns
```

24. DC02 Post-Promotion Validation

Run on DC02 (and repeat on DC01)

```
PS> Get-ADDomain
PS> Get-ADForest
PS> Get-ADDomainController -Filter * | Format-Table Name,IPv4Address,IsGlobalCatalog,Site,OperatingSystem
PS> Get-Service DNS,NTDS,Netlogon,DFSR,KDC | Format-Table Name,Status
PS> nslookup corp.nilebank.lab
PS> nslookup -type=SRV _ldap._tcp.dc._msdcs.corp.nilebank.lab
PS> Get-DnsServerZone
PS> Get-ADComputer DC02 | Select DistinguishedName
PS> net share
PS> netdom query fsmo
```

Check	Expected result
Get-ADDomain / Get-ADForest	Same DNSRoot corp.nilebank.lab; ReplicaDirectoryServers lists **DC01 and DC02**; GlobalCatalogs lists both.
Get-ADDomainController -Filter *	**Two rows**: DC01 10.10.30.10 and DC02 10.10.30.11, both IsGlobalCatalog True, same site.
Services	DNS, NTDS, Netlogon, DFSR, KDC Running.
nslookup corp.nilebank.lab	Returns **both** 10.10.30.10 and 10.10.30.11.
SRV lookup	Lists dc01 and dc02.
Get-DnsServerZone	corp.nilebank.lab, _msdcs.corp.nilebank.lab and the reverse zones present on DC02 (AD-integrated, replicated).
Get-ADComputer DC02	CN=DC02,**OU=Domain Controllers**,DC=corp,DC=nilebank,DC=lab
net share	SYSVOL and NETLOGON present (initial SYSVOL replication complete). In the DFS Replication log look for Event ID **4604**.
netdom query fsmo	All five roles on DC01 (default). In a single-domain forest with all DCs as GC, this is acceptable.

25. Active Directory Replication

25.1 Concepts

Term	Explanation
Replication	Multi-master copy of directory changes between DCs. Every DC accepts writes (except RODC) and replicates them to partners using USNs and up-to-dateness vectors.
Inbound / Outbound replication	Inbound = changes a DC *receives* from a partner; outbound = changes it *sends*. repadmin shows both directions; a DC may be healthy one way but failing the other.
Replication partners	DCs connected by **connection objects** created by the KCC (Knowledge Consistency Checker). Within a site, change notification occurs in about 15 seconds; across sites on a schedule.
Directory partitions	Schema, Configuration, Domain (corp), DomainDnsZones, ForestDnsZones (DNS). All replicate between DC01 and DC02.
SYSVOL	Folder with GPO files and logon scripts, replicated by **DFS Replication** (DFSR), not by AD replication.
NETLOGON	Share (backed by SYSVOL\scripts) published only when the DC is advertising and Netlogon is healthy.
DNS replication	AD-integrated zone data replicates in the DomainDnsZones/ForestDnsZones partitions, so both DCs serve identical records.

25.2 Commands

Run from either DC

```
CMD> repadmin /replsummary
CMD> repadmin /showrepl
CMD> repadmin /showrepl * /csv > C:\Temp\showrepl.csv
CMD> repadmin /syncall DC01 /AdeP
CMD> repadmin /queue
CMD> dcdiag /test:dns /v
CMD> dcdiag /q
CMD> dfsrmig /getglobalstate
PS> Get-ADReplicationFailure -Target DC01,DC02
PS> Get-ADReplicationPartnerMetadata -Target DC01 | Format-Table Server,Partner,LastReplicationSuccess,LastReplicationRes
```

Command	Meaning
repadmin /replsummary	Summarizes inbound/outbound replication health for all DCs: largest delta, fail count, % errors.
repadmin /showrepl	Per-partition detail: neighbors, last attempt time, success/failure and error codes.
repadmin /syncall /AdeP	Forces synchronization of all partitions to all partners (A = all partitions, d = identify by DN, e = enterprise across sites, P = push).
repadmin /queue	Pending replication operations (should be 0).
dcdiag /test:dns /v	Verifies DNS: authentication, basic connectivity, forwarders, delegations, dynamic update, record registration.
dcdiag /q	Quiet mode: prints only failures. No output = healthy.
dfsrmig /getglobalstate	Confirms SYSVOL uses DFSR: 'Eliminated' state (State 3).

25.3 What healthy looks like

Example repadmin /replsummary (healthy)

```
Replication Summary Start Time: 2026-xx-xx xx:xx:xx
Beginning data collection for replication summary, this may take awhile:
  ......
Source DSA          largest delta    fails/total %   error
 DC01                  05m:12s    0 /   5    0
 DC02                  04m:50s    0 /   5    0

Destination DSA     largest delta    fails/total %   error
 DC01                  04m:50s    0 /   5    0
 DC02                  05m:12s    0 /   5    0
```

- **fails/total = 0 / n** for every DC; **% (percent) = 0**; no error text.
- **largest delta** small (seconds to a few minutes; a very large delta means no replication). Anything beyond 60 minutes intrasite deserves investigation.
- `repadmin /showrepl`: every partition 'Last attempt @ ... was successful'.
- `dcdiag /test:dns /v`: **passed** for DC01 and DC02 in every sub-test (Auth, Basic, Forwarders, Delegation, Dynamic update, RecordRegistration).
- Create a test object on DC01 (`New-ADUser test.repl`) and read it on DC02 within ~15–60 s (`Get-ADUser test.repl -Server DC02`); delete it afterwards. Drop a file into SYSVOL\...\Policies is **not** advised; instead check that GPO versions match in GPMC → Group Policy Objects → Status tab.

Common replication error	Typical cause	Fix
8524 DSA operation unable to proceed because of a DNS lookup failure	DNS SRV/CNAME for DC GUID missing or client DNS wrong	Fix DNS client settings; run `ipconfig /registerdns` and `nltest /dsregdns`; restart Netlogon.
1722 RPC server unavailable	Firewall/network/VLAN, RPC service, DC offline	Test ports 135 and dynamic RPC; check VLAN and RTR01 rules.
8453 Replication access denied	Security descriptor/permissions; time skew	Check time, secure channel (`nltest /sc_verify:corp.nilebank.lab`).
1908 / 5 time / Kerberos	Time skew > 5 min	Sync time with PDC emulator (w32tm /resync).
DFSR 4012 / SYSVOL not shared	DC offline longer than MaxOfflineTimeInDays or DFSR dirty shutdown	Authoritative/non-authoritative DFSR restore (OPTIONAL / ADVANCED).

26. Domain Controller Failover Test

> **LAB vs PRODUCTION:** This is a controlled exercise in an **isolated lab**. In a bank, DC failover tests follow an approved change request, a maintenance window, rollback plan and monitoring; they are never improvised on production DCs.

26.1 Prerequisites

- Section 25 shows healthy replication (`repadmin /replsummary` zero failures).
- CLIENT01 is domain-joined and DHCP provides DNS 10.10.30.10 and 10.10.30.11 (or set manually as below).
- FILE01 online and shares work. A test user (s.nabil) has **never logged on to CLIENT01** (so no cached credentials).
- Test timeline is recorded in a notes table (Section 26.4).

26.2 Procedure

1. **Confirm CLIENT01 works with DC01.** `nltest /dsgetdc:corp.nilebank.lab` returns DC01; `echo %LOGONSERVER%` → \\DC01; `gpupdate /force` succeeds.
2. **Confirm DNS.** `nslookup corp.nilebank.lab 10.10.30.10` and `nslookup file01.corp.nilebank.lab` succeed.
3. **Confirm DC02 has the AD DNS zone.** On DC02: `Get-DnsServerZone` and `Get-DnsServerResourceRecord -ZoneName corp.nilebank.lab -Name file01`. Also `nslookup file01.corp.nilebank.lab 10.10.30.11` from CLIENT01.
4. **Configure CLIENT01 to prefer DC02 DNS** (for the test): `Set-DnsClientServerAddress -InterfaceAlias Ethernet -ServerAddresses 10.10.30.11,10.10.30.10`; `ipconfig /flushdns`. (Normally the client lists both servers and fails over automatically after a short timeout.)
5. **Make DC01 unavailable.** Record the time T0. Host: `Stop-VM DC01 -TurnOff` (simulates power loss; use plain `Stop-VM DC01` for a graceful shutdown scenario). Do **not** delete DC01 or its disk.
6. **Test DNS resolution** from CLIENT01: `ipconfig /flushdns`, `nslookup corp.nilebank.lab`, `nslookup file01.corp.nilebank.lab`, `nslookup www.microsoft.com`. Expect answers from dc02 (10.10.30.11).
7. **Test authentication.** Sign out and sign in as **s.nabil** (never used on this client, so it must contact a DC). Success proves DC02 handled Kerberos. Check `echo %LOGONSERVER%` → \\DC02 and `klist` for a TGT issued by DC02.
8. **Test domain resources.** Open \\FILE01\Finance (Kerberos service ticket from DC02), run `gpupdate /force`, change a password with Ctrl+Alt+Del, add a user on DC02: `New-ADUser failover.test`.
9. **Restore DC01.** Record T1. `Start-VM DC01`. Wait until the console shows the logon screen (≈2–3 min), then run `Get-Service NTDS,DNS,Netlogon,DFSR` on DC01.
10. **Verify replication.** On DC02: `repadmin /replsummary`; `repadmin /showrepl`; `repadmin /syncall /AdeP`. On DC01 `Get-ADUser failover.test` must exist (change made while DC01 was down replicated back).
11. **Confirm health.** `dcdiag /q` on both DCs and `dcdiag /test:dns /v`. Restore CLIENT01 DNS (`10.10.30.10,10.10.30.11`) and delete the test user. Record T2 = replication healthy.

26.3 Reverse test

Repeat with DC02 off and DC01 on to prove symmetry. Also test the PDC emulator effect: while DC01 (PDC Emulator) is off, password *changes* succeed on DC02, but *account lockout propagation* and urgent replication for password changes are delayed, and time authority is lost until the role is moved or DC01 returns. Optionally transfer the role gracefully with `Move-ADDirectoryServerOperationMasterRole -Identity DC02 -OperationMasterRole PDCEmulator` (transfer, never seize, unless DC01 is permanently lost).

26.4 Documentation template

Item	Record
Failure	DC01 powered off (hard) at T0 = hh:mm:ss
Impact	Expected: no outage for users; short delay (1–15 s) when the client first tries the unavailable DNS/DC; new logons and file access continue via DC02. PDC emulator functions unavailable.
Detection	CLIENT01 nslookup showing DC02 as responding server; Zabbix alert 'DC01 unreachable'; `repadmin /replsummary` showing failure to DC01; System log Event 1722/5805 style Netlogon messages.
Recovery	DC01 started at T1; services up at T1+n; replication re-established at T2
Recovery time (RTO observed)	T2 − T0 = ____ min (measure and write the real figure)
Data loss (RPO observed)	0 objects lost: changes created on DC02 during outage appear on DC01 (confirm).
Lessons learned	Two DNS entries on every client/server are mandatory; DNS on DC01 only would have caused failures; time source role; FSMO transfer planning; monitoring must alert before a user reports.

NOTE: Do not copy the 'expected' values into your evidence as results. Record what actually happened in your lab with timestamps and screenshots.

27. Backup and Disaster Recovery

27.1 Architecture

Backup storage lives under `D:\NileBank-Lab\Backups\` on the host. **BACKUP01** (Windows Server 2022, 10.10.30.50 proposed, domain-joined, 1.5 GB) acts as the backup/recovery server with a data VHDX stored in that folder. Host-level VM exports also go there. **Lab limitation:** the backups are on the same physical host and disk as the VMs, so a host disk failure destroys both. Copy the repository to an external USB drive or cloud storage periodically to meet the 3-2-1 rule (3 copies, 2 media types, 1 off-site).

Backup layout

```
D:\NileBank-Lab\Backups\
├── BACKUP01\BACKUP01-Repo.vhdx     (repository disk attached to BACKUP01)
├── VM-Exports\<VM>\<yyyy-mm-dd>\   (Export-VM output)
├── Config\{GPO,DHCP,DNS,Print,AD}\ (configuration exports)
└── Logs\                            (backup logs, restore evidence)
```

BACKUP01 repository

```
PS> New-VHD -Path 'D:\NileBank-Lab\Backups\BACKUP01\BACKUP01-Repo.vhdx' -SizeBytes 200GB -Dynamic
PS> Add-VMHardDiskDrive -VMName BACKUP01 -Path 'D:\NileBank-Lab\Backups\BACKUP01\BACKUP01-Repo.vhdx'
# On BACKUP01 (after domain join): initialize as D:, then share
PS> New-Item D:\Repo\DCBackups,D:\Repo\FileBackups,D:\Repo\Config -ItemType Directory -Force
PS> New-SmbShare -Name Repo -Path D:\Repo -FullAccess 'NILEBANK\GG-IT-Admins','NILEBANK\GG-Servers-Backup' -EncryptData $
PS> icacls D:\Repo /inheritance:r /grant 'NT AUTHORITY\SYSTEM:(OI)(CI)F' 'NILEBANK\GG-IT-Admins:(OI)(CI)F' 'NILEBANK\GG-S
PS> Install-WindowsFeature Windows-Server-Backup
```

27.2 What is backed up

Asset	Method	Schedule	Retention	Type
DC01 and DC02 System State (AD, SYSVOL, registry, boot files)	`wbadmin start systemstatebackup` to a local backup disk (E:) then copy to \\BACKUP01\Repo\DCBackups	Daily 01:00 (DC01), 01:30 (DC02)	7 daily / 4 weekly / 3 monthly	Full
FILE01 data (D:\Shares)	VSS shadow copies for quick restore + Windows Server Backup of volume D: to \\BACKUP01\Repo\FileBackups	Shadow copies 07:00 & 12:00; WSB daily 02:00 (full on Sunday, incremental via VSS snapshots Mon–Sat)	Shadow copies 14 days; WSB 14 daily/4 weekly	Full + Incr emental
VMs (all)	`Export-VM` to D:\NileBank-Lab\Backups\VM-Exports (VM powered off or live export)	Weekly (Sunday) per VM; before major changes	Last 3	Full (VM level)
Configuration: GPO, DHCP, DNS, Print, scripts	Backup-GPO, Backup-DhcpServer, dnscmd/Export-DnsServerZone, PrintBRM, copy of this documentation	Daily 03:00 and before each change	30 days	Configurati on
Linux (monitoring DB, configs)	`mysqldump`, tar of /etc/zabbix, rsync to BACKUP01	Daily	14 days	Full

Full vs incremental: a *full* backup copies everything and restores fastest but uses most space/time; an *incremental* copies only changes since the last backup, saving space but needing the chain to restore. Windows Server Backup uses VSS snapshot-based block-level 'incremental' images, so every run appears as a full that stores only changed blocks.

27.3 Configure the jobs

Backup configuration

```
# A. DC System State (run on each DC; add a 30 GB backup VHDX 'E:' to each DC first)
PS> Install-WindowsFeature Windows-Server-Backup
PS> wbadmin start systemstatebackup -backupTarget:E: -quiet
PS> wbadmin get versions
PS> robocopy E:\WindowsImageBackup \\BACKUP01\Repo\DCBackups\DC01 /MIR /R:2 /W:5 /LOG:E:\copy.log

# Schedule on the DC (Task Scheduler via PowerShell)
PS> $a = New-ScheduledTaskAction -Execute 'wbadmin.exe' -Argument 'start systemstatebackup -backupTarget:E: -quiet'
PS> $t = New-ScheduledTaskTrigger -Daily -At 1:00am
PS> Register-ScheduledTask -TaskName 'NileBank-SystemState' -Action $a -Trigger $t -User 'SYSTEM' -RunLevel Highest

# B. File server: enable Previous Versions
PS> vssadmin add shadowstorage /for=D: /on=D: /maxsize=15GB
PS> vssadmin create shadow /for=D:
#  GUI schedule: D: → Properties → Shadow Copies → Settings → Schedule (07:00 and 12:00 Mon–Fri)
PS> wbadmin enable backup -addtarget:\\BACKUP01\Repo\FileBackups -include:D: -schedule:02:00 -user:NILEBANK\svc-backup -p

# C. VM export from the host (weekly)
PS> $d = Get-Date -Format 'yyyy-MM-dd'
PS> 'DC01','DC02','FILE01','PRINT01','DHCP01' | ForEach-Object { Export-VM -Name $_ -Path "D:\NileBank-Lab\Backups\VM-Exp

# D. Configuration
PS> Backup-GPO -All -Path 'D:\NileBank-Lab\Backups\Config\GPO'
PS> Backup-DhcpServer -ComputerName DHCP01 -Path 'D:\NileBank-Lab\Backups\Config\DHCP'
PS> dnscmd /zoneexport corp.nilebank.lab corp.nilebank.lab.bak      # file written in C:\Windows\System32\dns
PS> & "$env:SystemRoot\System32\spool\tools\PrintBrm.exe" -B -S \\PRINT01 -F D:\NileBank-Lab\Backups\Config\Print\print.p
PS> Enable-ADOptionalFeature 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -Target corp.nilebank.lab    # one-way
```

WARNING: Never restore a **checkpoint** or a VM copy of a domain controller as a normal recovery method. Hyper-V 2016+ exposes *VM-GenerationID*, which lets Windows Server 2012+ domain controllers detect snapshot reverts and protect AD, but backup and recovery with System State and good replication partners remains the supported method. Also never run a domain controller restore across the **tombstone lifetime (180 days)**.

27.4 RPO and RTO

RPO (Recovery Point Objective) is the maximum acceptable amount of data loss measured in time: with daily backups, the RPO is up to 24 hours; with shadow copies at 07:00 and 12:00 it is about 5 hours for files. **RTO (Recovery Time Objective)** is the maximum acceptable time to restore service. Lab targets below are measured and written down after each test.

Service	Lab RPO	Lab RTO target	Why
AD / DNS (with DC02)	~0 (multi-master replication)	< 5 min (clients fail over)	Redundancy removes the need to restore for a single DC failure.
File data	≤ 5 h (shadow copies), 24 h (WSB)	< 30 min for a file, < 2 h for the volume	VSS for instant file restore; WSB for volume.
Print server	24 h	< 1 h	Configuration only; rebuild from PrintBRM export.
DHCP	24 h	< 1 h	Restore Backup-DhcpServer, re-authorize.
Monitoring (LINUX01)	24 h	< 4 h	Not business-critical.

LAB vs PRODUCTION: A bank defines RPO/RTO by business service criticality (payments may require RPO near zero and RTO of minutes with synchronous replication and a second site), tests recovery on a schedule, keeps immutable/air-gapped copies, and reports results to risk and audit.

27.5 Recovery scenarios and restore tests

#	Scenario	Recovery steps	Evidence to capture
1	Deleted file	On CLIENT01: right-click folder → **Properties → Previous Versions** → Restore. Or on FILE01: `robocopy \\BACKUP01\Repo\FileBackups\... D:\Shares\Finance /E`.	Before/after screenshots, shadow copy list `vssadmin list shadows`.
2	Corrupted file	Open Previous Versions, choose an earlier timestamp, Copy to a new location, compare, then replace.	Hash comparison `Get-FileHash` of old/new.

#	Scenario	Recovery steps	Evidence to capture
3	Failed VM (non-DC)	Hyper-V: `Import-VM -Path '...\FILE01\2026-xx-xx\FILE01\Virtual Machines\<GUID>.vmcx' -Copy -GenerateNewId` (or -Register to reuse IDs when the original is gone). Start and verify services and domain trust (`Test-ComputerSecureChannel -Repair` if needed).	Import wizard, VM running, shares accessible.
4	Failed DC	**If DC02 is healthy:** clients continue. Remove DC01 metadata: `Remove-ADDomainController` not possible offline → in ADUC delete the DC computer object (Delete this DC), or `ntdsutil → metadata cleanup`. Build a new Windows Server, promote as in Section 23, transfer/seize FSMO if needed (`Move-ADDirectoryServerOperationMasterRole -OperationMasterRole 0,1,2,3,4 -Force` to seize). **If all DCs lost:** restore System State from the latest backup into DSRM (below).	dcdiag, repadmin, FSMO listing, new DC in Get-ADDomainController.
5	DNS failure	Restart service (`Restart-Service DNS`); if zone damaged, delete zone on one DC and let AD replicate it back, or restore System State. Re-register records: `ipconfig /registerdns; nltest /dsregdns`. Rebuild reverse zones from the exported config.	nslookup results before/after; DNS event log.
6	Accidental configuration change	GPO: `Restore-GPO -Name 'GPO-Users-Restrictions' -Path D:\...\GPO`; DHCP: `Restore-DhcpServer -Path ...`; deleted AD object: `Get-ADObject -Filter 'isDeleted -eq $true' -IncludeDeletedObjects | Restore-ADObject` (Recycle Bin).	gpresult before/after; restored object visible.

System State restore (all domain controllers lost) — OPTIONAL / ADVANCED

System State restore sequence

```
1. Reinstall Windows Server with the same hostname, join nothing, install Windows Server Backup.
2. Boot Directory Services Restore Mode:   bcdedit /set safeboot dsrepair   → restart
3. Log on with the DSRM password (.\Administrator).
4. wbadmin get versions -backupTarget:E:
5. wbadmin start systemstaterecovery -version:<version-id> -backupTarget:E: -quiet
6. (Authoritative restore of deleted objects only)  ntdsutil "activate instance ntds" "authoritative restore" "restore su
7. bcdedit /deletevalue safeboot   → restart normally.
8. repadmin /replsummary ; dcdiag
```

27.6 Restore test log

Test	Date/time	Result	Time to restore	Screenshot file
File deletion + Previous Versions				30-Restore-File.png
GPO restore				31-Restore-GPO.png
VM import from export				32-Restore-VM.png
AD object restore (Recycle Bin)				33-Restore-ADObject.png
System State restore (optional)				34-Restore-SystemState.png

Leave the blanks until you have **actually performed** each restore. An untested backup is only a hope.

28. Security Hardening

28.1 Why a bank needs defense in depth

No single control stops every attack. A bank layers independent controls so that when one fails (a phished password, a missed patch, a misconfigured share) others contain the damage: network segmentation, firewalls, least privilege, strong authentication, endpoint protection, patching, logging and monitoring, backups and tested recovery, and trained people with documented processes. Regulators and auditors expect layered, evidenced controls.

28.2 Controls implemented in this lab

Control	Implementation (lab)	Production banking expectation
Least privilege	AGDLP groups; no user ACLs; share/NTFS minimum rights; Modify not Full Control	Role-based access control with periodic recertification, JIT elevation
Separate admin accounts	adm.a.hassan (Domain Admin) vs a.hassan (standard); admin member of Protected Users; PSO with stricter policy	Tiered administration (Tier 0/1/2), Privileged Access Workstations, PAM vault
Standard user accounts	Users are not local admins; GPO-Users-Restrictions	Application control, no local admin, EDR
Strong passwords + lockout	14+ chars, 24 history, lockout 5/15 min	MFA everywhere for administrators and remote access
Windows Defender	GPO-Computers-Defender; check `Get-MpComputerStatus`	Enterprise EDR with SOC monitoring
Windows Firewall	On all profiles via GPO; RDP only from Management VLAN (below)	Host + network firewalls, micro-segmentation
Server patching	Windows Update with schedule; DCs patched one at a time	Change-controlled patch rings, vulnerability management
Restricted administrative access	Only GG-IT-Admins in Administrators on servers (Restricted Groups); admin logon denied on workstations	Bastion hosts, session recording
Service account separation	Service Accounts OU; gMSA; deny interactive logon	gMSA/PAM, secret rotation
Auditing	Advanced Audit Policy; Security log 1 GB; DC logon/directory service changes	Central SIEM with 24x7 SOC, retention per regulation
Secure management	SSH keys, RDP with NLA, WinRM only from VLAN 10, SMB signing/encryption	Out-of-band management network
Backup protection	Repo ACL restricted; service account; copy offline	Immutable/air-gapped backups, ransomware-resilient design

28.3 Hardening commands

Windows hardening

```
# Firewall: RDP only from Management VLAN on all servers
PS> Get-NetFirewallRule -DisplayGroup 'Remote Desktop' | Set-NetFirewallRule -RemoteAddress 10.10.10.0/24
PS> Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled True
PS> New-NetFirewallRule -DisplayName 'WinRM from Mgmt only' -Direction Inbound -Protocol TCP -LocalPort 5985,5986 -Remote

# Disable legacy protocols
PS> Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol -NoRestart
PS> Set-SmbServerConfiguration -EnableSMB1Protocol $false -RequireSecuritySignature $true -Force
PS> Set-ItemProperty 'HKLM:\SYSTEM\CurrentControlSet\Control\Lsa' -Name LmCompatibilityLevel -Value 5

# Auditing
PS> auditpol /set /subcategory:"Logon" /success:enable /failure:enable
PS> auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
PS> auditpol /set /subcategory:"Directory Service Changes" /success:enable
PS> wevtutil sl Security /ms:1073741824          # 1 GB Security log
PS> Get-WinEvent -FilterHashtable @{LogName='Security';Id=4625} -MaxEvents 20     # failed logons

# Defender status
PS> Get-MpComputerStatus | Select AMServiceEnabled,RealTimeProtectionEnabled,AntivirusSignatureLastUpdated
PS> Update-MpSignature; Start-MpScan -ScanType QuickScan

# Windows LAPS (OPTIONAL / ADVANCED, Windows Server 2022 with April 2023+ updates)
PS> Update-LapsADSchema
PS> Set-LapsADComputerSelfPermission -Identity 'OU=Servers,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
#   GPO: Administrative Templates → System → LAPS → Configure password backup directory = Active Directory
```

Strict inter-VLAN firewall (replaces the baseline in Section 5.4)

Port-specific ruleset

```
# /etc/nftables.conf on RTR01 (forward chain) — Users -> Servers: only required services
chain forward { type filter hook forward priority 0; policy drop;
  ct state established,related accept
  ip protocol icmp accept                                    # keep ICMP for troubleshooting (restrict in production)
  iifname "vlan10" accept                                     # Management -> everywhere
  # Users (VLAN20) -> Domain controllers
  iifname "vlan20" oifname "vlan30" ip daddr { 10.10.30.10, 10.10.30.11 } \
        tcp dport { 53, 88, 135, 389, 445, 464, 636, 3268, 3269, 49152-65535 } accept
  iifname "vlan20" oifname "vlan30" ip daddr { 10.10.30.10, 10.10.30.11 } udp dport { 53, 88, 123, 389, 464 } accept
  # Users -> File / Print servers (SMB + RPC for printing)
  iifname "vlan20" oifname "vlan30" ip daddr { 10.10.30.20, 10.10.30.21 } tcp dport { 135, 445, 49152-65535 } accept
  # Users -> DHCP relay handled by RTR01 itself (input chain), not forwarded
  # Servers -> Internet (updates) and servers -> DHCP clients denied
  iifname "vlan30" oifname "eth0" accept
  iifname "vlan20" oifname "eth0" tcp dport { 80, 443 } accept
}
```

NOTE: Inter-VLAN DHCP relay is handled in RTR01's **input/output** path: allow UDP 67/68 to the router on VLAN 20 and permit relay → 10.10.30.12 in the output chain (default accept). If the DHCP lease fails after tightening, check the rule order first.

28.4 Event Viewer security queries

Security review

```
PS> Get-WinEvent -FilterHashtable @{LogName='Security';Id=4624,4625,4740,4728,4732,4756} -MaxEvents 30
PS> Get-ADUser -Filter {LockedOut -eq $true} | Select Name,SamAccountName
PS> Search-ADAccount -LockedOut
PS> Get-ADGroupMember 'Domain Admins' | Select Name
```

Event ID	Meaning
4624 / 4625	Successful / failed logon
4740	Account locked out
4720 / 4726	User created / deleted
4728 / 4732 / 4756	Member added to global / local / universal security group
1102	Security log cleared (always alert on this)

WARNING: Every setting in this section is for the isolated lab. Do not paste lab hardening into a real bank environment without change approval, regulatory review, testing and rollback.

29. Monitoring

Monitoring answers three questions: **Is it up? Is it healthy? Is it about to fail?** The platform is Zabbix on LINUX01 (Section 22) with Zabbix Agent 2 on Windows servers; Prometheus + Grafana is an acceptable alternative. Netdata or Windows Admin Center can supplement for real-time views.

29.1 Monitoring matrix

Component	Metrics / checks	Method
All servers	CPU %, RAM %, disk free %, disk latency, network in/out, uptime, pending reboot	Zabbix agent templates
Domain Controllers	Services NTDS, Netlogon, KDC, DFSR, W32Time; LDAP port 389/636; Kerberos 88; replication failures (repadmin); SYSVOL/NETLOGON shares; AD logon time; Security events 4625/4740	Agent + custom script (`repadmin /replsummary`), event log items
DNS	DNS service, resolution of corp.nilebank.lab, SRV record availability, query latency, forwarder health	`net.dns.record` checks from LINUX01, service check
DHCP	DHCPServer service, scope utilisation %, lease availability, authorization	Service + PowerShell script `Get-DhcpServerv4ScopeStatistics`
File Server	LanmanServer, disk free on D:, shadow copy age, share availability (445), open sessions	Agent, `net.tcp.service`
Print Server	Spooler service, queue length, error jobs	Service + PowerShell
Linux	CPU/RAM/disk/network, systemd services (ssh, zabbix-server, mysql, apache2), uptime, failed logins	Local agent
Network / RTR01	Interface status, throughput, ICMP loss to each VLAN gateway, ip_forward, nftables service	ICMP ping, agent/SNMP
Event logs	Critical/Error in System, Application, Directory Service, DNS Server; Event 1102 (log cleared)	Zabbix eventlog[] or forwarding
Backups	Last successful System State/WSB time; repo free space; failed jobs	Script that checks `wbadmin get versions` timestamp
Hyper-V host	Free RAM, free space on D:, VM state, VM CPU wait	Windows exporter/Agent on host or PowerShell

29.2 Alert rules

Severity	Trigger	Action
Disaster / High	DC unreachable > 2 min; NTDS/DNS/Netlogon stopped; replication failure > 1 h; Security log cleared (1102); disk free < 5%	Notify immediately (email/Telegram/Slack), start runbook
Average	CPU > 90% for 10 min; RAM > 90% for 10 min; disk free < 15%; scope utilisation > 85%; Spooler stopped; backup older than 26 h; time offset > 2 min	Ticket + email
Warning	Disk free < 25%; certificate expiring < 30 days; pending reboot > 7 days; > 10 failed logons in 5 min for one user	Dashboard + daily digest
Information	Service restarted, new DHCP lease burst, successful backup	Log only

Principle: alert on symptoms that need a human decision, with thresholds tied to the RPO/RTO; avoid alert fatigue by tuning thresholds and requiring a duration before firing. Lab SMTP: Zabbix → Administration → Media types → Email (use a test mailbox or a local MailHog/MAIL01).

30. Troubleshooting Runbook

General method: **1) Define the problem and scope (one user, one VLAN, all?) → 2) Check recent changes → 3) Test network → 4) Test DNS → 5) Test authentication/time → 6) Check services → 7) Read event logs → 8) Fix one thing → 9) Verify → 10) Document.** Follow the OSI-style order: physical/VM state → IP → DNS → services → permissions.

30.1 AD login failure

Diagnostic sequence

```
CMD> ipconfig /all
CMD> nslookup corp.nilebank.lab
CMD> ping DC01
CMD> whoami
CMD> gpresult /r
CMD> nltest /dsgetdc:corp.nilebank.lab
CMD> nltest /sc_verify:corp.nilebank.lab
CMD> w32tm /query /status
PS>  Get-ADUser s.nabil -Properties LockedOut,Enabled,PasswordExpired,LastBadPasswordAttempt
```

Check	How	Typical finding / fix
DNS	ipconfig /all, nslookup corp.nilebank.lab	Client uses 1.1.1.1 or router DNS → set DNS to 10.10.30.10/.11 (DHCP option 006).
Network	ping gateway, ping DC01, Test-NetConnection DC01 -Port 389	Wrong VLAN ID, RTR01 down, firewall rule.
Domain membership / secure channel	nltest /sc_verify, Test-ComputerSecureChannel	Broken trust → `Test-ComputerSecureChannel -Repair -Credential …` or rejoin.
Time synchronization	w32tm /query /status; compare with DC	Skew > 5 min breaks Kerberos → `w32tm /resync`; check Hyper-V time sync.
Account status	Get-ADUser … LockedOut/Enabled/PasswordExpired; Search-ADAccount -LockedOut	Unlock-ADAccount; reset password; check logon hours/workstations.
GPO	gpresult /r, gpresult /h	Logon scripts/policy denying logon ("Deny log on locally"), corrupted GPO.
Event logs	Security 4625 (Status/SubStatus codes) on DC; System Netlogon on client	0xC000006A bad password; 0xC0000234 locked out; 0xC0000070 outside logon hours.

30.2 DNS failure

Diagnostic sequence

```
PS>  Get-Service DNS
PS>  Get-WindowsFeature DNS
CMD> nslookup
     > server 10.10.30.10
     > corp.nilebank.lab
     > set type=SRV
     > _ldap._tcp.dc._msdcs.corp.nilebank.lab
CMD> ipconfig /flushdns
CMD> ipconfig /registerdns
PS>  Restart-Service DNS; Restart-Service Netlogon
PS>  dcdiag /test:dns /v
PS>  Get-DnsServerZone ; Get-DnsServerForwarder
```

1. **Scope.** Does it fail for one name or all names, for one client or all clients, internal or external names only?
2. **Client configuration.** `ipconfig /all` — correct DNS servers? DHCP-assigned or static?
3. **Reachability.** `Test-NetConnection 10.10.30.10 -Port 53`. If it fails: network/VLAN/firewall.
4. **Service.** `Get-Service DNS`; start it; read the 'DNS Server' log for event 4000/4013 (AD not ready) or zone load errors.
5. **Data.** Does the record exist? Wrong IP? Stale record? Remove/add; confirm scavenging and secure dynamic update settings.

6. **Forwarders.** Internal names OK but Internet names fail → test forwarders; `nslookup www.microsoft.com 1.1.1.1` from the DC; check RTR01 NAT.
7. **Cache.** `ipconfig /flushdns` (client), `Clear-DnsServerCache` (server).
8. **Registration.** `ipconfig /registerdns`, `nltest /dsregdns`, restart Netlogon to re-register SRV records.

30.3 Domain join failure

Check	Command / action	Fix
DNS	nslookup corp.nilebank.lab on the joining computer	Set DNS to the DC. 'The specified domain either does not exist or could not be contacted' is almost always DNS.
Network	ping/Test-NetConnection DC01 -Port 389,445,88,135	VLAN/RTR01/firewall; DC reachable?
Time	w32tm /query /status	Skew < 5 min.
Credentials	Use NILEBANK\adm.a.hassan; account enabled; password current	Standard users can join 10 computers by default (ms-DS-MachineAccountQuota).
Firewall	Windows Firewall / RTR01 ruleset	Open required AD ports.
AD availability	Get-Service NTDS,DNS on DC; dcdiag /q	Start DCs; resolve replication.
Duplicate/stale computer object	Get-ADComputer CLIENT01; same name or old SID	Delete stale object or reset: `Reset-ComputerMachinePassword`.
Event Viewer	On client: `C:\Windows\debug\NetSetup.log`; System log; DC Security/Directory Service	Read the specific error code.

30.4 File access failure

Diagnostic sequence

```
PS> Get-ADPrincipalGroupMembership <user> | Select Name
PS> Get-SmbShareAccess -Name Finance                          # on FILE01
PS> icacls D:\Shares\Finance                                  # NTFS ACL
PS> (Get-Acl D:\Shares\Finance).Access | ft IdentityReference,FileSystemRights,AccessControlType,IsInherited
PS> Test-NetConnection FILE01 -Port 445
PS> nslookup file01.corp.nilebank.lab
```

Check	What to confirm
User membership	User is in the correct global group; token refreshed (log off/on).
Security group nesting	Global → Domain Local nesting is correct (AGDLP) and groups are Security (not Distribution).
Share permissions	Share ACL allows the group (Full/Change/Read).
NTFS permissions	NTFS ACL grants the group Modify/Read; Effective Access tab for the user.
Explicit Deny	Search for Deny ACEs; Deny beats Allow.
Inheritance	Sub-folder broke inheritance or inherited an unexpected ACE.
Server availability	FILE01 powered, LanmanServer running, disk online, 445 reachable, DNS record correct.
Client	Mapped drive disconnected; stale credentials: `net use * /delete`; `klist purge`.

30.5 Printer failure

Check	Command / action
Printer queue	Print Management → Open Printer Queue; cancel stuck jobs.
Print Spooler	`Get-Service Spooler`; restart; clear C:\Windows\System32\spool\PRINTERS.
Driver	Print Management → Drivers; reinstall class driver; Event 372.
Network	`ping PRINT01`, `Test-NetConnection PRINT01 -Port 445`; printer port address reachable.
Permissions	Printer Security tab; DL-Print-* group membership.
Print server	PRINT01 up, joined to domain, File and Printer Sharing allowed.
Client connectivity	Remove/re-add printer; `Get-Printer`; check Point and Print GPO.

31. Event Viewer

Open with `eventvwr.msc` or query with `Get-WinEvent`. Use **Custom Views → Administrative Events** first, then filter by Event ID and time. Always correlate the time of the user's complaint with the event time and read the **whole** message text, not just the ID.

Log	Location	Use / key events
System	Windows Logs → System	Service failures (7000–7034), unexpected shutdown 41/6008, disk 7/11/153, Netlogon 5719/5805, Time-Service 129/134.
Application	Windows Logs → Application	Application crashes (1000), VSS 8193, .NET errors.
Security	Windows Logs → Security	Logon 4624/4625, lockout 4740, account/group change 4720–4756, privilege use 4672, log cleared 1102, GPO/object changes.
Directory Service	Applications and Services Logs → Directory Service	AD health: 1311/1925 replication topology/link, 2087/2088 DNS lookup failure, 1000/1001 start/stop, 2092 FSMO.
DNS Server	Applications and Services Logs → DNS Server	4000/4013 AD not available at start, 4015 critical AD error, 6702 zone errors, dynamic update failures 8015.
DFS Replication	Applications and Services Logs → DFS Replication	SYSVOL replication: 4602 / 4604 initialized, 4012 stopped too long, 2213 dirty shutdown.
DHCP	Applications and Services Logs → Microsoft → Windows → DHCP-Server → Admin / Operational	1063 server authorized/unauthorized, 1012 scope exhausted, lease events.
Group Policy	Applications and Services Logs → Microsoft → Windows → GroupPolicy → Operational	Processing start/end 4001–5016, errors 1058 (cannot access gpt.ini), 1030, 1085 (CSE failure), 8001/8004 summary.
PrintService	Applications and Services Logs → Microsoft → Windows → PrintService → Admin / Operational	Driver and queue errors 372, 808; Operational shows job lifecycle (enable log first).

Useful queries

```
PS> Get-WinEvent -FilterHashtable @{LogName='System';Level=1,2;StartTime=(Get-Date).AddHours(-24)} | Select TimeCreated,I
PS> Get-WinEvent -LogName 'Directory Service' -MaxEvents 30 | Where Level -le 3
PS> Get-WinEvent -LogName 'DNS Server' -MaxEvents 20
PS> Get-WinEvent -LogName 'Microsoft-Windows-GroupPolicy/Operational' -MaxEvents 20
PS> wevtutil epl System C:\Temp\System.evtx            # export log for evidence
```

How an IT Support / System Administrator uses Event Viewer

- Start from the symptom and time, not from random log browsing.
- Filter by **Level** (Critical, Error, Warning) and **Source**; open the Details tab for error codes.
- Search the Event ID plus provider name in vendor documentation.
- Correlate across machines: user's PC (System/GroupPolicy), DC (Security/Directory Service), server (Application).
- Export evidence (.evtx) for tickets and post-incident reviews; forward logs to central monitoring (SIEM in a bank).

32. Failure and Recovery Exercises

Run each exercise on the lab only, after a recent backup or with the exported VMs ready. Record start/end times and capture screenshots. Restore the normal state after each test and validate.

32.1 DNS unavailable

Field	Detail
Problem	DNS service stopped on both DCs (test one DC first).
Symptoms	Clients cannot resolve names; 'domain cannot be contacted'; shares by name fail; nslookup times out.
Detection	Zabbix DNS check fails; nslookup timeout; DNS Server event 4000 series.
Root Cause	Service stopped or crashed; zone failed to load; wrong DNS on clients.
Troubleshooting	Get-Service DNS; Test-NetConnection -Port 53; read DNS log; verify DC DNS settings.
Resolution	Start-Service DNS; correct DNS client settings; ipconfig /registerdns; nltest /dsregdns.
Validation	nslookup corp.nilebank.lab succeeds; domain logon works; dcdiag /test:dns passes.
Prevention	Two DNS servers on every host; monitoring on DNS; service recovery options set to restart.

32.2 DC01 unavailable

Field	Detail
Problem	DC01 powered off (Section 26).
Symptoms	Short delay on new logons; PDC emulator functions missing; replication errors to DC01.
Detection	Zabbix host down; repadmin /replsummary shows failures; Netlogon 5719.
Root Cause	Host/VM failure or planned maintenance.
Troubleshooting	Confirm DC02 services; nltest /dsgetdc; check FSMO holders (netdom query fsmo).
Resolution	Start DC01; or transfer/seize FSMO if permanent loss; rebuild if necessary.
Validation	repadmin /replsummary 0 fails; dcdiag /q clean; users log on.
Prevention	Two or more DCs, FSMO documentation, monitoring, tested backups.

32.3 DC02 unavailable

Field	Detail
Problem	DC02 powered off.
Symptoms	No user impact if DC01 up; loss of redundancy.
Detection	Monitoring alert; replication failure to DC02.
Root Cause	VM failure/maintenance.
Troubleshooting	Verify DC01 serves all roles; check clients using DC02 first in DNS list.
Resolution	Restart DC02; check replication.
Validation	repadmin /replsummary; Get-ADDomainController -Filter * shows both reachable.
Prevention	Treat single remaining DC as a high-priority risk; patch DCs one at a time.

32.4 DHCP unavailable

Field	Detail
Problem	DHCP01 powered off or DHCPServer stopped.
Symptoms	New/renewing clients get 169.254.x.x; existing leases work until expiry (8 days).
Detection	ipconfig shows APIPA; Zabbix DHCP check; DHCP event log silent.
Root Cause	Server down, service stopped, scope inactive, relay down, authorization lost.
Troubleshooting	ipconfig /renew; ping 10.10.30.12; check relay on RTR01 (systemctl status isc-dhcp-relay); Get-DhcpServerInDC.
Resolution	Start service/VM; activate scope; re-authorize; Restore-DhcpServer from backup.

Field	Detail
Validation	Client receives 10.10.20.1xx lease with correct options.
Prevention	DHCP failover partner, long lease time, monitoring of scope utilisation, backups.

32.5 File server unavailable

Field	Detail
Problem	FILE01 powered off or LanmanServer stopped.
Symptoms	Mapped drives show red X; 'network path not found'.
Detection	Zabbix port 445 check fails; users report drive errors.
Root Cause	VM off, disk offline, service stopped, DNS record stale.
Troubleshooting	ping/nslookup FILE01; Test-NetConnection -Port 445; check disk status on FILE01.
Resolution	Start VM, bring disk online, start services; if destroyed, restore from backup (Section 27).
Validation	Open \\FILE01\Finance as m.farouk; files intact.
Prevention	Backups, shadow copies, DFS Namespace + second server for HA (OPTIONAL / ADVANCED).

32.6 Print Spooler stopped

Field	Detail
Problem	Stop-Service Spooler on PRINT01.
Symptoms	Printing fails; printers show Offline/Error; Print Management cannot connect.
Detection	Spooler service alert; PrintService events.
Root Cause	Service crash, bad driver, or stuck job.
Troubleshooting	Get-Service Spooler; clear spool folder; check drivers.
Resolution	Start-Service Spooler; remove problem job/driver.
Validation	Test page prints; queue empty.
Prevention	Standard class drivers, service recovery set to restart, monitoring.

32.7 Wrong NTFS permission

Field	Detail
Problem	A mistake grants HR modify on Finance or removes Finance access.
Symptoms	HR sees Finance data (security breach) or Finance gets Access Denied.
Detection	Access tests fail/succeed unexpectedly; audit events 4663 if object auditing enabled.
Root Cause	ACL edited directly or inheritance broken; wrong group nesting.
Troubleshooting	icacls D:\Shares\Finance; Effective Access; Get-ADPrincipalGroupMembership.
Resolution	Re-apply ACL via the Set-ShareAcl function; remove stray ACEs; document the change.
Validation	Repeat test matrix in Section 20.5.
Prevention	Permissions only through groups; change control; periodic ACL review.

32.8 Broken GPO

Field	Detail
Problem	A GPO blocks everything (e.g. 'Prohibit access to Control Panel' applied to IT, or firewall blocks all).
Symptoms	Users cannot work; admins locked out of tools; logon takes long.
Detection	gpresult /r shows the GPO; GroupPolicy Operational log errors.
Root Cause	Mis-scoped GPO or wrong setting.
Troubleshooting	gpresult /h; GPMC → Group Policy Results; unlink GPO to test.
Resolution	Unlink/disable; Restore-GPO from backup; fix scope/filtering; gpupdate /force.
Validation	gpresult no longer lists the GPO; settings normal.

Field	Detail
Prevention	Test GPOs on a pilot OU first, back up before edits, change approvals.

32.9 Domain join failure

Field	Detail
Problem	New VM points to router DNS and cannot join.
Symptoms	'An Active Directory Domain Controller for the domain could not be contacted'.
Detection	User error message; NetSetup.log.
Root Cause	Wrong DNS, VLAN or firewall; time skew; stale computer object.
Troubleshooting	Section 30.3 checklist.
Resolution	Set DNS to DC, fix VLAN, correct time, delete stale object, retry join.
Validation	systeminfo shows domain; computer in correct OU.
Prevention	Standard build checklist, DHCP options for DNS, documentation.

32.10 Incorrect DNS configuration

Field	Detail
Problem	Workstation or server set to 1.1.1.1 as only DNS.
Symptoms	Internet works but AD logon, domain join, GPO and shares fail; Netlogon errors.
Detection	nslookup corp.nilebank.lab fails; Event ID 5719.
Root Cause	Manual misconfiguration.
Troubleshooting	ipconfig /all; nslookup _ldap._tcp.dc._msdcs.corp.nilebank.lab.
Resolution	Set DNS to 10.10.30.10/.11 or return to DHCP; ipconfig /flushdns; gpupdate /force.
Validation	AD logon and domain lookups succeed.
Prevention	DHCP option 006; GPO/baseline checks; DNS client config audit.

33. Documentation and Evidence

33.1 Naming convention

Name screenshots `NN-Component-Description.png`: two-digit sequence, component, what it shows; no spaces. Store them in `Screenshots/<Phase>/` in the repository. Crop to the relevant window; include the VM name, time and command prompt where possible; redact real passwords and personal data.

Examples

```
01-HyperV-DC01.png          02-DC01-IPConfig.png        03-ADDS-Installed.png
04-DNS-Zone.png              05-DC02-Replication.png     06-GPO-Results.png
```

33.2 Evidence checklist

#	File name	Shows
01	01-HyperV-Manager-VMList.png	Hyper-V Manager with all VMs and state
02	02-HyperV-VSwitch-Manager.png	NILEBANK-LAB and NILEBANK-WAN switches
03	03-HyperV-DC01-Settings.png	DC01 settings: CPU, RAM, VLAN 30, Secure Boot
04	04-DC01-IPConfig.png	ipconfig /all showing 10.10.30.10
05	05-ServerManager-DC01.png	Server Manager dashboard
06	06-ADDS-Installed.png	Get-WindowsFeature AD-Domain-Services
07	07-ADDS-Promotion-Summary.png	Promotion wizard review page
08	08-ADDomain-ADForest.png	Get-ADDomain / Get-ADForest
09	09-ADUC-OUStructure.png	OU tree
10	10-ADUC-UsersGroups.png	Users and security groups
11	11-DNS-Manager-Zones.png	Forward/reverse zones
12	12-DNS-SRV-Records.png	SRV records in _msdcs
13	13-CLIENT01-DomainJoin.png	Domain join success
14	14-CLIENT01-WhoamiGpresult.png	whoami, systeminfo, gpresult /r
15	15-GPMC-Overview.png	GPMC with linked GPOs
16	16-GPO-Report.png	gpresult /h report
17	17-DHCP-Scope.png	DHCP console with scope and options
18	18-DHCP-Lease.png	Lease for CLIENT01
19	19-FILE01-Shares.png	SMB shares
20	20-FILE01-NTFS-Finance.png	NTFS permissions of Finance
21	21-FILE-Test-Allow.png / 22-FILE-Test-Deny.png	Allow and Deny test results
23	23-PRINT01-PrintManagement.png	Printers, drivers, sharing
24	24-CLIENT01-PrinterTest.png	Printer connected from CLIENT01
25	25-LINUX01-IPandSSH.png	ip addr, SSH session, ufw status
26	26-Monitoring-Dashboard.png	Zabbix/Grafana dashboard
27	27-Monitoring-Alert.png	Triggered alert
28	28-DC02-Promotion.png	DC02 promotion wizard (existing domain)
29	29-DC02-GetADDomainController.png	Both DCs listed
30	30-DC02-Replication.png	repadmin /replsummary and showrepl
31	31-DC02-DNSZone.png	DNS zone on DC02
32	32-Failover-DC01-Off.png	DC01 off, CLIENT01 working
33	33-Failover-Recovery.png	Replication healthy after DC01 start
34	34-Backup-Config.png	Backup job and schedule
35	35-Backup-Success.png	wbadmin get versions

#	File name	Shows
36	36-Restore-Test.png	File/GPO/AD object restore
37	37-Security-Audit.png	Audit policy/Security events
38	38-Failure-Tests.png	Each failure scenario (one per file)

33.3 Evidence log template

Phase	Date	What was done	Evidence file(s)	Result

34. Project Acceptance Checklist

Tick an item only when it is implemented **and** validated, with evidence stored.

- ■ Hyper-V configured
- ■ NILEBANK-LAB switch created (plus NILEBANK-WAN)
- ■ RTR01 routing between VLANs working
- ■ DC01 created
- ■ DC01 static IP configured (10.10.30.10)
- ■ AD DS installed
- ■ Domain created (corp.nilebank.lab)
- ■ DNS working (forward + reverse zones, forwarders)
- ■ OUs created
- ■ Users created
- ■ Security groups created (AGDLP)
- ■ CLIENT01 joined
- ■ GPO implemented (8 GPOs)
- ■ DHCP implemented (DHCP01 authorized, scope active, relay working)
- ■ FILE01 implemented
- ■ NTFS permissions tested (allow/deny matrix)
- ■ PRINT01 implemented
- ■ LINUX01 implemented
- ■ DC02 implemented
- ■ DNS redundancy verified
- ■ AD replication verified
- ■ Failover tested (DC01 off, DC02 serves)
- ■ Backup configured
- ■ Restore tested (with evidence)
- ■ Monitoring implemented
- ■ Security hardening completed
- ■ Troubleshooting runbook completed
- ■ Documentation completed

Acceptance criteria

Area	Criterion	Evidence
Network	CLIENT01 reaches DC01/DC02 across VLANs; ping + DNS + logon OK	13, 14
Identity	All users log on; groups give expected access; admin separation proven	10, 14, 20–22
Redundancy	Both DCs return 0 replication failures; logons succeed with DC01 off	30, 32, 33
Services	DHCP lease received, printers/drives mapped by policy	17, 18, 24
Resilience	At least one file, GPO and AD-object restore proven	36
Operations	Monitoring shows all hosts; at least one alert fired and cleared	26, 27
Security	No user ACLs; Defender/Firewall on; audit events visible	37

35. Complete Command Reference

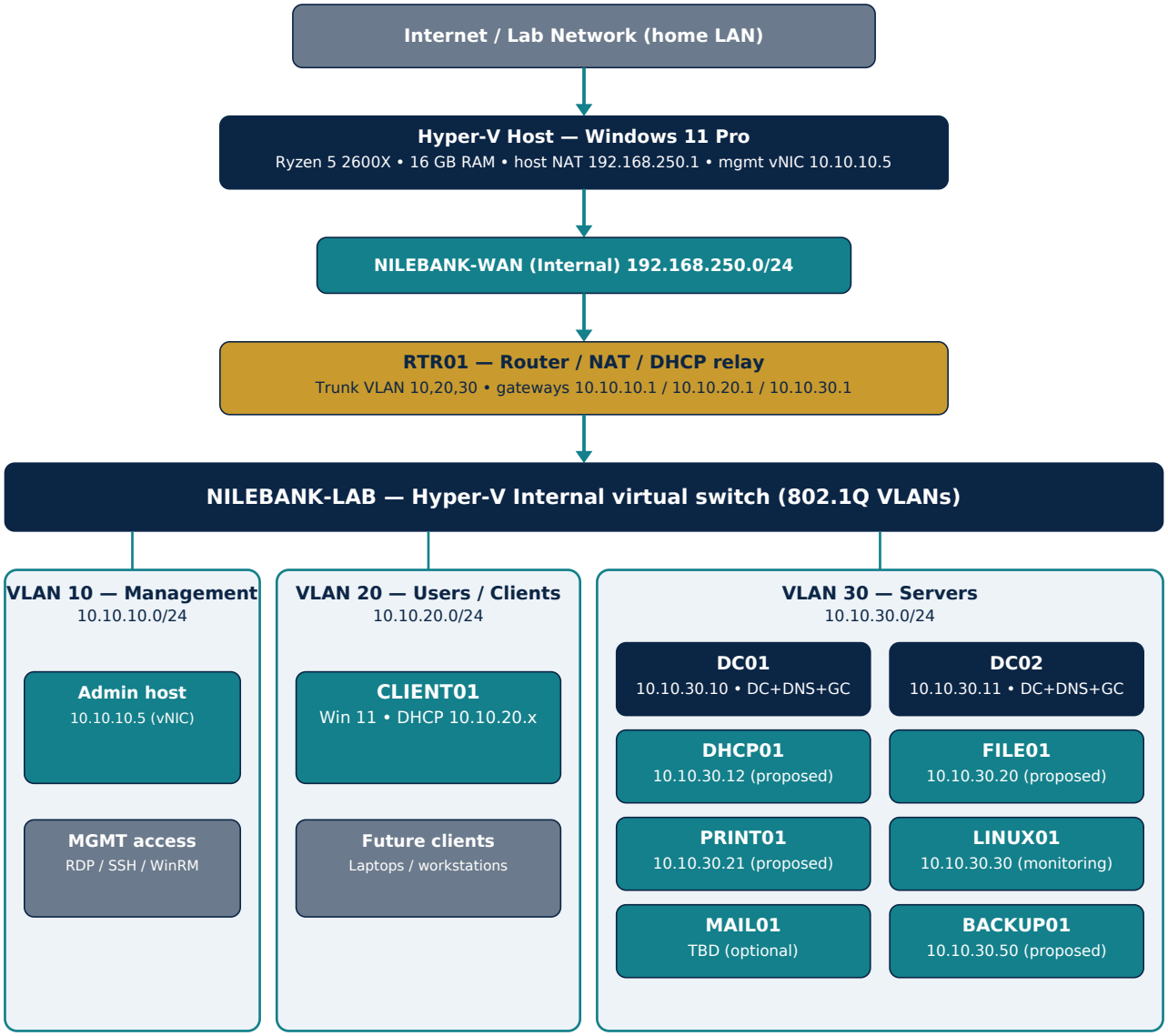
Command	Explanation	Typical use
ipconfig /all	Shows full IP configuration for every adapter: IP, mask, gateway, DNS, DHCP, MAC.	First step in any network problem.
ipconfig /flushdns	Clears the client DNS resolver cache.	After DNS record changes.
ipconfig /registerdns	Re-registers the computer's DNS records and renews DHCP leases for DNS.	Fix missing A/PTR record.
hostname	Prints the computer name.	Confirm identity.
whoami	Shows the current user (DOMAIN\user). `whoami /groups` lists group SIDs; `whoami /fqdn` the DN.	Confirm context and group membership.
systeminfo	OS version, domain, hotfixes, boot time, hardware summary.	`systeminfo | findstr /B /C:"Domain"` to confirm membership.
nslookup	Queries DNS servers directly. Interactive mode: `server`, `set type=SRV`.	Test name resolution and SRV records.
ping	ICMP echo reachability test.	Basic connectivity; may be blocked by firewall.
tracert	Traces the route (hops) to a destination.	`tracert 10.10.30.10` from CLIENT01 should show 10.10.20.1 then the target.
Get-WindowsFeature	Lists installed/available roles and features.	`Get-WindowsFeature AD-Domain-Services`.
Get-Service	Lists services and states.	`Get-Service DNS,NTDS,Netlogon`.
Get-ADDomain	Properties of the domain (modes, FSMO roles, DNSRoot).	Verify domain design.
Get-ADForest	Properties of the forest (GCs, schema master, domains).	Verify forest.
Get-ADDomainController -Filter *	All domain controllers with IP, site, GC.	Confirm DC01 + DC02.
Get-ADUser	Reads user objects: `-Identity`, `-Filter`, `-Properties *`.	Account status checks.
Get-ADGroup	Reads groups; pair with `Get-ADGroupMember`.	Check AGDLP nesting.
Get-ADComputer	Reads computer accounts.	Verify OU placement.
Get-DnsServerZone	Lists DNS zones on the server with type and replication scope.	Confirm AD-integrated zones.
gpupdate /force	Reapplies Group Policy immediately.	After GPO changes.
gpresult /r	Text summary of applied/denied policy for user and computer.	GPO troubleshooting.
gpresult /h C:\Temp\gpresult.html	Detailed HTML policy report.	Evidence and deep analysis.
repadmin /replsummary	Summary of AD replication health.	Daily health check.
repadmin /showrepl	Detailed replication per partition and partner.	Find the failing link and error code.
dcdiag	Runs DC health tests (connectivity, services, replication, FSMO, DNS…).	`dcdiag /q` shows only failures.
dcdiag /test:dns /v	Verbose DNS test of the DC.	Validate DNS before/after promotion.
Get-NetIPConfiguration	PowerShell summary of adapters with IP, gateway, DNS.	Fast overview.
Get-NetAdapter	Network adapter name, status, speed, MAC.	Check link state.
Test-NetConnection	Tests TCP ports and routes: `-ComputerName DC01 -Port 389`; `-TraceRoute`.	Verify service ports through firewalls.

Additional useful commands

More

```
netdom query fsmo            nltest /dsgetdc:corp.nilebank.lab     nltest /sc_verify:corp.nilebank.lab
klist / klist purge          w32tm /query /status                  w32tm /resync
net use                      net share                             Get-SmbShare / Get-SmbShareAccess
icacls <path>                Get-Acl <path>                        Get-DhcpServerv4Scope / Get-DhcpServerv4Lease
Get-Printer / Get-PrintJob   Get-WinEvent -LogName <name>          Backup-GPO / Restore-GPO
wbadmin get versions         Export-VM / Import-VM                 Get-VM | ft Name,State,MemoryAssigned
```

36. Architecture Diagram



Reading the diagram: all VMs attach to one Hyper-V internal switch, separated into three VLANs by 802.1Q tags. RTR01 is the only path between VLANs and to the Internet; it enforces the firewall rules in Sections 5 and 28. The host's management vNIC is the administrator's entry point on VLAN 10. MAIL01 is optional and powered off unless needed.

37. IP Address Table

Status values: **Fixed** = specified by the project; **Proposed** = recommended in this document and to be confirmed (and recorded) at implementation; **DHCP** = leased. No address is used twice.

Hostname	Role	IP	Subnet	Gateway	DNS	VLAN	Status
RTR01 (LAN side)	Router / NAT / DHCP relay	10.10.10.1 / 10.10.20.1 / 10.10.30.1	/24 each	192.168.250.1 (WAN)	1.1.1.1 (WAN)	Trunk 10,20,30	Proposed
Host vNIC	Hyper-V host management	10.10.10.5	255.255.255.0	none	10.10.30.10, .11	10	Proposed
DC01	Primary DC + DNS + GC	10.10.30.10	255.255.255.0	10.10.30.1	10.10.30.11, 10.10.30.10	30	**Fixed**
DC02	Secondary DC + DNS + GC	10.10.30.11	255.255.255.0	10.10.30.1	10.10.30.10, 10.10.30.11	30	**Fixed**
DHCP01	DHCP server	10.10.30.12	255.255.255.0	10.10.30.1	10.10.30.10, .11	30	Proposed (TBD)
FILE01	File server	10.10.30.20	255.255.255.0	10.10.30.1	10.10.30.10, .11	30	Proposed (TBD)
PRINT01	Print server	10.10.30.21	255.255.255.0	10.10.30.1	10.10.30.10, .11	30	Proposed (TBD)
LINUX01	Linux monitoring	10.10.30.30	255.255.255.0	10.10.30.1	10.10.30.10, .11	30	Proposed (TBD)
MAIL01	Mail / Exchange lab (optional)	10.10.30.40	255.255.255.0	10.10.30.1	10.10.30.10, .11	30	Proposed (TBD), powered off by default
BACKUP01	Backup / recovery	10.10.30.50	255.255.255.0	10.10.30.1	10.10.30.10, .11	30	Proposed (TBD)
CLIENT01	Windows 11 client	10.10.20.100–199 (DHCP)	255.255.255.0	10.10.20.1	10.10.30.10, .11 (DHCP)	20	DHCP (reservation optional .50)
Virtual printer	PRN-FIN-01 / PRN-GEN-01 port	10.10.30.250	255.255.255.0	10.10.30.1	n/a	30	Proposed (reserved)

NOTE: Reserved ranges: VLAN 30 .1 gateway, .10–.19 identity, .20–.49 file/print/app, .50–.99 operations, .250 printers. VLAN 20 .1 gateway, .2–.99 static/reserved, .100–.199 DHCP. VLAN 10 .1 gateway, .2–.99 static admin, .100–.119 DHCP.

38. VM Resource Table

VM	vCPU	RAM (startup / min–max)	Disk	Network	Role	Startup priority
RTR01	1	512 MB / 256–768 MB	8 GB	WAN + LAN trunk	Router / NAT / relay	1
DC01	2	2 GB / 1–3 GB	60 GB	VLAN 30	DC, DNS, GC, FSMO	2
DC02	2	2 GB / 1–3 GB	60 GB	VLAN 30	DC, DNS, GC	3
DHCP01	1	1 GB / 768 MB–1.5 GB	50 GB	VLAN 30	DHCP	4
FILE01	2	1.5 GB / 1–2 GB	50 GB OS + 100 GB data	VLAN 30	SMB shares	5
PRINT01	1	1.5 GB / 1–2 GB	50 GB	VLAN 30	Print server	5
LINUX01	2	1 GB / 512 MB–2 GB	40 GB	VLAN 30	Monitoring	6
BACKUP01	1	1.5 GB / 1–2 GB	50 GB OS + 200 GB repo	VLAN 30	Backup/recovery	6 (on demand)
CLIENT01	2	3 GB / 2–4 GB	60 GB	VLAN 20	Windows 11 client	7
MAIL01 (optional)	2	6 GB static	80 GB	VLAN 30	Exchange lab	On demand only

How resources were selected: the 16 GB host reserves ~4 GB for Windows 11, Hyper-V and tools. The remaining ~10–11 GB is divided so that the core identity set (RTR01 + DC01 + DC02 + CLIENT01) uses ~7.5 GB at maximum and the optional services are added in profiles (Section 2.5). Domain controllers receive 2 GB because AD DS caches the database in RAM; routers and DHCP need little; the client has the most because Windows 11 is the heaviest guest. Dynamic Memory lets idle VMs shrink. Dynamic VHDX files use space only as needed; total maximum disk use is approximately 700 GB, within the ~1.1 TB capacity, but monitor free space on D:. Four vCPUs more than the six cores can be over-committed safely on this light workload.

39. Real-World Banking Context

WARNING: This lab is a **simulated learning environment**. It is NOT an actual banking production system, handles no real customer data, and does not meet any regulatory standard.

Area	In this lab	In a real enterprise bank
Centralized identity	Single AD forest/domain, 2 DCs	Multiple DCs across data centers/sites, tiered admin model, PAM, MFA, federation (Entra ID/SSO), HR-driven joiner/mover/leaver automation.
Access control	AGDLP groups, NTFS/SMB	Role-based access with periodic recertification, segregation of duties, approval workflow, dual control for sensitive access.
Network segmentation	3 VLANs via one vSwitch + Linux router	Dozens of zones (DMZ, core banking, payments, SWIFT, ATM, management), next-gen firewalls, NAC, micro-segmentation.
Redundancy	DC02, single DHCP/file/print	Active-active data centers, clustered services, DHCP failover, DFS-N/R, load balancers, redundant links and power.
Monitoring	Zabbix/Prometheus	Enterprise monitoring + SIEM/SOC 24x7, NOC, capacity management, SLAs.
Backup	wbadmin/VSS/exports on same host	Enterprise backup platform, immutable/air-gapped copies, offsite vaulting, scheduled restore tests, retention per regulation.
Disaster recovery	Documented scenarios, manual restores	DR site with replication, defined RTO/RPO per service, regular DR exercises and board reporting.
Security	Hardening GPOs, firewall, Defender	Defense in depth, EDR/XDR, vulnerability management, PKI, encryption, DLP, penetration testing, compliance (PCI DSS, ISO 27001, local central-bank rules).
Auditing	Windows audit policy, Event Viewer	Immutable central logging, access reviews, internal/external audit evidence.
Change management	This document + notes	ITIL change advisory board, approval, test, rollback, CMDB.
Incident response	Troubleshooting runbook, failure exercises	SOC playbooks, severity matrix, on-call, forensics, regulator notification.
Business continuity	Failover tests, restore tests	BCP/DR plans tested with business units, crisis management, third-party dependencies.

40. Career Value: IT Support → System Administrator → Cloud Engineer

The lab demonstrates the progression from supporting users (troubleshooting, permissions, printing) to administering infrastructure (AD, DNS, DHCP, GPO, backup) to engineering platforms (automation, redundancy, monitoring, and the concepts behind cloud identity and networking).

Lab component	Skill demonstrated	Role relevance
Active Directory	Identity management, users, groups, OU design, delegation	IT Support, SysAdmin, Cloud (Entra ID foundation)
DNS	Network services, name resolution, troubleshooting	All roles; Cloud (Azure DNS, private zones)
DHCP	Network services, IP planning, scopes, relay	IT Support, SysAdmin
Group Policy	Endpoint management and standardization	SysAdmin (Intune/MDM equivalent)
Hyper-V	Virtualization, resource planning, virtual networking, VLANs	SysAdmin, Cloud (IaaS concepts)
Linux	Cross-platform administration, SSH, firewall, services	SysAdmin, Cloud (most cloud workloads run Linux)
File/Print	Permissions (AGDLP), share/NTFS, troubleshooting	IT Support, SysAdmin
Backup/DR	Business continuity, RPO/RTO, restore testing	SysAdmin, Cloud (Azure Backup/Site Recovery)
Monitoring	Operations, alerting, capacity	SysAdmin, SRE/Cloud (Azure Monitor)
Security hardening	Infrastructure security, least privilege, auditing	All roles; Cloud (Defender, Sentinel concepts)
PowerShell	Automation and repeatability	SysAdmin, Cloud (IaC foundation)
Documentation	Technical writing, runbooks, evidence	All roles

41. CV Section

Enterprise Banking Infrastructure Lab — Personal Project / Home Lab

Designed and implemented a simulated enterprise banking infrastructure using Windows Server, Hyper-V, Active Directory, DNS, DHCP, Group Policy, Windows/Linux administration, file and print services, monitoring, backup/DR and domain-controller redundancy.

- Built an enterprise-style Active Directory environment with structured OUs, users, security groups and Group Policy.
- Configured Windows Server infrastructure including AD DS, DNS and DHCP in a segmented Hyper-V lab.
- Implemented and validated a secondary domain controller with DNS and Active Directory replication for redundancy.
- Designed and tested SMB/NTFS permissions and domain-joined Windows clients.
- Implemented infrastructure monitoring and documented troubleshooting procedures.
- Designed backup and disaster-recovery scenarios and performed recovery validation.
- Applied security hardening principles including least privilege, administrative separation and controlled access.

WARNING: Label this clearly as a **Personal Project / Home Lab**. Never present it as real production banking infrastructure. Claim only what you actually implemented and tested: delete any bullet whose phase you have not completed (for example, remove 'performed recovery validation' until a restore test has been done and evidenced).

CV claim	Only claim when you have…
AD with OUs/users/groups/GPO	Phases 3, 4, 6 complete with screenshots 08–10, 15–16
AD DS, DNS, DHCP in segmented Hyper-V lab	Phases 1–3, 7, VLAN tests complete
Secondary DC with replication	Phases 11–12 complete, repadmin output saved
SMB/NTFS permissions	Allow/deny test matrix passed (Section 20.5)
Monitoring	Phase 13 dashboard + alert evidence
Backup and DR validation	Phases 14–15 with at least one successful restore
Security hardening	Phase 16 controls verified

42. LinkedIn Project Description

Enterprise Banking Infrastructure Lab (Personal Project / Home Lab)

A hands-on home lab simulating a segmented on-premises banking IT environment on Hyper-V. I designed a three-VLAN network (Management, Users, Servers) behind a Linux router and built a Windows Server infrastructure: Active Directory with a structured OU model and AGDLP groups, AD-integrated DNS, DHCP with relay, Group Policy baselines, SMB/NTFS file services with tested permissions, a print server and a Linux monitoring server. I added a second domain controller, validated replication with repadmin/dcdiag, and ran failover, failure and recovery exercises supported by backup and restore tests and a troubleshooting runbook.

This is a learning environment — not production — documented end to end with architecture diagrams, runbooks and PowerShell automation. *Skills: Windows Server, Active Directory, DNS, DHCP, Group Policy, Hyper-V, Linux, PowerShell, backup & disaster recovery, monitoring, infrastructure security.*

> **NOTE:** Remove any technology from the description that you did not implement. Add the GitHub link and 3–4 evidence screenshots.

43. Interview Answers

NOTE: Written in the first person for convenience. Replace example details with what you actually built, measured and learned; interviewers will ask follow-up questions on every claim.

43.1 30-second answer

“I built a simulated bank infrastructure on Hyper-V with Windows Server and Linux. It has a segmented network, Active Directory and DNS with two domain controllers for redundancy, DHCP, Group Policy, file and print services, monitoring and backup. I tested failover by shutting down a DC, and I documented the troubleshooting and recovery procedures. It's a home lab, but it follows enterprise design principles.”

43.2 Two-minute answer

“I wanted hands-on experience beyond a single-server lab, so I designed Nile Bank, a simulated banking environment on a 16 GB workstation. I created three VLANs — management, users and servers — routed by a small Linux router. The core is a corp.nilebank.lab Active Directory forest with a primary and secondary domain controller, both running DNS and Global Catalog. I organised OUs by administration and policy, used AGDLP so permissions are assigned to groups rather than users, and built eight Group Policy objects such as password, lockout, screen lock and security baseline. On top of that I deployed DHCP with a relay, a file server with share and NTFS permissions that I tested with different users, a print server and a Linux monitoring server. I proved redundancy by shutting down DC01 and confirming that DNS, logon and file access still worked through DC02, then verified replication after recovery. I also set up backups and practised restores. The biggest lessons were that DNS underpins everything and that documenting and testing recovery matters as much as building.”

43.3 Detailed technical answers

Question	Answer
Why did you build this?	To practise the full lifecycle an infrastructure team handles: design, build, secure, monitor, break, fix and document. A bank context forced me to think about segmentation, least privilege, redundancy and recovery rather than just installing roles.
What architecture did you design?	A Hyper-V host with two internal switches. NILEBANK-LAB carries VLAN 10/20/30 using 802.1Q tags; an Ubuntu router VM provides inter-VLAN routing, NAT and DHCP relay. Servers sit in VLAN 30, clients in VLAN 20, administration from VLAN 10. Firewall rules limit client access to AD, DNS, SMB and print ports.
Which technologies?	Hyper-V, Windows Server 2022, AD DS, DNS, DHCP, Group Policy, SMB/NTFS, Print Services, Ubuntu, Zabbix or Prometheus/Grafana, Windows Server Backup/VSS, PowerShell.
How did you implement AD?	Installed AD DS on DC01, promoted a new forest corp.nilebank.lab with integrated DNS and GC, set DSRM password, created OUs by object type, groups using AGDLP, separate admin accounts, and a service-accounts OU with a gMSA. Verified with Get-ADDomain, Get-ADForest and dcdiag.
Why DC02?	A single DC is a single point of failure for authentication and DNS. DC02 provides redundancy, load distribution and a recovery source. It runs DNS and GC so name resolution and universal-group lookups continue.
How did you test redundancy?	Confirmed healthy replication with repadmin, powered off DC01, flushed DNS on a client, resolved names via DC02, logged in with a user not cached on the client, accessed shares, created an object on DC02, restarted DC01 and verified the object replicated and dcdiag was clean. I recorded the timings.
How did you troubleshoot DNS?	Followed a sequence: client DNS settings → connectivity to port 53 → DNS service state → record existence → forwarders → caches → record registration (ipconfig /registerdns, nltest /dsregdns) → Event Viewer and dcdiag /test:dns.
How did you handle file permissions?	AGDLP: users in role global groups, those nested into domain local resource groups, and only domain local groups on NTFS ACLs. Share permissions are broad to those groups, NTFS enforces. I tested allow and deny cases, including an explicit deny, and used Effective Access to verify.
Backup and recovery?	System State backups of DCs, VSS shadow copies and WSB for files, VM exports, config exports (GPO, DHCP, DNS, print), AD Recycle Bin. I defined RPO/RTO per service and practised file, GPO and AD object restores. I noted that backups on the same host are a lab limitation.
What did you learn?	DNS and time are the root of most AD problems; design OUs for policy; test recovery, do not assume; and write the runbook while you troubleshoot.

Question	Answer
How does it prepare you for SysAdmin/Cloud?	The same concepts map to cloud: identity (Entra ID), virtual networks and NSGs (VLANs and firewall), IaaS VMs (Hyper-V), backup/monitoring (Azure Backup/Monitor), and automation with PowerShell leads to IaC with Terraform.

44. GitHub Repository Structure

Recommended structure

```
NileBank-Enterprise-Infrastructure-Lab/
├── README.md
├── Architecture/        ├── Network/          ├── Active-Directory/
├── DNS/                 ├── DHCP/             ├── Group-Policy/
├── File-Server/         ├── Print-Server/     ├── Linux/
├── Monitoring/          ├── Backup-DR/        ├── Security/
├── Troubleshooting/     ├── Scripts/          ├── Screenshots/
└── Documentation/
```

Folder	Contents
README.md	Project summary, diagram image, technologies, status, how to navigate, disclaimer (simulated lab), link to PDF.
Architecture/	Architecture diagram (PNG/draw.io source), design decisions, VM resource table, IP table.
Network/	VLAN plan, Hyper-V switch configuration, RTR01 netplan and nftables files, firewall matrix.
Active-Directory/	Domain design, OU tree, user/group matrix (no passwords), AGDLP model, promotion notes for DC01/DC02.
DNS/	Zone design, forwarders, test outputs, DNS troubleshooting notes.
DHCP/	Scope design, relay configuration, backup export (no secrets).
Group-Policy/	GPO list, settings, GPO reports (HTML), backups (sanitised).
File-Server/	Share/NTFS design, permission test matrix and results.
Print-Server/	Printer design, deployment method, troubleshooting.
Linux/	LINUX01 setup, hardening, monitoring configuration.
Monitoring/	Monitoring matrix, alert rules, dashboards (JSON exports).
Backup-DR/	Backup plan, RPO/RTO, scripts, restore test logs.
Security/	Hardening checklist, audit policy, baseline notes.
Troubleshooting/	Runbooks, failure exercise reports.
Scripts/	PowerShell/Bash scripts from Section 45 with README for each.
Screenshots/	Evidence organised by phase, using the naming convention.
Documentation/	Final PDF, final report, lessons learned, CV and LinkedIn text.

WARNING: Never commit passwords, DSRM secrets, private keys, real company names or personal data. Add a .gitignore and review every file before pushing.

45. PowerShell Automation (Optional)

All scripts are **lab-safe**: they read secrets interactively, check whether objects exist before creating them, support `-WhatIf` where it modifies state, and never delete anything. Save in `D:\NileBank-Lab\Scripts\` (or the repository Scripts folder). Run from an elevated PowerShell on DC01 with the ActiveDirectory module. Review each script before running.

45.1 New-NileBankOUs.ps1

Does: creates the NILEBANK OU tree from Section 14 if missing. Protected from accidental deletion. Safe to re-run (idempotent).

New-NileBankOUs.ps1

```
[CmdletBinding(SupportsShouldProcess)]
param([string]$Domain = (Get-ADDomain).DistinguishedName)
Import-Module ActiveDirectory
function Ensure-OU($name, $path) {
    $dn = "OU=$name,$path"
    if (-not (Get-ADOrganizationalUnit -Filter "DistinguishedName -eq '$dn'" -ErrorAction SilentlyContinue)) {
        if ($PSCmdlet.ShouldProcess($dn, 'Create OU')) { New-ADOrganizationalUnit -Name $name -Path $path -ProtectedFromA
    } else { Write-Host "Exists  $dn" }
}
Ensure-OU 'NILEBANK' $Domain
$top = "OU=NILEBANK,$Domain"
'Users','Computers','Servers','Groups','Service Accounts' | % { Ensure-OU $_ $top }
'IT','Finance','HR','Operations','Management'    | % { Ensure-OU $_ "OU=Users,$top" }
'Workstations','Laptops'                          | % { Ensure-OU $_ "OU=Computers,$top" }
'Domain Controllers','Application','File','Print' | % { Ensure-OU $_ "OU=Servers,$top" }
```

45.2 New-NileBankGroups.ps1

Does: creates the global role groups and domain local resource groups and nests them (AGDLP). Existing groups are skipped.

New-NileBankGroups.ps1

```
[CmdletBinding(SupportsShouldProcess)]
param([string]$Path = 'OU=Groups,OU=NILEBANK,DC=corp,DC=nilebank,DC=lab')
Import-Module ActiveDirectory
$global = 'IT-Users','IT-Admins','Finance-Users','HR-Users','Operations-Users','Management-Users','AllStaff'
$local  = 'Share-Finance-RW','Share-Finance-R','Share-HR-RW','Share-HR-R','Share-IT-RW','Share-Operations-RW','Share-Oper
foreach ($g in $global) { $n="GG-$g"; if (-not (Get-ADGroup -Filter "Name -eq '$n'")) { if ($PSCmdlet.ShouldProcess($n,'C
foreach ($l in $local)  { $n="DL-$l"; if (-not (Get-ADGroup -Filter "Name -eq '$n'")) { if ($PSCmdlet.ShouldProcess($n,'C
$map = @{ 'DL-Share-Finance-RW'='GG-Finance-Users'; 'DL-Share-Finance-R'='GG-Management-Users'; 'DL-Share-HR-RW'='GG-HR-U
          'DL-Share-Operations-RW'='GG-Operations-Users'; 'DL-Share-Operations-R'='GG-Management-Users'; 'DL-Share-Public
          'DL-Print-General'='GG-AllStaff' }
foreach ($k in $map.Keys) { if ($PSCmdlet.ShouldProcess("$k <- $($map[$k])",'Add member')) { Add-ADGroupMember -Identity
```

45.3 New-NileBankUsers.ps1

Does: reads users from a CSV (`GivenName,Surname,Sam,Department,Title,Group`), creates them in the department OU with a password entered at runtime and adds them to the role group. Skips existing accounts. Contains no passwords.

New-NileBankUsers.ps1

```
[CmdletBinding(SupportsShouldProcess)]
param([Parameter(Mandatory)][string]$CsvPath)
Import-Module ActiveDirectory
$base = 'OU=NILEBANK,DC=corp,DC=nilebank,DC=lab'
$pw = Read-Host -AsSecureString 'Initial password for new users'
foreach ($u in Import-Csv $CsvPath) {
    if (Get-ADUser -Filter "SamAccountName -eq '$($u.Sam)'") { Write-Host "Skip $($u.Sam) (exists)"; continue }
    if ($PSCmdlet.ShouldProcess($u.Sam,'Create user')) {
        New-ADUser -GivenName $u.GivenName -Surname $u.Surname -Name "$($u.GivenName) $($u.Surname)" -SamAccountName $u.S
            -UserPrincipalName "$($u.Sam)@corp.nilebank.lab" -Department $u.Department -Title $u.Title `
            -Path "OU=$($u.Department),OU=Users,$base" -AccountPassword $pw -Enabled $true -ChangePasswordAtLogon $true
        if ($u.Group) { Add-ADGroupMember -Identity $u.Group -Members $u.Sam }
        Write-Host "Created $($u.Sam)"
    }
}
```

users.csv example

```
GivenName,Surname,Sam,Department,Title,Group
Mona,Farouk,m.farouk,Finance,Accountant,GG-Finance-Users
Layla,Mostafa,l.mostafa,HR,HR Specialist,GG-HR-Users
```

45.4 Check-NileBankServices.ps1

Does: read-only. Checks critical services on each server over WinRM and prints a pass/fail table.

Check-NileBankServices.ps1

```
$checks = @{
  'DC01'='NTDS','DNS','Netlogon','KDC','DFSR','W32Time'; 'DC02'='NTDS','DNS','Netlogon','KDC','DFSR','W32Time';
  'DHCP01'='DHCPServer'; 'FILE01'='LanmanServer'; 'PRINT01'='Spooler' }
$result = foreach ($h in $checks.Keys) {
  foreach ($s in $checks[$h]) {
    try { $svc = Get-Service -ComputerName $h -Name $s -ErrorAction Stop; [pscustomobject]@{Host=$h;Service=$s;Status=$sv
    catch { [pscustomobject]@{Host=$h;Service=$s;Status='Unreachable';OK=$false} }
  } }
$result | Sort-Object OK,Host | Format-Table -AutoSize
if ($result.OK -contains $false) { Write-Warning 'One or more checks failed.' }
```

45.5 Check-NileBankDNS.ps1

Does: read-only. Resolves key names and SRV records against each DC's DNS and compares answers.

Check-NileBankDNS.ps1

```
$servers = '10.10.30.10','10.10.30.11'
$names = 'corp.nilebank.lab','dc01.corp.nilebank.lab','dc02.corp.nilebank.lab','file01.corp.nilebank.lab','www.microsoft.
foreach ($s in $servers) { foreach ($n in $names) {
  try { $r = Resolve-DnsName -Name $n -Server $s -ErrorAction Stop | Where-Object Type -eq 'A' | Select-Object -First 1;
  catch { "{0,-14} {1,-30} FAIL {2}" -f $s,$n,$_.Exception.Message } } }
foreach ($s in $servers) { try { (Resolve-DnsName -Type SRV -Name '_ldap._tcp.dc._msdcs.corp.nilebank.lab' -Server $s -Er
```

45.6 Check-NileBankADHealth.ps1 and Check-NileBankReplication.ps1

Does: read-only. Lists DCs, FSMO holders, runs a quiet dcdiag, and summarizes replication failures and last success times. Exports a timestamped text report to the Backups\Logs folder.

Check-NileBankADHealth.ps1 (includes replication)

```
$out = "D:\NileBank-Lab\Backups\Logs\ADHealth-$(Get-Date -Format yyyyMMdd-HHmm).txt"
New-Item (Split-Path $out) -ItemType Directory -Force | Out-Null
& {
  '=== Domain Controllers ==='; Get-ADDomainController -Filter * | Format-Table Name,IPv4Address,Site,IsGlobalCatalog -Au
  '=== FSMO ===';  netdom query fsmo
  '=== dcdiag (failures only) ==='; dcdiag /q
  '=== repadmin /replsummary ==='; repadmin /replsummary
  '=== Replication failures ==='; Get-ADReplicationFailure -Target (Get-ADDomainController -Filter *).HostName | Format-T
  '=== Partner metadata ==='; Get-ADReplicationPartnerMetadata -Target (Get-ADDomainController -Filter *).HostName |
       Format-Table Server,Partner,Partition,LastReplicationSuccess,LastReplicationResult -AutoSize | Out-String
} *>&1 | Tee-Object -FilePath $out
Write-Host "Report saved to $out"
```

45.7 Export-NileBankSystemInfo.ps1

Does: read-only. Collects inventory (OS, CPU, RAM, disks, IPs, hotfixes) from listed computers into a CSV for documentation.

Export-NileBankSystemInfo.ps1

```
$hosts = 'DC01','DC02','DHCP01','FILE01','PRINT01'
$rows = foreach ($h in $hosts) { try {
  $os=Get-CimInstance Win32_OperatingSystem -ComputerName $h -ErrorAction Stop; $cs=Get-CimInstance Win32_ComputerSystem
  $ip=(Get-CimInstance Win32_NetworkAdapterConfiguration -ComputerName $h | Where IPEnabled).IPAddress -join ','
  [pscustomobject]@{Host=$h;OS=$os.Caption;Build=$os.BuildNumber;RAM_GB=[math]::Round($cs.TotalPhysicalMemory/1GB,1);IP=$
} catch { [pscustomobject]@{Host=$h;OS='Unreachable'} } }
$rows | Export-Csv "D:\NileBank-Lab\Backups\Logs\Inventory-$(Get-Date -Format yyyyMMdd).csv" -NoTypeInformation
$rows | Format-Table -AutoSize
```

46. Project Phases

Nineteen phases, in order. Do not start a phase until the previous phase's validation passes. Detailed procedures live in the referenced sections; the tables below give objective, prerequisites, steps, commands, validation, common errors, troubleshooting and evidence for each.

Phase 1 — Hyper-V

Item	Detail
Objective	Prepare the host, storage, switches, VLANs and the router.
Prerequisites	Windows 11 Pro, SVM enabled, ISOs downloaded.

Steps

1. Enable Hyper-V and reboot (Sec 2.2).
2. Create D:\NileBank-Lab structure (Sec 3).
3. Create NILEBANK-LAB and NILEBANK-WAN, host NAT, VLAN 10 vNIC (Sec 4).
4. Build RTR01, netplan, nftables, DHCP relay (Sec 5.4).

Commands
`Get-VMSwitch; Get-NetNat; ip -br addr (RTR01); ping 192.168.250.1`

Item	Detail
Validation	Switches exist; RTR01 reaches Internet; host reaches 10.10.10.1.
Common Errors	Hyper-V role not installed (Home edition / BIOS virtualization off); NAT already exists; Linux NIC names differ from eth0/eth1.
Troubleshooting	Check Get-WindowsOptionalFeature, Get-NetNat, ip link, Hyper-V NIC MAC addresses.
Evidence	01, 02, RTR01 ip addr.

Phase 2 — DC01

Item	Detail
Objective	Create and install the first server.
Prerequisites	Phase 1 complete.

Steps

1. Create VM DC01 (Sec 8).
2. Install Windows Server, rename, patch, static IP 10.10.30.10 (Sec 9).

Commands
`ipconfig /all; ping 10.10.30.1; hostname; systeminfo`

Item	Detail
Validation	Static IP works; gateway reachable.
Common Errors	No boot (missed key press); wrong VLAN; no network because RTR01 off.
Troubleshooting	Boot order, Get-VMNetworkAdapterVlan, VM state.
Evidence	03, 04.

Phase 3 — AD DS + DNS

Item	Detail
Objective	Create the forest and AD-integrated DNS.
Prerequisites	Phase 2; DNS set to 10.10.30.10.

Steps

1. Install AD DS (Sec 11).
2. Promote: new forest corp.nilebank.lab, DNS + GC, DSRM.
3. Verify (Sec 12) and configure DNS (Sec 13).

Commands
`Get-ADDomain; Get-ADForest; Get-ADDomainController; dcdiag /test:dns /v; nslookup corp.nilebank.lab`

Item	Detail
Validation	Domain/forest properties correct; SYSVOL and NETLOGON shared; SRV records exist.
Common Errors	Weak password rejected; delegation warning (expected); DNS pointing to public resolver; prerequisite check fails (static IP or password).
Troubleshooting	Read dcpromo log C:\Windows\debug\dcpromo.log, event logs (Directory Service, DNS).
Evidence	05–08, 11, 12.

Phase 4 — OUs + Users + Groups

Item	Detail
Objective	Build the directory structure and AGDLP.
Prerequisites	Phase 3.

Steps

1. Create OUs (Sec 14).
2. Create users (Sec 15.2), groups and nesting (15.3), service accounts (Sec 16).

Commands
`Get-ADOrganizationalUnit -Filter *; Get-ADUser -Filter *; Get-ADGroupMember DL-Share-Finance-RW -Recursive`

Item	Detail
Validation	OU tree complete; each user in correct group.
Common Errors	Password does not meet policy; wrong OU path; OU protected from deletion blocks removal.
Troubleshooting	Check DN strings, run scripts with -WhatIf.
Evidence	09, 10.

Phase 5 — CLIENT01

Item	Detail
Objective	Create and domain-join the client.
Prerequisites	Phases 3–4; RTR01 routing.

Steps

1. Create VM (VLAN 20, TPM) and install Windows 11 (Sec 17).
2. Set static IP/DNS, join domain, log on as a domain user.

Commands
`whoami; systeminfo | findstr /B /C:"Domain"; nslookup corp.nilebank.lab; gpresult /r`

Item	Detail
Validation	Logon as NILEBANK\m.farouk; computer in Workstations OU.
Common Errors	'Domain could not be contacted' (DNS); TPM missing; time skew.
Troubleshooting	Section 30.3.
Evidence	13, 14.

Phase 6 — GPO

Item	Detail
Objective	Create the eight GPOs and test them.
Prerequisites	Phase 5.

Steps

1. Create and link GPOs (Sec 18.3).
2. Configure settings, filtering, exceptions, inheritance exercises (18.4).
3. Backup-GPO.

Commands
`gpupdate /force; gpresult /r; gpresult /h C:\Temp\gpresult.html`

Item	Detail
Validation	Settings apply to expected users/computers only.
Common Errors	GPO linked to wrong OU; filtering excludes Authenticated Users; setting in wrong half.
Troubleshooting	gpresult denied list, GPMC Results.
Evidence	15, 16.

Phase 7 — DHCP01

Item	Detail
Objective	Provide dynamic addressing.
Prerequisites	Phases 3, 5; RTR01 relay.

Steps

1. Build DHCP01 (Sec 19.2).
2. Install, authorize, create scope/options/reservations.
3. Switch CLIENT01 to DHCP.

Commands
`Get-DhcpServerInDC; Get-DhcpServerv4Scope; ipconfig /renew; Get-DhcpServerv4Lease`

Item	Detail
Validation	CLIENT01 receives correct lease and DNS.
Common Errors	Server not authorized; relay not running; scope inactive; VLAN mismatch.
Troubleshooting	Section 32.4.
Evidence	17, 18.

Phase 8 — FILE01

Item	Detail
Objective	Deliver departmental shares.
Prerequisites	Phases 4–5.

Steps

1. Build FILE01, add data disk (Sec 20.1).
2. Install role, create folders, NTFS, shares.
3. Run the allow/deny test matrix (20.5).

Commands
`Get-SmbShare; icacls; Get-SmbShareAccess; net use`

Item	Detail
Validation	All 9 tests pass.

Item	Detail
Common Errors	User still denied after group change (old token); Everyone left on ACL; share-vs-NTFS confusion.
Troubleshooting	Section 30.4.
Evidence	19–22.

Phase 9 — PRINT01

Item	Detail
Objective	Deliver shared printing.
Prerequisites	Phase 8 pattern.

Steps

1. Build PRINT01 (Sec 21).
2. Add driver, port, printers, share, publish, permissions.
3. Deploy to CLIENT01.

Commands
`Get-Printer; Add-Printer -ConnectionName; Get-Service Spooler`

Item	Detail
Validation	Test page queued; access control works.
Common Errors	Wrong driver architecture; permissions; spooler stuck.
Troubleshooting	Section 21.6.
Evidence	23, 24.

Phase 10 — LINUX01

Item	Detail
Objective	Build the monitoring server.
Prerequisites	Phase 1.

Steps

1. Create VM, install Ubuntu, static IP, hardening (Sec 22).
2. Install Zabbix (or Prometheus/Grafana).

Commands
`ip -br addr; ufw status; systemctl status zabbix-server`

Item	Detail
Validation	Web UI reachable from VLAN 10; host added.
Common Errors	Secure Boot template not set for Linux; DB password errors; firewall blocking.
Troubleshooting	journalctl -u zabbix-server; ufw status.
Evidence	25.

Phase 11 — DC02

Item	Detail
Objective	Add the second domain controller.
Prerequisites	Phases 3–5, DNS healthy.

Steps

1. Build DC02, join, install AD DS, promote to existing domain (Sec 23).
2. Update DNS client settings on all servers.

Commands
`Get-ADDomainController -Filter *; nslookup corp.nilebank.lab; net share`

Item	Detail
Validation	Two DCs listed, both GC; SYSVOL/NETLOGON present.
Common Errors	Selected 'new forest' by mistake; DC01 not reachable; DNS pointing elsewhere; time skew.
Troubleshooting	dcpromo log; Sec 24.
Evidence	28, 29, 31.

Phase 12 — Replication + Failover

Item	Detail
Objective	Prove health and redundancy.
Prerequisites	Phase 11.

Steps

1. Run replication checks (Sec 25).
2. Run the failover test (Sec 26) and record timings.

Commands
`repadmin /replsummary; repadmin /showrepl; dcdiag /test:dns /v; Stop-VM DC01; Start-VM DC01`

Item	Detail
Validation	0 failures; logons work with DC01 off.
Common Errors	Client lists only DC01 as DNS; stale cached sessions mask failure; replication delay.
Troubleshooting	Sec 25.3 errors table.
Evidence	30, 32, 33.

Phase 13 — Monitoring

Item	Detail
Objective	Monitor all hosts and alert.
Prerequisites	Phases 10–11.

Steps

1. Install agents; add hosts and templates (22.5, 29).
2. Create alert triggers; test by stopping a service.

Commands
`Get-Service ZabbixAgent2; Test-NetConnection LINUX01 -Port 10051`

Item	Detail
Validation	All hosts green; test alert fires and clears.
Common Errors	Agent firewall rule missing; wrong server IP in agent config.
Troubleshooting	Agent log; netstat -an | find "10050".
Evidence	26, 27.

Phase 14 — Backup

Item	Detail
Objective	Implement backup jobs.
Prerequisites	Phases 8, 11.

Steps

1. Build BACKUP01 and repository (27.1).
2. Configure System State, file, VM export, config backups (27.3).

Commands
`wbadmin get versions; vssadmin list shadows; Get-ScheduledTask NileBank-SystemState`

Item	Detail
Validation	At least one successful backup of each type.
Common Errors	Not enough space; backup target on same disk; credentials.
Troubleshooting	wbadmin logs, Event Viewer Backup.
Evidence	34, 35.

Phase 15 — Disaster Recovery

Item	Detail
Objective	Practise restores.
Prerequisites	Phase 14.

Steps

1. Run scenarios 1–6 (27.5).
2. Fill the restore test log with real results and timings.

Commands
`Restore-GPO; Import-VM; Restore-ADObject; Previous Versions`

Item	Detail
Validation	Each scenario restored and validated.
Common Errors	Restoring a DC copy as a clone; wrong backup version; Recycle Bin not enabled before deletion.
Troubleshooting	Section 27 warnings.
Evidence	36.

Phase 16 — Security Hardening

Item	Detail
Objective	Apply and verify controls.
Prerequisites	Phases 6, 11.

Steps

1. Apply Section 28.3 hardening and strict firewall.
2. Verify admin separation, auditing and logging.

Commands
`auditpol /get /category:*; Get-NetFirewallProfile; Get-MpComputerStatus; nft list ruleset`

Item	Detail
Validation	Controls verified by test.
Common Errors	Hardening breaks DHCP/relay or shares; overly strict GPO.
Troubleshooting	Roll back via Restore-GPO / previous nftables file.
Evidence	37.

Phase 17 — Failure Testing

Item	Detail
Objective	Run the 10 failure scenarios (Sec 32).

Item	Detail
Prerequisites	Phases 1–16.

Steps

1. Execute each failure on the lab only.
2. Fill in the eight-field table with real observations.

Commands
`As per each scenario`

Item	Detail
Validation	Normal operation restored and validated after each test.
Common Errors	Failing to restore the normal state; testing several faults at once.
Troubleshooting	Reverse the single change; restore from backup.
Evidence	38 series.

Phase 18 — Documentation

Item	Detail
Objective	Finish and review the documentation.
Prerequisites	Phases 1–17.

Steps

1. Complete evidence log (33), acceptance checklist (34).
2. Update IP and resource tables with actual values; final report (47).

Commands
`Review all screenshots; spell-check`

Item	Detail
Validation	No placeholders remain; all values match the lab.
Common Errors	Stale IPs, missing screenshots, passwords visible.
Troubleshooting	Peer review.
Evidence	All.

Phase 19 — Portfolio / CV / LinkedIn

Item	Detail
Objective	Publish the project.
Prerequisites	Phase 18.

Steps

1. Create GitHub repository (Sec 44).
2. Add CV bullets and LinkedIn text (41–42) truthfully.
3. Prepare the interview answers (43).

Commands
`git add; git commit; git push`

Item	Detail
Validation	Repository public, no secrets.
Common Errors	Claiming untested work; committing secrets.
Troubleshooting	Review every claim against evidence.
Evidence	README screenshots.

47. Final Project Report

> **NOTE:** Complete the italic fields with your real results. A report with unfilled brackets is a draft, not a deliverable.

47.1 Executive summary

Nile Bank is a simulated enterprise banking infrastructure built on a single Hyper-V host. It demonstrates identity, name services, endpoint management, segmentation, file/print services, monitoring, redundancy, backup and recovery, and security hardening, with a documented methodology and tested procedures. It is a learning project, not a production system.

47.2 Business scenario

Nile Bank (fictional) operates a head office with Finance, HR, IT, Operations and Management staff. The business requires centralised authentication, departmental data segregation, reliable name resolution and addressing, managed endpoints, shared printing, availability despite a server failure, recoverability after data loss, and evidence of control for audit.

47.3 Requirements

ID	Requirement	Addressed by
R1	Central identity and access control	AD DS, OUs, AGDLP, GPO
R2	Segmented network	VLAN 10/20/30, RTR01 firewall
R3	Name and address services	AD-integrated DNS, DHCP
R4	Departmental file and print services	FILE01, PRINT01
R5	High availability of authentication/DNS	DC02, replication
R6	Monitoring and alerting	Zabbix/Prometheus
R7	Backup and recovery	wbadmin, VSS, exports, DR scenarios
R8	Security hardening and auditing	Section 28
R9	Run on a 16 GB host	Dynamic memory, profiles, phases

47.4 Architecture, network, server, AD and security design

Architecture and network: see Sections 4–6 and 36. **Server design:** Sections 6, 37, 38. **Active Directory:** Sections 7, 14–16 — one forest/domain, two DCs, OUs by object type, AGDLP, separate admin accounts, gMSA. **Security:** Sections 18, 28 — layered controls with VLAN firewalling, GPO baselines, least privilege, auditing and protected backups.

47.5 Implementation, testing and failure testing

Area	Test	Result (fill in)
Networking	CLIENT01 → DC01 ping, DNS, logon	*pass/fail, date*
AD	dcdiag /q, repadmin /replsummary	*output summary*
Permissions	Allow/deny matrix (9 tests)	*x/9 passed*
Failover	DC01 off; DNS, logon, file access	*recovery time = __ min*
Failure tests	10 scenarios	*x/10 documented*

47.6 Backup, DR, monitoring and troubleshooting

Backups: Section 27 (System State, VSS, VM exports, configuration exports; RPO/RTO table; six recovery scenarios). Monitoring: Section 29. Troubleshooting: Sections 30–32. *Record the dates and results of each restore test in Section 27.6.*

47.7 Lessons learned

- DNS correctness is the foundation of AD; most failures trace to incorrect DNS client settings.
- Time sync (Kerberos) and DC redundancy must be designed, not assumed.
- Group-based permissions (AGDLP) make changes manageable and auditable.
- Restore tests reveal gaps that backup success messages hide.
- Limited RAM forces disciplined resource planning; phased profiles work well.
- Documentation written during implementation is more accurate than after the fact.

47.8 Future improvements, career value, conclusion

See Section 48 for extensions. Career value is summarised in Section 40. **Conclusion:** the lab provides a reproducible, documented, and tested foundation of enterprise infrastructure skills on commodity hardware, ready to be extended into hybrid and cloud scenarios.

48. Future Improvements

WARNING: Everything in this section is a **future extension**. It is NOT part of the current on-premises implementation and must not be claimed on a CV unless it has actually been built and tested.

Extension	Idea	Value
Azure	Create a free-tier subscription; resource groups, naming, tagging, RBAC	Cloud foundation
Microsoft Entra ID (Azure AD)	Cloud tenant; sync users with Entra Connect (hybrid identity); SSO, MFA, Conditional Access	Hybrid identity
Azure Virtual Network	VNet/subnets/NSGs mirroring VLAN design	Cloud networking
Azure VPN	Site-to-site or point-to-site VPN from RTR01/host to Azure	Hybrid connectivity
Azure Backup	Back up FILE01/DC System State to a Recovery Services vault (MARS agent)	Off-site backup
Azure Monitor	Log Analytics workspace, Azure Monitor Agent, alerts and dashboards	Cloud observability
Microsoft Defender	Defender for Endpoint/Cloud onboarding of lab servers	Security operations
Terraform	Provision Azure network and VMs as code	Infrastructure as Code
PowerShell automation / IaC	Desired State Configuration, automated builds with unattended answer files, packer/Hyper-V templates	Repeatability
CI/CD	GitHub Actions to lint scripts, validate Terraform, publish documentation	DevOps skills
Hybrid identity	Password hash sync, seamless SSO, group writeback	Enterprise identity
Azure Arc	Project lab servers to Azure for central management/policy	Hybrid management
Microsoft Sentinel	SIEM: ingest Windows security logs, build analytics rules	SOC skills

OPTIONAL / ADVANCED: Also consider: DHCP failover, DFS Namespaces + Replication, a Windows Server Core member server, WSUS, an internal PKI (AD CS), Exchange/MAIL01, RODC simulation and a second AD site with a simulated WAN link.

Project Completion Statement

> “This project demonstrates the design, implementation, administration, troubleshooting, security, redundancy, backup, disaster recovery and documentation of a simulated enterprise banking infrastructure environment.”

Prepared by:	Saleh Fadl
Role:	IT Support Engineer | System Administrator | Cloud Engineer
Project Type:	Personal Project / Home Lab
Environment:	Hyper-V + Windows Server + Linux
Status:	Enterprise Infrastructure Lab